EMIGRATION RESTRICTIONS AND ECONOMIC DEVELOPMENT: EVIDENCE FROM THE ITALIAN MASS MIGRATION TO THE U.S. *

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Abstract

Between 1920 and 1921, Italian emigration to the United States dropped by 85% after the Emergency Quota Act, a severely restrictive immigration law, was passed by the US Congress. Using newly digitized data from Italian historical censuses in a difference-in-differences setting, we leverage variation in exposure across Italian districts to this large restriction on human mobility. More exposed districts display a sizable population increase. Moreover, the policy substantially hampered the adoption of labor-saving technology. Consistent with directed technology adoption theory, manufacturing employment increased markedly, and evidence suggests that "missing migrants," whose migration was inhibited by the Act, drive this result.

Keywords: Age of Mass Migration, Emigration, Technology Adoption.

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INTRODUCTION

Technology adoption is a key driver of economic growth in developing countries, which operate far from the technology frontier (Suri, 2011; Bryan, Chowdhury and Mobarak, 2014).[1] This paper examines how out-migration, a typical feature of developing countries, influences the incentive for firms to adopt productivity-enhancing technologies. Specifically, we investigate how restrictions to human mobility imposed by immigration countries influence technology adoption in the emigrants' countries of origin.[2]

The effects of out-migration on technology adoption are *ex-ante* ambiguous and potentially conflicting. On the one hand, emigration entails a loss of human capital–a "brain drain"–that may hamper the ability of countries to adopt new technologies (Kwok and Leland, 1982; Gibson and McKenzie, 2011). On the other hand, however, higher emigration rates may incentivize the adoption of labor-saving technologies by increasing the relative cost of labor (e.g., see Habakkuk, 1962; Hicks, 1932). From the directed technical change theory perspective, one can interpret immigration restriction policies (henceforth, IRPs) as "passive" labor market policies that increase the labor supply in targeted countries. These positive labor supply shocks would, in turn, prompt the substitution of capital with more abundant–hence cheaper–labor, thus depressing investment in capital-intensive technologies. Which effect prevails is, ultimately, an empirical question.

We study the Italian mass migration, the largest episode of voluntary migration from a single country in recorded history (Choate, 2008). Between 1876 and 1924, approximately fourteen million emigrants left Italy (nearly 45% of the average Italian population in 1900); about half never returned. Italy had one of the highest emigration rates and, since the 1890s, it was the leader in sheer emigration numbers (Hatton and Williamson, 1998). On average, 40% of emigrants headed toward the United States, the most common destination country and the focus of this paper. The Italian mass migration to the United States abruptly ended in 1921 when Congress passed the first of two restrictive IRPs that we collectively refer to as the "Quota Acts." The Quota Acts defined numerical quotas on yearly arrivals from European countries, which drastically reduced the number of Italians entering the United States.[3] Between 1920 and 1921, the inflow of Italians in the US dropped by 85% and never recovered (see Figure I). A follow-up tightening policy was imposed in 1924. We exploit variation arising from this sharp and massive restriction to human mobility.

Throughout this period, a group of countries, including Italy, started experiencing industrializa-

[1] Economic historians have long recognized that countries that industrialized relatively late, such as Germany or Italy, relied heavily on innovation produced abroad to catch up with the core industrial countries (Gerschenkron, 1962; Rosenberg, 1982). Recent literature on endogenous growth theory embeds technology diffusion dynamics and quantifies their substantial contribution to productivity growth (e.g. Eaton and Kortum, 1999; Buera and Oberfield, 2020).

[2] For brevity, we refer to those policies as immigration restriction policies (IRPs). Data from de Haas, Natter and Vezzoli (2015) suggest that IRPs have become increasingly common since the 1970s and currently account for 40% of the entire corpus of migration laws.

[3] The 1921 Emergency Quota Act restricted the annual number of immigrants admitted into the US to no more than 3% of the number of residents from that country, as recorded in the 1910 census. The 1924 Johnson-Reed Act reduced the quota to 2% and pegged the reference date to the 1890 census. These laws explicitly targeted Southern and Eastern European countries, which until the early 1900s hardly took part in the Age of Mass Migration and whose immigrants were perceived by the public as a threat to America's economic welfare and cultural values (Higham, 2002).

tion and structural change as part of a broader transformational phenomenon known as the "Second Industrial Revolution" (Mokyr, 1998). The Italian economy, in particular, outperformed the leading industrial countries for the first time between 1895 and 1913. The postwar years, especially the Fascist period, however, were marked by economic stagnation and languishing productivity growth. The economic divide between Northern and Southern areas, which had started to narrow during the economic boom, severely widened during the 1920s and 1930s (Cohen and Federico, 2001). Previous scholarship documents that insufficient investments, especially in the South, hampered the adoption of productivity-enhancing technologies. In this paper, we argue that scarce investments partly resulted from the post-1921 restrictive US immigration policy, which, therefore, plausibly contributed to the widening economic gap between the North and the South of the country and, more generally, to the disappointing performance of the Italian economy in the 1920s and 1930s.

To identify the effect of the American immigration restriction policy shock on Italian economic development, we define a district in Italy as more exposed to the Quotas if a larger proportion of its emigrants had moved to the United States before 1921, conditional on the overall number of international emigrants. In a difference-in-differences setting, we thus compare districts located in provinces with similar emigration outflows and leverage variation among destination countries.[4] Formally, our identification assumption requires that districts located in provinces with similar volumes of international emigrant outflows but whose emigrants headed toward different destinations would not have undergone diverging development trajectories in the absence of the Quota Acts. We provide a battery of checks to ensure the plausibility of this assumption, but ultimately, we cannot test the conditional exogeneity of treatment intensity. We thus adapt the shift-share instrumental variable design developed by Card (2001) to construct plausibly random variation in the intensity of exposure to the Quotas across Italian districts. The instrument predicts district-level emigration to the US over time by interacting the shares of initial migration flows between Italian districts and US counties with subsequent non-Italian immigration inflows into US counties. The "naïve" and the instrumented difference-in-differences designs yield quantitatively consistent results.

The historical setting allows us to overcome several limitations of contemporary scenarios. First, emigration seldom flows into only a few destinations; hence, observing significant restrictive policy shifts is challenging. Second, migration dynamics are often affected by co-evolving regulations enacted by both receiving and sending countries, which were absent during the period we study (Abramitzky and Boustan, 2017). Third, it is often difficult to retrieve information on emigrants in their home country (Dustmann, Frattini and Rosso, 2015).

Existing data from official statistics are not suitable for our analysis because (i) digitized US and Italian censuses and complementary historical statistics do not report the origin of Italian migrants at a granular level of spatial aggregation, and (ii) disaggregated indicators of economic performance for Italy remain scarce during this period. We thus construct a dataset that links the administrative records of Italian emigrants who arrived at Ellis Island between 1892 and 1930 to their district of

[4] This intensity-of-treatment design is closely related to the conceptual framework adopted by Abramitzky, Ager, Boustan, Cohen and Hansen (2023) to study the effect of the Quota Acts on the US labor market.

origin, we match a subset of these records to the US full-count population census, and we digitize comprehensive, new data from Italian industrial and population censuses.

The empirical analysis proceeds in three steps. First, we establish that the Quota Acts had a tangible impact on Italian districts. Theoretically, it is possible that those who would have migrated to the United States in the absence of the policy shift could move to a different country. Looking at aggregate emigration numbers (Figure I), this seems implausible. Emigration to the United States completely dried up after 1921, but emigration to other countries did not. The total number of emigrants in the 1920s is roughly comparable to that in the 1880s and early 1890s before the US migration gained momentum.

We formally test this "imperfect substitution" argument through the DiD design and find that the population in districts that were conditionally more exposed to the Quota Acts increased after 1921. The effect of the Quotas on the Italian population is sizable in magnitude. A 1% increase in the number of US emigrants leads to a 0.05% increase in population. Equivalently, moving from the 50th to the 75th percentile of the distribution of exposure to the policy yields 8,800 additional population, relative to an average pre-Quota population of approximately 247,000. The change in population is equivalent to almost 80% of the number of US emigrants between 1892 and 1921. While this figure should be interpreted with caution, for it includes foregone migrants as well as their children who would not have been born if they had migrated, it nonetheless suggests that a substantial share–plausibly, at least 50%–of those who would have migrated remained in Italy after the United States closed the border. This finding is consistent with previous studies documenting the spatial persistence of Italian emigration flows (Gould, 1980b; Brum, 2019; Spitzer, Tortorici and Zimran, 2025). More qualitative cross-country evidence confirms that emigration outflows toward countries that did not promulgate IRPs did not increase. Hence, districts supplying relatively more US emigrants before 1921 ended up with more "missing" migrants, i.e., people who would have migrated in the absence of the Quota Acts. This mechanism generates a spatially segmented positive labor supply shock.

Our second finding is that manufacturing firms in provinces more exposed to the Quota Acts substantially decreased investment in capital goods. We estimate negative effects across several measures of capital goods, ranging from firms with at least one engine to the number of installed engines and the power they generate. For instance, a 1% increase in US emigration yields a 0.17% decrease in the horsepower generated by mechanical engines and a 0.23% drop in the energy generated by electrical ones. All sectors within manufacturing exhibit similar decreases in capital investment. How can we reconcile the increasing population with the decreasing adoption of productivity-enhancing production technologies? We propose interpreting these findings through the lens of directed technological change theory (Acemoglu, 2002, 2007). As labor becomes a more abundant production input, firms are incentivized to forego investment in capital goods and employ more labor. In Appendix D, we develop a simple theoretical framework in the spirit of Zeira (1998) and San (2023) to show that investment in labor-saving technologies would decrease in response to a positive labor supply shock.

To test this interpretation, we study how employment across sectors reacted to the Quota Acts. First, we explore how agriculture and manufacturing employment responded to the policy shock at the district level. We find that manufacturing employment increases in districts that are more exposed to the IRP shock. Quantitatively, a 1% increase in instrumented US emigration yields a 0.084% increase in manufacturing employment while the effect on agriculture employment–a 0.016% decrease–is not statistically significant. Historical evidence suggests that Italian agriculture in the 1920s was organized as a heavily labor-intensive sector (Cohen and Federico, 2001). It thus seems plausible that manufacturing could be the primary beneficiary of the substantial labor supply shock. Second, we look at how different sectors within manufacturing absorbed the population increase. Analogous to the capital results, we estimate comparable increases in manufacturing employment across industries. Thus, the aggregate and sector-level employment and capital responses to the IRP shock are consistent with the predictions of the directed technical change conceptual framework.

We find little empirical support for several alternative mechanisms that may explain our results: internal migrations, gender-biased structural change triggered by skewed sex ratios among US emigrants, monetary remittances, brain drain, and brain gain dynamics. While we do not reject the possibility that other mechanisms may have operated alongside directed technical adoption incentives, the weight of the empirical evidence in favor of our proposed mechanism appears compelling.

Overall, this paper documents that policies enacted by immigration countries to curtail migrant inflows bear important consequences on countries sending migrants. Foregone emigration, in fact, generates a positive labor supply, which, in turn, dampens technology adoption, thus potentially hampering long-run prospects of economic growth. Despite the historical setting, the Italian emigration to the United States shares several similarities with contemporary migration flows between developing and developed countries. We thus view our results as informative for the current migration policy debate and for a more comprehensive evaluation of the consequences of immigration restriction policies.

Related Literature This paper is related to three streams of literature. First, we speak to several contributions investigating the impact of emigration on sending countries, as opposed to the larger literature studying the economic and social effects of immigration (Clemens, 2011). Emigration has been shown to impact economic development (Khanna, Murathanoglu, Theoharides and Yang, 2022), wages (e.g., Dustmann et al., 2015), human capital formation (Dinkelman and Mariotti, 2016), attitudes towards democracy and voting (Spilimbergo, 2009; Batista and Vicente, 2011; Barsbai, Rapoport, Steinmayr and Trebesch, 2017) and political change (Chauvet and Mercier, 2014; Kapur, 2014; Karadja and Prawitz, 2019), the diffusion of novel knowledge (Coluccia and Dossi, 2025), entrepreneurship (Anelli, Basso, Ippedico and Peri, 2023), and social (Beine, Docquier and Schiff, 2013; Bertoli and Marchetta, 2015; Tuccio and Wahba, 2018) and cultural (Knudsen, 2024; Godlonton and Theoharides, 2025) norms. We inform this literature by showing that emigration fosters the adoption of labor-saving technologies. We emphasize that this channel operates plausibly independently from human capital accumulation. In addition, we document that this mechanism arises in response to restrictive immigration policies enacted by receiving countries.

Second, we contribute to the literature that studies the relationship between technology adoption and the supply of production inputs. Following the seminal contributions by Hicks (1932) and

Habakkuk (1962), Hornbeck and Naidu (2014), Clemens, Lewis and Postel (2018), and Hanlon (2015) all study historical settings where changes in the availability of labor and other factors of production altered the direction of innovation activity. Lewis (2011) offers similar evidence in a modern setting. Our paper is closest in spirit to Andersson, Karadja and Prawitz (2022), who show that labor-saving innovation emerged in response to migration-induced labor shortages in 19th-century Sweden. We present several novel findings relative to their paper. First, we show that immigration restriction policies generate a positive labor supply shock in emigration countries because potential emigrants do not fully substitute the restricted destination with other countries. Second, we document that a *positive* labor supply shock depresses investment in capital technology. In principle, since a positive labor supply shock may sustain aggregate demand, its impact on investment is ambiguous. Our results, therefore, highlight the centrality of directed technical adoption incentives for firms' decision-making. Third, instead of innovation, we focus on technology adoption, which is the core driver of economic growth and modernization in developing countries (Suri, 2011; Bryan et al., 2014; Juhász, Squicciarini and Voigtländer, 2020). Finally, the episode we study allows us to implement a difference-in-differences estimation strategy whose identification assumption can be evaluated transparently.

Third, by its setting, this paper adds to the literature on technical change and the diffusion of novel technologies during the Age of Mass Migration. A growing number of studies examine the short-run (Arkolakis, Peters and Lee, 2020; Diodato, Morrison and Petralia, 2022; Moser, San and Parsa, 2025) as well as the long-run (Akcigit, Grigsby and Nicholas, 2017; Sequeira, Nunn and Qian, 2020; Burchardi, Chaney, Hassan, Tarquinio and Terry, 2025) implications of immigration on US innovation. More specifically, we contribute to the literature examining the archetypal Italian case of mass migration. Among others, Hatton and Williamson (1998) studies the aggregate determinants of Italian emigration. Spitzer et al. (2025) validate the Gould (1980) theory, whereby social networks exerted substantial influence on Italian emigration dynamics. Pérez (2021) compares the assimilation dynamics of Italian emigrants to the United States with those who moved to Argentina. Our contribution to this literature is twofold. Methodologically, we present newly digitized district-level data from Italian population and industrial censuses. In terms of new findings, we show that the mass migration was unlikely to have hampered the structural shift toward manufacturing. Our results suggest that the opposite impact prevailed: the immigration *restriction* likely hampered economic modernization in Italy.

Outline of the paper We structure the paper as follows. Section I describes the Italian mass migration, the policies that shaped it, and the fundamental economic characteristics of early 20$^{\text{th}}$-century Italy. Section II discusses our data-collection contribution and the sources. Section III describes the difference-in-differences and instrumental variable strategies. In Section IV, we present our three sets of results. Section V concludes.

I HISTORICAL BACKGROUND

I.A The Italian Mass Migration

The Italian mass migration (1876–1925) was the largest episode of voluntary migration in recorded history (Choate, 2008). Over this period, approximately 14 million, corresponding to 45% of the Italian population in 1900, emigrated. Most emigrants headed toward continental Europe and the Americas. Along with Ireland, Italy had the highest per capita emigration rate (Taylor and Williamson, 1997). Even though Bandiera, Rasul and Viarengo (2013) document that returns rates were among the highest in Europe, the Italian mass emigration has long been recognized as a focal feature of the country's development process (Hatton and Williamson, 1998).

I.A.1 A Short History of the Italian Mass Migration

Italy was a latecomer to large-scale mass migration. Northern European countries had been experiencing substantial population outflows since the 1840s. By contrast, Italy and other Southern and Eastern European countries did not start experiencing mass emigration until the 1880s. The country's migration patterns over 1870–1925 display substantial time variation. Until the 1880s, emigration rates remained relatively modest, and most migrants hailed from Northern regions. Prohibitively high transportation costs and prevailing poverty in rural Southern areas largely inhibited migration from the *Mezzogiorno*. During the 1880s, Northerners chiefly moved to neighboring countries on a temporary, seasonal basis (Sori, 1979). The widespread adoption of steamships and an agrarian crisis in the late 1880s kicked off the Southern mass emigration (Keeling, 1999). By the early 20th century, the script had flipped: most migrants were now coming from Southern regions. Though the share of migrants from Northern regions declined as the share from Southern regions grew, emigration rates from *both areas* rose steadily from 1870 to 1913 (Hatton and Williamson, 1998). By the 1890s, Italy had become the global leader in sheer numbers of emigrants and in emigration rate, which grew from 5‰ in 1880 to a peak of 25‰ in 1913.

Italian emigration collapsed during World War I (WWI) but quickly regained momentum in the years immediately following the war. The epoch ended in the early 1920s when the U.S. Congress enacted restrictive immigration policies that effectively halted mass emigration to the United States. Emigration to other transoceanic and European destinations nonetheless endured until the outbreak of World War II. In Table B.2, we tabulate data from official statistics on regional out-migration to the United States and other international destinations to gauge the geographical evolution of the Italian mass migration over time.

Internal migration is one last component of labor mobility in Italy during the Age of Mass Migration that remains largely overlooked in the historical literature. Data limitations hinder a quantitative study of internal migrations from 1870 to 1925. In the main analysis, we abstract from explicitly accounting for internal migrations for three reasons (beyond data availability). First, Gallo (2012) shows that internal migrants were easily outnumbered by international migration flows, particularly during the Age of Mass Migration. We provide a quantitative assessment of this claim in Section IV.E.2, where we confirm that the number of international emigrants is one order of magnitude larger than that of internal migrants. Second, internal mobility was largely temporary and seasonal, inherently different from transoceanic migration (Gallo, 2012). Third, internal migrations reflected historically deep-rooted, persistent economic relationships between regions unlikely to influence our results on economic modernization in the 1930s. We nonetheless explore the dynamics of internal migrations after the Quota Acts in Section IV.E.

I.A.2 Composition and Determinants of the Migratory Movements

In the 1880s, Italy was a young nation rife with regional disparities spanning cultural and economic dimensions (Smith, 1997). The observed geographic segmentation in migratory patterns largely reflected this heterogeneity and provides our empirical strategy's backbone. Until the early 1890s, most migrants from Northern regions moved to European countries, while the rest steamed across the Atlantic to Argentina and Brazil. Since the mid-1890s, however, most emigrants originated from Southern Italy and, by contrast, primarily headed toward the United States.

To explain why destinations with low relative wage gaps, such as Argentina and Brazil, received sizable migration inflows, Gould (1980b) hypothesizes that local emigration dynamics were driven by information diffusion. Information about emigration opportunities required time to spread across the country, and this diffusion accelerated as the volume of emigration increased. Gould (1980b) provides evidence that declining regional emigration-rate inequality is consistent with this mechanism. An indirect consequence of the Gould hypothesis is that local emigration rates displayed relatively little sensitivity to economic and demographic conditions, and instead featured high persistence (Hatton and Williamson, 1998). Spitzer and Zimran (2023) provide causal quantitative evidence consistent with Gould's diffusion hypothesis. They show that emigration began in a few districts in the 1870s and 1880s and subsequently spread to nearby districts over time through immigrants' social networks. In Appendix A.III.7, we present some evidence that points in the same direction.

I.B Migration Policy in Italy and the United States

An appealing feature of this context is that migratory flows from Europe to the United States remained essentially unregulated until 1921 (Abramitzky and Boustan, 2017). This section describes the key features of the historical Italian and U.S. systems of migration laws.

I.B.1 Italian Emigration Policy

Newly unified Italy had virtually no emigration policy until 1873. Occasional, largely ineffective provisions were enacted between 1873 and 1887 that reflected the perceived need to deal with labor agents and recruiters, the so-called *padroni*, but did not form a comprehensive corpus of migration law (Gabaccia, 2013). The first such attempt was the 1888 Crispi-De Zerbi law, which introduced and regulated the emigration contract between the migrant and the migration agent. However, the law was manifestly inadequate to deal with the waves of migration that unfolded starting in the 1890s (Foerster, 1919).

As emigration to the United States gained momentum, a new law was passed in 1901, reflecting the renewed understanding that emigration could have a positive impact on Italy (Foerster, 1919). As such, the law sought to protect migrants from exploitation rather than restricting their movement.

The law established a Commissioner-General of Emigration to oversee the institutions that were set up to protect migrants and collect data. Only companies licensed by the Commissioner-General could sell tickets, whose rates were reset every three months. Comparatively minor subsequent legislation further protected remittances (1901), strengthened the authority of the Commissioner-General (1910), and regulated citizenship (1913) (Rosoli, 1998).

I.B.2 American Immigration Policy

The United States maintained an open border between 1775 and the early 1920s, interrupted only by isolated outbreaks of anti-immigration policy interventions. During the Age of Mass Migration, some 30 million migrants entered the United States. By 1910, 22% of the labor force was foreign-born, the highest share ever since (Abramitzky and Boustan, 2017). In 1907, the United States Congressional Joint Immigration Commission, also known as the "Dillingham Commission," was formed to study immigrants' economic and social conditions and their impact on the native population. The Commission's 41-volume report favored "old" immigration countries such as England and Germany over "new" ones, mainly in Southern and Eastern Europe.

When immigration ramped up again after WWI, nativist demands for restrictions surged, and the Emergency Quota Act was passed in 1921. It was modified by the 1924 Immigration Act, which further tightened immigration restrictions on second-wave countries. The 1921 Emergency Quota Act envisaged a (temporary) annual quota of 360,000 immigrants from Europe.[5] Importantly, for our identification, entry quotas were assigned to each country as 3% of that country's nationals living in the United States in 1910, as recorded in that year's census. The 1924 Immigration Act made the quota system permanent, lowered the inflow from 3% to 2%, and shifted the census baseline year to 1890. The last provision, in particular, targets Southern and Eastern European countries that participated in the mass migration starting in the late 1890s. Abramitzky et al. (2023) note that the impact of the 1924 Immigration Act on immigration was highly heterogeneous across sending countries. Flows from Southern and Eastern Europe were heavily curtailed because the share of foreign-born individuals from those countries who lived in the United States in 1890 was very modest. Since the 1890s, the United States had been absorbing 30% to 40% of all Italian emigration, so the Quota Acts represented a significant policy shock for Italy.

I.C Technology Adoption and Economic Growth in Italy

Italy entered the Age of Mass Migration in the 1880s. The country was amid an agrarian crisis that followed two decades of stagnation (Toniolo, 2014). The period from 1895 to 1913 was the only time until the 1950s "economic miracle" when Italy outperformed and narrowed the income gap with the leading industrial nations. Still in the 1920s and 1930s, however, Italy remained a mainly agricultural country, featuring low income per capita and languishing productivity growth (Cohen and Federico, 2001). During the first half of the Fascist Regime, economic policy was aimed primarily at fiscal and monetary consolidation. Agricultural policy, which formed an integral part of Fascist propaganda, centered on boosting agricultural productivity, which had been stagnating since World War I, and

[5]U.S. immigration peaked in 1907 at 1,285,349 entrants. The number of entrants during the 1910s averaged around 800,000.

draining marshlands. However, sheer numbers attest that agrarian policies resulted in neither substantial intervention nor sizeable progress (Zamagni, 1993). Growth slowed after 1925, and regional disparities widened (Cohen and Federico, 2001).

We relate the migration shock to diminished investment in capital goods, especially technologically advanced ones, and a shift to labor-intensive production routines. Italy operated far from the technological frontier throughout the period, and skill premia declined from the 1890s onward (Vasta, 1999). Similar to contemporary developing countries, Italy lagged behind advanced industrial nations in research and development expenditures, and it imported substantial amounts of foreign technology, both patents and machinery (Cohen and Federico, 2001). A large pool of unskilled workers made it more profitable for Italian entrepreneurs to adopt labor-intensive technologies relative to the highly capital-intensive German and British ones.

II DATA

Our analysis spans the years 1881 to 1936. We collect data from several sources; data are organized by census years and analyzed at the *circondario* (henceforth, "district") level if available. Otherwise, the units of observation are provinces.[6] In 1921, there were 216 districts and 69 provinces, each consisting of a variable number of municipalities (see Appendix A.I for a complete description of the data). Because districts were abolished in 1927, all subsequent district-level data are collected at the municipality level and aggregated at the 1921 district boundaries. We adopt the geographical cross-walk procedure described by Eckert, Gvirtz, Liang and Peters (2020) to ensure geographic consistency at 1921 district and province borders. Table I reports summary statistics for the variables in our final dataset. Table B.1 lists the source and coverage of each variable in the final dataset.

II.A Emigration Data

Italian official emigration statistics provide insufficient information because out-migration flows by destination were recorded at the province level (Hatton and Williamson, 1998). This limitation poses a major challenge because it precludes the use of official statistics for a spatially detailed econometric exercise. Data from official statistics may also not be reliable, as official publications relied on issued passports, which were not required to emigrate to the US. We nonetheless digitize province-level emigration outflows from official statistics and use them to construct overall emigration rates and validate the series we derive from the individual dataset we assemble.

To overcome these issues, we collect the records of Italians who entered the United States between 1892 and 1924 through the Ellis Island immigration station.[7] To the best of our knowledge, the Ellis Island repository is the most comprehensive source spanning the whole Age of Mass Migration for

[6]Population censuses were taken in 1881, 1901, 1911, 1921, 1931, and 1936. Manufacturing censuses were taken in 1911, 1927, and 1936. Districts were instituted in 1859 as the middle administrative unit between municipalities and provinces. They had mainly statistical and judicial purposes and were granted little administrative autonomy. Districts were organized in provinces, which encompassed one to five districts. In the Appendix A.I, we discuss the sources we digitized in more detail and present a visual summary of all the variables we analyze. In Appendix A.II, we describe how the final estimation samples are constructed.

[7]These records are freely available at `heritage.statueofliberty.org`.

Italy at this level of aggregation.[8] Our final dataset matches the coverage of the data constructed by Gray, Narciso and Tortorici (2019), who also rely on the Ellis Island passenger roster files. We enrich these records by applying state-of-the-art natural language processing to names and surnames, and by complementing them with confidential census information, as explained below. Spitzer and Zimran (2018) relied on a small subset of the overall Ellis Island data in their seminal contribution. Spitzer and Zimran (2023) construct a municipality-level emigration dataset from tabulations contained in Italian official statistics.

We search for the universe of surnames of Italian immigrants that appear in the US census (Ruggles, Fitch, Goeken, Hacker, Nelson, Roberts, Schouweiler and Sobek, 2021), yielding an individual-level dataset of 4,194,728 uniquely identified immigrants we identify as Italians.[9] Historians estimate that between 4 and 4.5 million Italians entered the US over this period; hence, the data coverage appears comprehensive.[10] Administrative records report, for most migrants, name and surname, year of arrival, age, municipality of origin, and sailing ship. We impute their gender, which we use when testing the possible mechanisms behind the main results, from their name, as detailed in Appendix A.III.4.

The key information we extract from the immigrant records is the immigration year and the municipality of origin. The immigration year is never missing in the data. Table B.4 displays the coverage of the municipality of origin variable. Municipalities are missing in approximately 12% of the records.[11] Missing municipalities are most common before 1895, during World War 1, and after 1921, when the total number of emigrants was generally lower, as evidenced in Figure C.2. We geo-reference the remaining municipalities using the Google Maps API, as explained in Appendix A.III.3. Approximately 32% of records are geo-coded to a location outside of Italy. In the vast majority of cases, these records list a location within the US as their last place of residence. It is impossible to distinguish reporting errors from migrants who had previously immigrated to the US and returned after a temporary stay in Italy, as documented by Wyman (1993). Ultimately, we geo-reference 2,339,406 immigrant records to the latitude and longitude coordinates of their municipality of origin in Italy. This is the sample that we employ in the analysis and whose reliability we check by comparing it with external data.[12]

A plausible concern is that, since immigrants were asked to report their last municipality of residence, they may have indicated their port of departure rather than their place of origin. If this were the case, then one would expect Genoa, Naples, and Palermo–the three ports of departure of transat-

[8] Brum (2019) produced a similar dataset for the pre-1890s period.

[9] The Ellis Island nationality information relies on the reported municipality of origin, so it is missing whenever the municipality is missing. In Appendix A.III.2, we explain our approach to impute the nationality of immigrants with no reported municipality using natural language processing, which we validate using census and geographic data.

[10] According to official US statistics, between 1892 and 1924, a total of 14,277,144 migrants entered the country through Ellis Island, out of a total immigration inflow of 20,003,041 (Unrau, 1984, p. 185). Thus, Ellis Island alone accounted for 71.4% of the total immigrant inflow. Some 95% of all Italian immigrants passed through Ellis Island.

[11] We consider a municipality as "missing" whenever (i) it does not appear in the Ellis Island dataset, or (ii) the listed municipality corresponds to a coarser geographical unit (e.g., "Italy" or "Sicily"). It is worth noting that the share of missing municipalities for other immigrant groups and periods appears to be considerably higher than for Italians between 1892 and 1924.

[12] The software fails to geo-reference 10,003 records, equal to 0.24% of the sample, because the recorded municipality cannot be matched to an existing municipality.

lantic ships–to display disproportionately high US emigration rates. In Figure C.5, we display the spatial distribution of US emigrants normalized by population in 1881, before the mass migration movement started. The map highlights the provinces of the three transatlantic port cities (in red) and their provinces. We do not find higher US emigration rates in port municipalities compared to their surrounding areas. In Table B.6, we report the results of a regression between the municipality-level US emigration rate and an indicator equal to one for municipalities with a port (columns 1–2) or located in the same province as a port (columns 3–4). We find no higher US emigration rates in such areas, suggesting that immigrants at Ellis Island were not more likely to report their port of departure as their municipality of origin.

A related concern is that migrants may have reported a large or significant city close to their place of origin, rather than their actual city of origin, to the Ellis Island enumerators. In Table B.7, however, we show that neither regional capital cities (column 1) nor provincial capital cities (column 2) display disproportionately high US emigration rates. Additionally, we do not find a significant correlation between city population and US emigration rates (columns 3 and 4). It thus seems plausible that migrants reported their origin municipality, rather than other possibly better-known cities, as their last place of origin in the Ellis Island records.

In Figure C.3, we report the correlation between the number of US emigrants in each region and year in the Ellis Island dataset and in tabulations contained in the annual reports of the Italian Commissioner General for emigration. While official statistics are based on the number of issued passports, which were not necessary for emigration to the United States, they remain informative for validating the coverage of the Ellis Island dataset. We find a strong and highly statistically significant correlation between the two datasets, both in the raw data (Figure C.3a) and when controlling for region and year fixed effects (Figure C.3b). The correlation remains high and statistically significant throughout the sample period, as shown in Figure C.6. Table B.5 tabulates the estimates. The unconditional correlation between the two datasets is 0.92 (column 1), indicating that the Ellis Island dataset accounts for a significant proportion of US emigration, as reported by Italian official statistics.

Figure I plots the country-level yearly inflow of immigrants who landed in Ellis Island from 1892 to 1930 (red solid line) compared to the aggregate number of international emigrants (blue dashed line). Emigration took off in the mid-1890s and peaked between 1905 and 1913. It collapsed during WWI, quickly regained momentum in 1920, and was definitively shut down by the Quota Acts in 1921 and 1924. Our data are consistent with comprehensive US immigration data and overall Italian migration patterns (Brum, 2019; Sequeira et al., 2020). In Figure II, we plot the geographical distribution of migrants across districts. Figure IIa displays the number of international emigrants across provinces. Figure IIb reports the volume of US out-migration across districts. Out-migration flows exhibit substantial dispersion across provinces and are not clustered in any specific area of the country. Emigration to the United States appears to be more clearly concentrated in Southern districts. Appendix A.III.7 presents further stylized facts that the data allows to document.

To construct a Bartik-type instrumental variable, we link the individual-level Ellis Island immigration records for 1895–1900 to the full-count US population census (Ruggles et al., 2021). To our knowledge, this is the first attempt to link census records with Ellis Island data. Concretely, we link individuals with records of Italians who appear in the 1900 census using their name, surname, age, and immigration year. We detail the procedure in Appendix A.III.8. Using the linked sample, we attach a municipality of origin to Italian migrants as they appear in the US census, allowing us to compute migration flows from Italian districts to US counties. We verify that the sample of linked migrants is balanced and hence representative of the full population (Figure C.10) and that, at the district-level, US out-migration computed on the full and the linked samples are strongly positively correlated (Figure C.11) for the years 1895–1900, when they overlap.

We construct the total number of international emigrants (irrespective of their country of destination) from official statistics compiled by the Commissioner General for Emigration. The data cover the same period as the Ellis Island dataset, but are available at the province level. Since more geographically disaggregated data do not exist, our analysis employs province-level emigration as a control variable, and, as a robustness exercise, we present the results obtained by attributing emigration at the district-level using two imputation methodologies in Section III.B.

II.B Population and Manufacture Censuses

The Italian Statistical Office (ISTAT) compiled the population of each municipality from the historical population censuses. We aggregate these tabulations by district and province for each census between 1881 and 1936.[13] We compute the k-urbanization rates as (i) the share of the population residing in cities with at least k-thousand inhabitants and (ii) the share of cities with at least k-thousand inhabitants. These tables also contain the altitude and area of each municipality and an indicator variable returning a value of one for towns near the sea. We collapse the first two at the district and province levels, weighting them by municipality population and tagging districts and provinces with access to the sea.

We construct manufacturing and agriculture employment series from disaggregated data listed in census records. These are available at the district level before 1921 and the municipality level afterward. For consistency, we recast them at the district level throughout the sample period. Employment in agriculture was not reported in the 1931 population census. Population censuses contain sector-level employment data for manufacturing firms at the district level until 1921. After 1921, the same variables are reported in the manufacturing census at the provincial level. Thus, we can observe sector-level manufacturing employment at the province level throughout the sample period.

Manufacturing censuses contain detailed information on the quantity and quality of capital employed in each province by manufacturing firms in 1911, 1927, and 1937. We collect within-manufacturing sector-level data on (i) the number of operating firms, (ii) the number of operating firms employing inanimate horsepower, (iii) the number of mechanical engines, (iv) the number of electrical engines, (v) the amount of horsepower generated by mechanical engines, and (vi) the amount of horsepower generated by electrical engines. We distinguish between electrical and mechanical engines because

[13]Population censuses were taken in 1881, 1901, 1911, 1921, 1927, and 1936. Manufacture censuses were taken in 1911, 1927, and 1936. Data in manufacturing censuses are only available at the provincial level.

the former were at the forefront of technological progress (David, 1990). This allows us to disentangle the possibly differential impact of the labor supply shock induced by the migration shock on different technology vintages.

II.C Other Data

Italy participated in World War I between 1915 and 1918. Because the war took place between two census years and ended just three years before the Emergency Quota Act, it can potentially confound our estimates. We collect WW1 death records to measure the geographical variation in the cost imposed by the war across districts.[14] The dataset provides rich information on Italian military personnel who died during WWI. Importantly for our analysis, it includes the municipality of origin of each soldier. Because we conduct the analysis at the district level, we collapse the dataset from municipalities to 1921 districts, and we measure the war's severity in a given district as the ratio between deaths and population in 1910.

In robustness checks, we use aggregate information on US GDP and industrial production as further controls. These are constructed by Maddison (2007) and Davis (2004), respectively.

To account for changes in market access, we digitize the entire Italian railway network over the 1839–1926 period.[15] The dataset is at the trunk level: for each trunk, we know the stations it connected, the date when it was built and opened to the public, the distance it covered, and the train's traction system. Stations are generally labeled according to the municipality in which they are located. Further details are included for stations located in municipalities with more than one station. To capture the evolution of the network over time, we take snapshots at a decade frequency.

We digitize internal migration data at the regional level from official statistics tabulations. The data are available at the decade level between 1901 and 1931, allowing us to track region-to-region bilateral migration flows. From the same data source, we digitize annual remittances transferred from Italian immigrants in the United States to Italy, and complement them with Italian nominal GDP estimates produced by Baffigi (2013).

III EMPIRICAL STRATEGY

This section presents the empirical strategy we adopt to identify the effects of the Quota Acts and provides evidence supporting the research design's validity.

III.A Measuring Exposure to the Quota Acts

The empirical analysis hinges on the observation that areas whose emigrants were more likely to settle in the United States before 1921 would be more exposed to the Quota Acts. This implicitly assumes

[14]Death records were compiled by the Fascist regime for propaganda purposes and are available at `cadutigrandeguerra.it`. This dataset is maintained by the *Istituto per la storia della Resistenza e della società contemporanea*. Acemoglu, De Feo, De Luca and Russo (2022) were among the first to use it in the economics literature.

[15]The data come from the volume *Sviluppo delle ferrovie italiane dal 1839 al 31 dicembre 1926*, edited by the Italian Statistical Office (*Ufficio Centrale di Statistica*) in 1927. To our knowledge, this is the first paper to use these data. Compared to Ciccarelli, Magazzino and Marcucci (2021), we do not have access to the geography of historical railway routes, as we only digitize the list of stations and the year each trunk was opened. The advantage is that we can reconstruct the network until 1924, which marks the end of the transatlantic emigration, while previous studies focus on the period until 1913.

that emigrants do not perfectly substitute the United States with other–internal or international–destinations. The first step of the analysis validates this assumption.

In principle, districts with more emigrants headed toward the United States before 1921 would be relatively more exposed to the Quota Acts shock. However, if we were to leverage unconditional variation in the number of US emigrants, we would compare districts with very different pre-treatment emigration intensities. If the decision to emigrate were correlated with other, possibly unobserved, characteristics, then the resulting estimates would conflate this underlying spurious association.

Our identification strategy tackles this issue explicitly. To estimate the impact of the Quota Acts on Italian economic development, we compare districts located in provinces with similar volumes of international emigrants and leverage variation in the volume of emigration flows towards the United States. Both variables are computed as the cumulative number of people who emigrated between 1892 and 1921. We set 1921 as the final year because, in 1921, the United States enacted the first restrictive Quota Act.[16] Formally, we leverage variation in the number of emigrants towards the United States, conditional on the overall intensity of international emigration outflows. Since district-level data on the *total* number of emigrants are not available, we control for the number of international emigrants at the province level.

III.B Baseline Difference-in-Differences Model

Throughout the paper, we estimate variations of the following difference-in-differences specification:

$$\begin{aligned} y_{it} = \alpha_i + \alpha_t + \mathbf{x}_{it}'\boldsymbol{\beta} + \sum_{\tau \neq 1911} \gamma_\tau \times \text{Emigrants}_{p(i)} \times I(t = \tau) + \\ + \sum_{\tau \neq 1911} \delta_\tau \times \text{US Emigrants}_i \times I(t = \tau) + \varepsilon_{it}, \end{aligned} \tag{1}$$

where α_i and α_t capture, respectively, unit and time fixed effects, and $I(\cdot)$ are indicators for each period since the Quota shock. The term (US Emigrants) is the log number of migrants who left district i and moved to the United States between 1892 and 1921. The baseline period is the year of the last census before the Quotas (1911). Depending on the outcome y, the included periods are either the population census years (1881, 1901, 1911, 1921, 1931, 1936) or the manufacturing census years (1911, 1927, 1937). All outcomes are expressed in log-terms. The term $\mathbf{x}$ denotes a set of unit time-varying controls. Throughout the paper, we also report the results of the mean-shift specification associated with regression (1), which substitutes the interaction between US emigration and the time-to-treatment dummies with the treatment (US Emigrants$_i$ $\times$ Post Quotas$_t$) term, where we interact US emigration with a categorical variable equal to one for the years after the Quotas (1921), and zero otherwise.

Importantly, equation (1) controls for an interaction term between the cumulative log number of emigrants over the period 1892–1921 (Emigrants$_{p(i)}$) and a set of indicator variables coding the years

[16] In several robustness checks, we further restrict the sample period to exclude the years 1892–1900, when municipalities in the Ellis Island dataset are recorded less frequently, the war years (1915–1918), and the post-war period (1915–1924). The results remain quantitatively unchanged through these different sample restrictions.

since the Quotas were enacted. Controlling for emigration at the province level allows us to mitigate the concern that transatlantic emigration from a given district could be compensated by internal mobility from the other districts in the same province. This ensures that the DiD estimators $\{\delta_\tau\}$ compare districts located in areas with comparable emigration intensities. In the baseline analysis, $\mathbf{x}$ contains an indicator variable for southern regions interacted with a post-Quota indicator to account for potential diverging North-South trends after the Quotas. In several robustness checks, however, we enrich the included controls. Since districts are heterogeneous in population, we weight them by their 1881 population in all regressions to ensure that very small areas do not drive the results. Standard errors are clustered at the level of the treatment relevant to each regression: districts for district-level regressions and provinces for province-level regressions.

The identification assumption that model (1) requires can be stated in terms of conventional parallel trends. Absent the Quotas, units with a conditionally higher number of US emigrants would not have displayed different patterns in y compared with districts with fewer relative US emigrants. While this assumption is not testable, we estimate pre-treatment coefficients that are never statistically different from zero for population (Figure IV) and manufacturing and agriculture employment (Figure VII). These exercises support the parallel trends assumption. We defer a more detailed discussion on the plausibility of the assumption of parallel trends to the following section.

In Table II, we provide one additional complementary exercise to gauge the validity of the research design. In each line, we report the correlation between district-level variables from population censuses (Panel A), province-level manufacturing employment by sector (Panel B), province-level capital variables (Panel C), the number of US emigrants (in columns 1–6), and the instrumental variable defined in (2). In columns (1–2) and (7–8), the variables are measured in 1901; in columns (3–4) and (9–10), they are measured in 1911. Importantly, identification in a difference-in-differences setting requires that the treatment does not correlate with *changes* in pre-determined characteristics, which may impact the outcomes of interest. In columns (5–6) and (11–12), we thus compute the correlation between the growth rate of each variable between 1901 and 1911 and the two treatment indicators. These indicate that observation units appear comparable across a majority of all observed variables in terms of growth rates, despite level differences. We cannot repeat this exercise for the capital variables since we do not observe two pre-treatment values. In Panel C, we thus report the correlation between the treatment and the capital variables at the 1911 level for completeness. The correlation between the district-level observable variables and the instrument (columns 11–12) is consistently smaller than the observed Quota exposure term (columns 5–6), and it is never statistically significant, except for the level of mechanical engines (at the 10% level).

To avoid the pitfalls of the DiD estimator with continuous treatments reported by Callaway, Goodman-Bacon and Sant'Anna (2024), we report estimates of model (1) where the intensive margin is coded as a binary variable returning value one for districts above the median value, and zero for districts below the median. Additionally, in Figure C.17, we employ the method developed by Rambachan and Roth (2023) to assess the robustness of our results to violations of the parallel trends assumption. We find that the baseline estimates withstand considerable deviations from parallel trends in the pre-treatment period.

In equation (1), we control for the total number of emigrants to compare areas with similar out-migration flows, and leverage variation in the emigrants' destination country (US *vis-à-vis* other countries). District-level overall emigration data are, however, not available, so we can only control for province-level out-migration flows. To gauge the relevance of this limitation for our results, in Table B.9, we impute district-level emigration by apportioning province-level data based on fixed district characteristics. Let $E_{p(i)}$ be the number of emigrants leaving province $p(i)$ of district i. Let x_i be the apportioning variable. We impute district-level out-migration $\tilde{E}_i$ by allocating the province-level total $E_{p(i)}$ in proportion to x_i, as follows: $\tilde{E}_i \equiv (x_i / \sum_{i' \in p(i)} x_{i'}) \times E_{p(i)}$. We use two apportioning variables. First, we employ the US emigration rate, so that we impute out-migration by assuming that the overall emigration rate was the same as the US emigration rate. Columns (1–3) display the OLS (Panel A) and IV (Panel B) estimates. The estimates obtained are comparable in size and magnitude to the baseline results (Table III–Table V). Second, we employ the share of agricultural employment, because historical evidence indicates that migrants typically worked in agriculture. Columns (4–6) report the results obtained when controlling for out-migration imputed using the relative agricultural employment share. As before, the estimates are qualitatively and quantitatively consistent with those obtained when controlling for province-level out-migration. These findings corroborate the use of province-level out-migration in the baseline research design.

III.C Instrumental Variable for Quota Exposure

The key identification concern of the DiD estimator is that there may be residual unobserved "push" shocks that correlate with the US emigration rate and influence the outcomes *after* the Quotas are enacted. The statistically insignificant pre-treatment coefficients we estimate substantially narrow the space for unobserved correlated shocks. The instrumental variable provides a complementary approach to controlling for them.

We construct a shift-share instrument to predict US emigration across Italian districts. Our approach mirrors a widely employed methodology to estimate the causal impact of immigration (Card, 2001; Tabellini, 2020), which has recently been adapted to the case of out-migration by Anelli et al. (2023). The instrument predicts district-level out-migration flows to the United States by interacting the 1895–1900 migration ties between Italian districts and US counties–the "shares"–with subsequent immigration into US counties from countries other than Italy–the "shifts." Formally, the instrument predicts US out-migration as

$$\text{US }\widehat{\text{Emigrants}}_{it} \equiv \sum_j \omega_{ij} \times \text{Immigrants}_{jt}^{-\text{Italy}}, \tag{2}$$

where ω_{ij} denotes the number of emigrants from district i into county j normalized by the total number of emigrants to county j between 1895 and 1900,[17] while the shift term ($\text{Immigrants}_{jt}^{-\text{Italy}}$) is the

[17] Formally, the exposure term is calculated as $\omega_{ij} \equiv \text{US Emigrants}^0_{i \to j} / \text{US Emigrants}^0_j$, where district-to-county migration ties

number of non-Italian immigrants who settle in county j in year t between 1900 and 1921. To compute the exposure share terms (ω_{ij}), we rely on our novel dataset that links Ellis Island immigrant records with the full-count US population census (Ruggles et al., 2021). We construct bilateral flows between districts and counties by attaching a municipality of origin to Italian immigrants recorded in the US census. In regression (1), we instrument the observed US emigration term with the shift-share instrument defined in (2) and aggregated at the district or province level. Moreover, as in (1), we control for the total number of international emigrants. We refer to regression (1) as OLS-DiD, and to the instrumented difference-in-differences specification as IV-DiD.

In Figure III, we check that actual and predicted US emigration flows are positively correlated. We compute the residualized values of observed and instrumented number of US emigrants by regressing each variable against province-fixed effects. We then plot the residuals and highlight the binned values in blue. This exercise reveals a strong, positive, and significant correlation between the instrument and observed US emigration. We provide more detailed evidence in Table B.10.

The intuition behind (2) is to leverage US county-level pull shocks that are not affected by Italian district-level (or province-level) unobserved factors to predict US out-migration. Because the "shift" term excludes Italian immigrants, it plausibly does not reflect idiosyncratic time-varying shocks that may influence local economic conditions in Italy. Formally, identification thus builds on the "shift-exogeneity" framework of Borusyak, Hull and Jaravel (2022). It requires that non-Italian immigration to US counties is not correlated with unobserved district-level factors that predict the number of emigrants to the United States and correlate with the outcomes. Importantly, this setting does *not* require the exogeneity of the shares (ω_{ij}), hence allowing for the possibility that emigrants initially sorted in a non-random fashion across US counties.

To understand why the IV helps mitigate residual endogeneity concerns associated with the OLS-DiD estimator, it is important to note that the identifying variation comes from fluctuations in the arrival of non-Italian immigrants across the US. These inflows were driven by shocks in origin countries such as Poland and Russia, and by local US pull shocks, rather than by contemporaneous developments in Italian districts. Because Italians are excluded from the shift, the instrument does not directly reflect either push or pull factors specific to Italian migration. What remains is county-level variation in immigration that is plausibly uncorrelated to unobserved changes in economic conditions in Italian districts. In other words, conditional on the included controls, the IV relies on the assumption that shocks attracting Poles to Chicago or Germans to Milwaukee are unrelated to evolving labor market conditions in, say, Sicily or Tuscany, except through their effect on the opportunities for Italians to migrate to those same US destinations.

Importantly, unlike in the standard IV setting of Anelli et al. (2023), the instrument may be correlated with the *levels* of the unobserved confounding factors. Validity of the IV-DiD estimator merely requires that the instrument does not predict *changes* in local unobserved variables that are not due to US out-migration. While this assumption cannot be tested, in Table II, we follow Borusyak et al. (2022) and show that the instrument does not systematically correlate with changes in the outcome

at baseline are given by US Emigrants$^0_{i\to j} \equiv \sum_{t=1895}^{1900}$ US Emigrants$_{i\to j,t}$ and US Emigrants$^0_j \equiv \sum_i \sum_{t=1895}^{1900}$ US Emigrants$^0_{i\to j}$.

variables before the Quotas. Additionally, throughout the paper, we estimate flexible IV-DiD specifications that allow us to quantify the correlation between the instrumented US emigration rate and the outcomes of interest before the Quotas. We always reject that their correlation is statistically significant at conventional confidence levels. These exercises support the validity of this research design.

A residual concern is that the instrument mechanically assigns a higher predicted US emigration to certain areas in Italy due to an underlying correlation between the shifts and unobserved local factors. In Table B.11 and Table B.12, we thus show that the baseline IV-DiD estimates remain unchanged when employing the method developed by Borusyak and Hull (2023), which constructs a "recentered" instrument by subtracting the predictable component of the shift-share from the baseline instrument (2).[18] This approach weakens the shock exogeneity identification assumption required in the baseline instrumental variable setting, because it ensures that the instrument is uncorrelated with exposure to unobservables driven by the share terms.

IV RESULTS

We organize the results in three logically distinct sections. First, we show that the immigration restriction policy generated "missing migrants" who remained in Italy. Second, we explore its effect on technology adoption and investment in physical capital. Third, we show that it generated a positive labor supply shock in manufacturing, dampening firms' incentive to invest in labor-saving technologies. We conclude by discussing the limitations of the analysis and possible avenues for future research.

IV.A Population Increases in Districts More Exposed to the Quota Acts

The first step of our argument requires that areas more exposed to the US Quota Acts experienced an unexpected population increase. Implicitly, this hinges on the presumption that not all those who would have migrated to the United States had the Quotas not been promulgated would settle in a different country. This "imperfect substitution" argument can be tested and quantified empirically. If the US and other countries were perfectly substitutable, we would expect to find no effect of exposure to the Quotas on population.

To assess the validity of the imperfect substitution hypothesis, we estimate model (1) using population as the outcome variable. Table III reports the results. We find that districts more exposed to the IRP shock display a larger population after 1921, both according to the OLS estimates (columns 1–3) as well as in the instrumental variable case (columns 4–6). Importantly, in columns (2) and (5), we include region-by-time fixed effects to account for time-varying unobserved heterogeneity at the district level. In columns (3) and (6), we repeat the estimation using a binary treatment for districts below and above the median US emigration rate to avoid issues related to continuous treatment DiD designs. All these specifications yield quantitatively consistent and statistically significant results. For instance, as shown in column (4), a 1% increase in the number of instrumented US emigration leads to a 0.05% increase in population. Equivalently, moving from the 50th to the 75th percentile of

[18] To compute the conditional expected value of the instrument, we take the average from 1,000 random permutations where we assign county-level immigration at random across districts.

the distribution of predicted US emigration yields 12,849 additional population, compared to a pre-treatment average equal to 243,000.[19] The increased population due to the Quotas amounts to almost 90% of the number of US emigrants between 1892 and 1921. This share should be interpreted with caution, for it includes foregone migrants *and* their children who would not have been born in Italy had they not migrated. Overall, however, we find that the share of those who would have migrated to the United States and remained in Italy was substantial, possibly exceeding 50%.

We also report the coefficient of an interaction between the number of international emigrants over the period 1890–1921 and a post-Quota indicator. While this term cannot be interpreted causally, it remains informative about the validity of the research study. The core observation is that emigration toward countries other than the United States was not restricted. We thus expect that the population in districts with relatively more intense out-migration flows would decrease following the Quotas. The empirical evidence confirms this claim. The coefficient of the interaction term between out-migration and the post-treatment indicator is negative and generally significant, indicating that emigration towards non-US destinations did not generate "missing migrants" in the years following the Quotas.

The validity of the DiD design hinges on a standard assumption of parallel trends. This requires that if the US had not promulgated the Quota Acts, the population would have evolved similarly in districts with relatively more or less intense US emigration. To gauge the empirical plausibility of this assumption, we estimate model (1) as a generalized instrumented difference-in-differences model. Figure IV reports the estimated dynamic treatment effects. We do not find any statistically significant correlation between predicted US emigration and population before the Quotas were enacted. This pattern supports the parallel trends assumption. We estimate a positive effect of the Quotas already in 1921, which persisted and grew in magnitude over time. The immediacy of the impact of the Quotas is not surprising and is motivated by two factors. First, the Quotas were enforced immediately, and immigration after 1921 swiftly decreased, as shown in Figure I. More importantly, however, it reflects that during World War I, international mobility collapsed. Areas that would be more exposed to the Quotas were also more exposed to the sharp drop in cross-country mobility due to the War. The 1921 population increase is thus plausibly a joint consequence of the War *and* the Quotas. The effect of the Quotas would, in turn, compound over time because every year, additional foregone migrants would remain in Italy and contribute to the increased population. The pattern of the dynamic response of the population to the Quotas, which increases over time throughout the sample period, is consistent with this prediction.

Gould (1980) and, more recently, Spitzer and Zimran (2023) document an S-shaped pattern in the evolution of US emigration at the municipality level. US emigration takes off slowly and rapidly increases until a saturation point, after which it remains relatively flat.[20] This pattern suggests that the district-level exposure to the Quota Act would depend on when they reached this inflection point. Areas that entered the transatlantic mass migration earlier would be less exposed to the Quotas than

[19]Throughout the paper, when performing back-of-the-envelope calculations to gauge the magnitudes of the estimated effects, we always refer to the two-stage least-squares (2SLS) estimates.

[20]Our data corroborate these empirical findings, as shown in Figure C.7.

districts with more recent out-migration flows. In Figure V, we evaluate the empirical standing of this prediction. In Figure Va, we thus estimate the baseline difference-in-differences regression and include an interaction term between the Quota exposure treatment and a set of indicators coding the terciles of US out-migration during the first half of the sample period, between 1892 and 1906. In Figure Vb, we perform the same exercise, but the terciles are computed over the distribution of US emigrants in the second half of the sample period (1907–1921).[21] Areas with relatively larger US out-migration flows in the early years of the sample period do not display an increased population following the Quotas. By contrast, in areas where US emigration was more intense in the years that more immediately preceded the Quotas, we estimate a larger population response. These patterns are highly consistent with the findings of Spitzer and Zimran (2023). The impact of the Quotas was larger in areas where US emigration was more intense and temporally closer to the enactment of the Act.

IV.B Capital Investments Decrease in Districts More Exposed to the Quota Acts

How did the increased population interact with technology adoption and investment? This section shows that investment in labor-saving and possibly productivity-enhancing technologies decreased in areas more exposed to the shock. To do so, we estimate model (1) using several proxies of capital investment as outcome variables. We distinguish between the overall number of firms, the number of firms with at least one installed engine, mechanical and electrical engines, and the mechanical and electrical horsepower generated by the respective installed engines. The data cover manufacturing firms only.

Table IV reports the estimated effect of exposure to the Quotas on capital investment. Panel A refers to the observed US emigration, whereas Panel B reports the two-stage least squares estimates. We find that investment in physical capital decreased in areas more exposed to the Quotas. This finding is confirmed across all the imperfect proxies available. In particular, we estimate larger treatment effects for electrical engines and horsepower. This is particularly informative, as David (1990), Mokyr (1998), and Reichardt (2024) note that electrical engines stood as a major defining technology of the Second Industrial Revolution, which could yield sizable productivity advantages. Gaggl, Gray, Marinescu and Morin (2021), moreover, show that electrification in the US was a catalyst for urbanization and structural transformation. In terms of magnitude, the estimated effects are sizable. Moving from the 50th to the 75th percentile of the distribution of predicted US emigration yields: 3,828 fewer firms, 629 fewer firms with engines, 566 fewer mechanical engines, 359 fewer electrical engines, 10,309 fewer mechanical horsepower, and 509 fewer electrical horsepower.

[21] Formally, denote with ξ_i the tercile of US emigrants over either sample period in district i. The empirical specification is:

$$y_{it} = \alpha_i + \alpha_t + \mathbf{x}_{it}'\boldsymbol{\beta} + \gamma \times \text{Emigrants}_{p(i)} \times \text{I}(t \geq 1921) + \delta_1 \times \text{US Emigrants}_i \times \text{I}(t \geq 1921) +$$
$$+ \sum_{k=2}^{3} \delta_2^k \times \text{US Emigrants}_i \times \text{I}(t \geq 1921) \times \text{I}(\xi_i = k) + \varepsilon_{it}.$$

We estimate the regression twice. In the first case, the terciles $\{\xi_i\}$ are computed on the distribution of US emigrants between 1892 and 1906. In the second, they refer to US emigration between 1907 and 1921. The Figure displays the coefficients $\{\delta_k^2\}_{k=2}^5$ for the two specifications, and the first tercile serves as the baseline category.

Figure VI displays the associated dynamic DiD coefficients. Unfortunately, the data are available for three points in time only. Of these, 1911 is the only pre-treatment observation. We thus cannot estimate pre-treatment coefficients. Instead, in Table II, we show that treated and control provinces were similar along all outcome variables except mechanical engines in 1911. We estimate decreasing treatment effects for all variables decreased in 1927: the effect remains persistent for electrical engines, while it appears to be shorter-lived for mechanical ones.

Thus far, we have grouped all manufacturing sectors. Nonetheless, the data allows us to undertake a more disaggregated analysis. We report the results in Figure C.12. Each "Capital" panel reports the estimated effect of the Quotas on each capital indicator for a given manufacturing sector. The reaction to the population shock does not exhibit sizable heterogeneity across industries. Capital investment decreases in all sectors except for chemical manufacturing.

IV.C The Quota Acts as Passive Labor Market Policies

Why did investments in capital and technology decrease in areas more exposed to the Quota Acts? We advance the hypothesis that the immigration restriction shock triggered directed technical adoption mechanics *à la* Zeira (1998) and Acemoglu (2002). Under the imperfect substitution hypothesis, which we documented in Section IV.A, areas that had been sending relatively more migrants to the United States are, in fact, subject to a larger labor supply shock because a relatively larger fraction of people who would have emigrated are prevented from doing so. This positive labor supply shock decreases wages, which triggers firms' incentive to substitute capital for labor. We formalize this argument in Appendix D.

To test this mechanism more formally, we study how manufacturing and agricultural employment reacted to the shock. We look at manufacturing and agriculture because they account for more than 80% of overall employment. In Section I, we noted that Italian agriculture in this period was labor-intensive and not mechanized (Cohen and Federico, 2001). Manufacturing firms, by contrast, still had scope for substituting machines with labor. The directed technical adoption narrative would thus predict that the labor supply shock generated by the Quota Acts would primarily flow into increased manufacturing employment.

In Table V, we explore the effect of the Quota Acts on manufacturing employment, while in Table VI, we focus on agriculture employment. We find that manufacturing employment increases in areas more exposed to the immigration restrictions shock, while employment in agriculture does not exhibit similarly detectable changes. In both tables, in columns (1–3), we report the OLS estimates, while in columns (4–6), we display the 2SLS estimates. Columns (3) and (6) refer to a binary treatment that codes as treated all districts with above-median US emigration, whereas in the other columns, we employ the continuous measure. We estimate a positive and statistically significant effect of exposure to the Quotas on manufacturing employment, as displayed in columns (1) and (4). Results remain similar in size when employing the instrumental variable, even after controlling for region-time fixed effects. By contrast, we find no detectable effect of exposure to the Quota Acts on agricultural employment across all specifications.

Since manufacturing employment increases while agriculture employment does not, the labor supply shock generated by the Quotas prompted a relative reallocation of labor from agriculture, a largely labor-intensive activity, to manufacturing. Overall, the evidence presented in the table provides solid evidence that manufacturing employment increased in areas more exposed to the Quotas. From a quantitative perspective, moving from the 50th to the 75th percentile of the distribution of predicted US emigration yields 3,440 more workers employed in manufacturing (compared to a pre-treatment average equal to 34,548) and 1,629 fewer agriculture workers (compared to a pre-treatment average of 52,786), although the latter effect is not statistically significant. The positive labor supply shock generated by the policy shift did not, however, foster structural change, as employment in agriculture in more exposed areas hardly reacted to the shock. Therefore, the increase in manufacturing employment plausibly reflects the fact that "missing migrants" take jobs in manufacturing rather than a reallocation of jobs from agriculture into manufacturing. The Quota Acts, therefore, did not enhance economic modernization in rural areas.

The historical literature views Italian emigration as a response to overpopulation in rural areas (Choate, 2008). We evaluate these claims by reporting the coefficient of the interaction between out-migration before the Quotas and a post-treatment indicator. While this term has no causal interpretation, it reveals important patterns related to the motives underlying the migration choice. Employment in manufacturing does not change in districts with relatively more emigrants after the Quota shock. By contrast, agricultural employment decreases. Since out-migration towards countries other than the United States was not curtailed, these patterns appear consistent with the historical literature. Out-migration was prevalent in rural agricultural areas, and thus, agricultural employment decreased in regions with more intense (non-US) out-migration flows after the Quotas.

We report the flexible difference-in-differences estimates in Figure VII. We do not estimate statistically significant pre-treatment coefficients for manufacturing employment, thus providing further support for the assumption of parallel trends. Manufacturing employment increased after the Quotas. The effect is relatively persistent as it lasted at least until 1931, although it peaked immediately after the immigration restriction shock. It is worth noting that this pattern is not inconsistent with the increasing effect of the Quotas on population over time documented in Figure IV. Since most emigrants were working-age men, it is plausible that the response of manufacturing employment to the Quotas would be more immediate than that of the overall population, which comprises the children of those who would have migrated and were born after 1921. We do not find any statistically significant response of agricultural employment, and we thus omit the event study.

We can break down manufacturing employment by sector, even though only at the coarser province level. As in Section IV.B, Figure C.12 displays that we do not uncover substantial heterogeneity across sectors. Employment increases in all industries except mining, where geographical factors may constrain the employment response. These findings confirm that a positive labor supply shock is associated with decreased capital investment and more labor-intensive production in all manufacturing sectors.

IV.D Robustness Checks

This section summarizes the checks we perform to assess the robustness of the baseline findings. We provide evidence that the baseline estimates are robust to alternative definitions of US emigration, to the inclusion of several covariates controlling for both push and pull factors, to the exclusion of specific parts of the sample, and to using different methods to estimate the standard errors.

Since observed US emigration may be subject to measurement error, in Table B.13, Table B.14, Table B.15, and Table B.16, we show that the results are robust to alternative definitions. The first concern is that our results might be driven either by remote migration patterns or by more recent migration closer to the introduction of the quotas. We thus construct two measures of US emigration that assign increasing or decreasing weights to more recent out-migration flows to the US. Additionally, we consider alternative definitions of the stock of US emigrants depending on the year when they migrated. First, we include all emigrants until 1924, i.e., until the inflow of Italians in the US completely dries up following the Johnson-Reed Act. Second, we construct the treatment using migration to the US before the first Quota Act in 1920. Last, as discussed in Section II, though emigration collapsed during WWI, it did not completely dry out. During the war, districts closer to emigration ports are disproportionately represented relative to previous shares, possibly because of their geographic position. To ensure that this pattern does not determine our results, we restrict the sample of US emigration to the years before the outbreak of World War I. In all cases, we find that all our baseline results hold.

In Table B.17, Table B.18, Table B.19, and Table B.20, we show that our baseline results are robust to the inclusion of several covariates measured before the Acts, interacted with a post-Quota indicator variable, as further controls. These unit-specific control variables are the literacy rate, measured as the share of people who could read relative to the district's baseline population; the urbanization rate at baseline, measured as the share of people living in towns with more than 5,000 inhabitants in the district; the altitude at which the district is located; an indicator variable that returns a value of one if the district has access to the railway network before 1901;[22] the number of deaths due to World War I expressed as a share of baseline population.[23] To control for pull factors, in some specifications, we explicitly include an interaction term between US GDP, which serves as an indicator of the business cycle, US emigration, and the post-Quota indicator. All results remain quantitatively unchanged.

In addition, we test that our results are robust to the exclusion of specific parts of the sample. In Figure C.14 and Figure C.16, we that the estimated treatment effects remain stable when excluding one region at a time from the estimation sample.

In Figure C.13 and Figure C.15, we compare the baseline estimated standard errors with a battery of alternative estimators. Among others, we employ the correction suggested by Conley (1999) to allow for spatial autocorrelation. The significance of the baseline results remains largely unaltered.

To further assess the validity of the instrumented difference-in-differences design, following Ram-

[22] For province-level analyses, we use the share of municipalities in the province that had access to the railway network before 1901.

[23] Shares are expressed relative to the population in 1911 for province-level regression, as 1911 is the first decade of observation for analyses at that level.

bachan and Roth (2023), in Figure C.17, we estimate bounds on the relative magnitude of deviations from parallel trends. This exercise addresses the possibility that there may be unobserved shocks prior to the Quotas that affected with differential intensity the districts that would be more exposed to the emigration restriction shock. We report the confidence intervals for different values of the $\bar{M}$ threshold for population (Figure C.17a) and manufacturing employment (Figure C.17b). In both cases, the estimated bounds indicate that the baseline estimates withstand considerable deviations from parallel trends in the pre-treatment periods. We interpret the results as corroborating the robustness of our findings.

IV.E Discussion and Limitations of the Analysis

In this section, we discuss alternative mechanisms that could be compatible with our findings, and we touch on how data limitations might preclude additional and potentially relevant analyses. We then briefly elaborate on the external validity of our results.

IV.E.1 Gender-Biased Structural Change

A first mechanism that may explain the drop in capital investment attributed to the Quota-induced labor supply shock is the sex imbalance that emerged as a consequence of Italian mass migration. Men were heavily overrepresented among emigrants (Del Boca and Venturini, 2005). According to our data, approximately 76% of emigrants were men (Figure C.4). If this gender imbalance was more pronounced among emigrants who settled in the US than in other destination countries, areas that were more exposed to the Quotas plausibly displayed more skewed sex ratios.

This circumstance could imply two consequences. On the one hand, a shortage of male workers could incentivize firms to invest in labor-saving capital, along the lines of previous scholarship on labor shortages caused by wars (Voth, Caprettini and Trew, 2023). If this were the case, decreasing capital investments in areas exposed to the Quotas after 1921 could be explained by previous higher investments due to skewed sex ratios. On the other hand, the relative abundance of women could shift economic specialization towards sectors requiring less physical labor input, such as textiles, rather than metallurgy or heavy machinery, in line with evidence by Ager, Goñi and Salvanes (2025). This shift could, in turn, influence household specialization, gender norms, and household specialization in the long run (Fernández, Fogli and Olivetti, 2004). Under this interpretation, lower capital investment after the Quotas could reflect pre-existing specialization in sectors that are less technology-intensive, rather than directed technological adoption. Both confounding alternatives, however, would only influence the DiD strategy if their effects would manifest at the same time as the Quota Acts.

We do not view either of these alternative mechanisms as capable of explaining our results. First, we find no evidence that areas with a higher stock of US emigrants displayed higher investment in capital technologies before the Quotas (see Panel C of Table II). Second, if skewed sex ratios impacted household specialization and gender norms, it is likely that they also influenced fertility choices. Instead, we find no evidence that districts with relatively higher exposure to the Quotas display different population dynamics (see Figure IV). If anything, the statistically insignificant difference between areas with high and low Quota exposure implies a relatively higher fertility rate in areas with higher US emigration rates, which would be inconsistent with a higher rate of female labor force participation. Third, in Panel B of Table II, we find no statistically significant correlation between the number of US emigrants and the composition of economic activity across sectors. Specifically, economic activity in areas with a higher number of US emigrants did not shift towards sectors that required lower male labor input.

To assess the empirical plausibility of these mechanisms more directly, we explore how the response to the Quota Acts differs in terms of the share of men among US emigrants. Figure C.4 reports the district-level distribution of the share of men among US emigrants. Men account for the majority of emigrants in each district, but the distribution features substantial dispersion, as the share of male US emigrants ranges from 55% to 90%. We leverage this heterogeneity to estimate heterogeneous responses to the Quotas in districts above and below the median share of male emigrants. If the observed decrease in capital investments after 1921 was due to sex ratios increasing technology adoption up to saturation before the Quotas, we would expect larger drops in capital investment in districts with above-median share of male US emigrants. Conversely, if the abundance of women shifted economic activity towards less technology-intensive sectors, districts with below-median shares of male emigrants would display larger responses.

Figure C.18 displays the treatment effect of the Quotas in districts where the share of men among US emigrants is above the median (in blue) and below the median (in red). We do not find evidence of treatment effect heterogeneity.[24] This pattern suggests that it is unlikely that our results are driven by gender imbalances that originated from the over-representation of men among US emigrants.

A possible related mechanism is that the increase in the share of women in districts with a higher US emigration rate could affect fertility rates and, hence, labor supply in the long run (Anelli and Balbo, 2021). This channel does not necessarily compete with the mechanism we propose, as it is plausible that the increase in labor supply stemming from shifted fertility preferences would have accentuated the increase in population driven by foregone migration. Nevertheless, two findings indicate that this mechanism was not first-order. First, we do not estimate different treatment effects depending on the share of women among US emigrants. Hence, districts that would have had different fertility preferences do not display different responses to the Quotas. Second, if differential US emigration affected fertility preferences, one would expect to estimate statistically significant differences between districts based on their exposure to the Quotas even *before* 1921, while we find no evidence in this direction.

IV.E.2 Internal Migrations

A potential concern is that internal migrations could replace international emigration following the Quota Acts. Despite efforts from the Fascist authorities to curtail internal human mobility and limit urbanization, historical studies document that internal migrations increased after World War 1 until the mid-1930s (Treves, 1976; Gallo, 2012).

[24]We estimate statistically different treatment effects only for agricultural employment, but none of them are statistically different from zero when taken singularly.

It is important to note that our results are entirely compatible with the possibility that *some* individuals who would have migrated to the US in the absence of the Quotas could move to other areas of the country.[25] The core of our argument is that substitution of US for other destinations was *imperfect*, implying that even if the labor supply shock experienced by Italian origin areas was not one-to-one with foregone US migration, labor still became more abundant because a fraction of foregone US migrants remained in their areas of origin. The first step of the empirical analysis explicitly tests for this hypothesis. The increase in population that we estimate in areas more exposed to the Quotas provides compelling evidence in favor of the imperfect substitutability hypothesis, implying that origin areas in Italy experienced a labor supply shock due to the Quota Acts.

To provide a more systematic analysis of the role of internal migrations, we digitize tabulations on internal human mobility compiled by the Italian statistical office. While the level of aggregation (regions) does not allow us to draw strong causal conclusions from this analysis, the data are nonetheless descriptively informative. Table B.3 provides a glance at the number of internal migrants who moved to other municipalities (column 2), provinces (column 3), or regions (column 4), compared to US emigration. In all regions, internal migration was vastly outnumbered by international migration. In all Southern regions, US emigration was more than seven times larger than cross-region migration. Figure C.20 displays the variation in internal migration rates over time. The data indicate an increase in the share of the population living in regions other than their origin region in 1931, although the change (7.5% in 1931 compared to 5% in 1921) is quantitatively modest.

In (column 1–3) of Table B.21, we show the share of the population who moved into other municipalities (1), provinces (2), and regions (3) evolved after the Quotas. This pre-post analysis shows that inter-regional migration increased (by 1.6%) after the Quotas. In columns (4–6), we estimate the baseline specification (1) on the same outcomes, region, and decade fixed effects, to explore whether changes in internal migrations were correlated with exposure to the Quota Acts. If this were the case, internal human mobility could attenuate the intensity of the labor supply shock driven by foregone US emigration. We, however, do not find evidence in this direction. The share of the population moving to different municipalities (column 4), provinces (column 5), or regions (column 6) does not appear to respond to exposure to the Quota Acts. Internal mobility thus plausibly played a very limited role in absorbing the foregone transatlantic migrations after 1921.

IV.E.3 Monetary Remittances

Remittances have traditionally been a major research topic within the emigration literature. Despite sizable global flows, Clemens (2011) argues that remittances have, at best, a second effect on economic growth in sending countries when compared to the welfare effects of immigration restriction barriers. Building on this insight and given data limitations, we abstracted from including remittances in our analysis.

Remittances nonetheless represent a competing mechanism. More exposed districts were receiving more remittances before the Quota Acts. Hence, they suffered the most from the border closure.

[25]This point is analogous to the possibility that we similarly allow for, that migrants could substitute the US with other destination countries.

Since within-household cash transfers result in aggregate savings, remittances may accrue to overall investment dynamics (Rapoport and Docquier, 2006). A large literature has nonetheless documented that remittances are largely spent on consumption and invested in human–rather than physical–capital (for a review, see Yang, 2011). A more sensible interpretation could be that remittances fostered literacy (e.g., Fernández, 2024). Exposed districts would have thus suffered a relative decline in skilled workers following the Acts, and the labor force would have shifted toward unskilled sectors. This channel does not conflict with the one we propose. If anything, a fall in the share of skilled workers due to lower human capital investments increases the relative supply of unskilled workers, thus triggering the same directed technical incentive to abandon investment in physical capital.

As is common in the literature, disaggregated remittances data are not available in our setting. Aggregate data, however, suggest that it is unlikely that our results are explained by a fall in remittances following the Quotas. In Figure C.21, we tabulate data on money transfers to Italy operated by the *Banco di Napoli* on behalf of Italian immigrants in the US.[26] Total nominal remittances, shown in blue, remained more than twice as large as in the pre-WWI period until 1925, while their level remains essentially stable (except for the war years) when normalizing by GDP, as displayed in red. Overall, it seems implausible to impute the drop in capital investment that we observe in areas more exposed to the Quotas already in 1927 to a drop in remittances that had yet to occur, at least at the national level.

IV.E.4 Human Capital Channel: Brain Drain or Brain Gain

Human capital spillovers ignited by out-migration have traditionally received sizable attention in the literature. The human-capital implications of out-migrations are debated in the literature. On the one hand, Spitzer and Zimran (2018) document that Italian emigrants to the United States were positively selected within Southern regions. Hence, emigration could exert a "brain drain" effect on Southern Italy. On the other hand, growing literature shows that out-migration may increase human capital and innovation in home countries (e.g., see Dinkelman and Mariotti, 2016; Coluccia and Dossi, 2025). Giffoni and Gomellini (2015) provides descriptive evidence in this direction for the case of Italy during the mass migration. The two arguments have opposing implications. The "brain drain" argument suggests that the Quotas reversed the loss of human capital generated by out-migration, thus proving beneficial for Italian development. The "brain gain" hypothesis, on the other hand, suggests that the IRP enacted by the US may have hindered the process of human capital accumulation in Italy.

Our findings are not consistent with the "brain drain" argument, as we estimate that regions where the supply of relatively skilled workers increased the most experienced the largest decline in physical capital investment. On the other hand, while the "brain gain" hypothesis does not conflict with the mechanism we propose, we consider it a secondary channel for two key reasons.

First, if out-migration to the US fostered human capital accumulation and investment in (complementary) machinery, then one would expect areas with larger out-migration flows to display higher levels of machinery investment before the Quotas. We find little to no evidence in this direction, as

[26]The data are comprehensive because *Banco di Napoli* was granted a monopoly over international monetary transfers from immigrants into Italy in 1901 (Barbiellini Amidei and Goldstein, 2007).

displayed in Panel C of Table II.

Second, we test the "brain gain" argument more formally by estimating heterogeneous treatment effects of the Quotas by the skill level of emigrants, as proxied by their occupation in the United States.[27] The "brain gain" argument suggests that the human capital remittances would be larger in areas sending relatively more skilled emigrants to the United States. We do not find evidence in this direction, as shown in Figure C.19. Specifically, we find that the treatment effects for districts with above- and below-median share of migrants employed in skilled occupations in the United States overlap for all the outcomes we consider. It is important to remark, however, that we do not–and cannot–reject the possibility that foregone out-migration hampered human capital accumulation at home. Our evidence, however, seems to be more plausibly aligned with the hypothesis that the positive labor supply shock triggered capital-saving incentives for firms, which could substitute capital with more abundant labor.

IV.E.5 A Comparison with Emigration to Argentina

One reason precluding a causal interpretation of our estimates is that, even when conditioning on the decision to emigrate, the choice of *where* to emigrate could be systematically correlated with factors that induce an underlying correlation with local economic development. We provide and discuss evidence against this interpretation throughout this paper. Historical scholarship, however, notes that assimilation patterns of Italian immigrants in the United States and Argentina during this period substantially differed (Klein, 1983).[28] If this was caused by pre-migration differences in characteristics, then our identification scheme may fail.

Using detailed data from censuses and passenger lists, Pérez (2021) nonetheless documents that the "success" of Italians in Argentina compared to Italians in the United States was unlikely to be caused by pre-migration differences in observable characteristics between the two groups. Emigrants to Argentina and the United States were essentially indistinguishable in terms of occupation and literacy rate, the only difference being that the former chiefly originated from Northern regions. In contrast, the latter mostly came from Southern areas. Selection patterns across the two groups do not display sizable differences, providing additional evidence in favor of our identification assumption.

IV.E.6 Limitations of the Analysis

First, data limitations prevent us from studying two additional, potentially interesting variables: wages and productivity. Studying wages would be informative because directed technical adoption hinges on relatively more abundant labor becoming relatively cheaper. An analysis of wages could reveal this pattern, which we can only assume. Unfortunately, geographically disaggregated data on wages do not exist. Productivity would, in turn, be key to investigating the welfare effects of the Quota Acts. However, disaggregated data on output were not recorded until 1936.

[27] We label an Italian immigrant in the US census as "skilled" if they perform an occupation (OCC1950) labeled as "Professional, Technical," "Managers, Officials, and Proprietors," or "Craftsmen." Approximately 18% of the sample is employed in a skilled occupation according to this classification. The results remain unchanged if we exclude craftsmen, who are essentially skilled manufacturing workers.

[28] Argentina and the United States were the two leading destinations for Italian emigrants in this period. Klein (1983), among others, noted that Italian immigrants in Argentina had higher home-ownership rates and were more likely to be employed in skilled occupations compared to Italians in the United States.

Second, it is not obvious that our results lend themselves to further generalization. Some similarities with contemporary settings nonetheless emerge. In terms of emigrant selection, the average Italian emigrant to the United States was locally positively selected, left a rural area, and took on unskilled industrial jobs once in the United States (Sequeira et al., 2020). This description is remarkably similar to contemporary emigration from poor and developing countries (e.g., Gibson, McKenzie and Stillman, 2011). While we do not claim that all our findings generalize to contemporary migration relationships, the evidence presented in this paper suggests that IRPs should be evaluated in terms of their joint effects on both sending and receiving countries, beyond remittances and human capital deprivation, especially when they target economically disadvantaged emigration countries.

V CONCLUSIONS

The adoption of foreign technology is a crucial driver of economic growth for developing countries, which typically operate far from the technology frontier (Eaton and Kortum, 1999; Suri, 2011). This paper explores the relationship between technology adoption and out-migration, a common feature of the development process.

We study the Italian mass migration to the United States between 1892 and 1936 using individual-level data on Italian immigrants and newly digitized census data. Between 1921 and 1924, the U.S. promulgated two immigration restriction policies, the "Quota Acts," which almost completely halted the inflow of Italian immigrants. Comparing districts located in areas with similar emigration rates but whose migrants moved to different destinations, we leverage variation in exposure to the Quota Acts to estimate the impact of immigration restriction laws in a difference-in-differences framework.

We document three facts. First, population increased in areas that were relatively more exposed to the Quota Acts. This finding supports an "imperfect substitution" hypothesis whereby immigration restriction policies generate foregone emigration because those who would have migrated do not (or cannot) substitute the restricted location with alternative destination countries. We thus interpret immigration restriction policies as "passive" labor market policies that increase the supply of labor in the countries they target. Second, we show that firms in treated locations decreased capital investment and technology adoption, particularly in relatively more advanced technology vintages. How can we reconcile a positive labor supply shock with depressed investment in productivity-enhancing capital? We argue that the immigration restriction policy triggered directed technical change incentives (Zeira, 1998; Acemoglu, 2002). Firms face an incentive to substitute capital with more abundant, hence relatively cheaper labor. To validate this hypothesis, we show that manufacturing employment increased in the districts more exposed to the Quotas.

This paper presents novel evidence that out-migration can act as a driver of technology adoption, thus potentially empowering long-run economic growth. We emphasize that this channel operates plausibly independently of "brain drain" effects. It also suggests that immigration restriction policies enacted by immigration countries may hamper the modernization of—typically, developing—emigration countries. Our findings thus inform policymakers about the potential long-term consequences of such policies.

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Table I. Descriptive Statistics

	Mean (1)	Std. Dev. (2)	Median (3)	Min. (4)	Max. (5)	Units (6)	Observations (7)
Panel A. District-Level Variables from Population Censuses (in 10,000 units unless otherwise specified)							
Population	1.703	1.542	1.278	0.250	15.041	212	1,239
US Emigrants	0.109	0.109	0.057	0.006	0.365	212	1,239
Emigrants	2.036	1.475	1.767	0.126	10.182	212	1,239
Manufacturing Employment Share (%)	10.820	6.189	9.186	0.278	38.287	212	1,239
Agriculture Employment Share (%)	27.213	8.771	26.833	0.953	77.549	212	1,028
Altitude	3.394	2.144	3.272	0.013	9.669	212	1,239
Area	1.334	0.830	1.097	0.105	4.982	212	1,239
Share of Cities Above 20,000 (%)	5.068	11.275	1.538	0.000	100.000	212	1,239
Share of Population in Cities Above 20,000 (%)	20.763	23.250	16.499	0.000	100.000	212	1,239
Panel B. Province-Level Manufacturing Employment by Sector (in 10,000 units)							
Agriculture	1.491	1.546	1.103	0.150	11.856	69	343
Chemicals	0.166	0.466	0.046	0.000	5.046	69	343
Construction	1.296	1.176	0.944	0.162	8.077	69	343
Metalworking	0.868	1.651	0.401	0.065	16.876	69	343
Mining	0.145	0.259	0.068	0.000	2.137	69	343
Textiles	2.030	2.736	1.151	0.070	19.988	69	343
Panel C. Province-Level Capital Variables (in 1,000 units)							
N. of Firms	8.231	7.817	6.162	0.000	57.109	71	208
N. of Firms with Engine	1.503	2.066	0.913	0.000	18.511	71	208
N. of Electrical Engines	5.267	15.082	1.555	0.000	168.288	71	208
Electrical Horsepower	39.577	94.606	10.408	0.000	893.338	71	208
N. of Mechanical Engines	0.572	0.416	0.430	0.000	2.079	71	208
Mechanical Horsepower	15.770	18.595	8.636	0.000	104.668	71	208

Notes. This Table reports summary statistics for the main variables used in the paper. Data in Panel A are tabulated from population censuses; data in Panels B and C are digitized from manufacturing censuses. Variables in Panels A and B are reported in 10,000 units unless otherwise specified, except population, which is expressed in 100,000 terms; variables in Panel C are reported in 1,000 units. All variables are cross-walked to consistent 1921 district (Panel A) and province (Panels B and C) borders. Referenced on page: 9.

Table II. Correlation between Treatment and Pre-period Observable Variables

	Observed Quota Exposure						Shift-Share Instrumental Variable					
	1901		1911		Growth Rate		1901		1911		Growth Rate	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Panel A. District-Level Variables from Population Census												
Population	0.480***	(0.046)	0.515***	(0.048)	0.009*	(0.005)	0.479***	(0.059)	0.511***	(0.062)	0.009	(0.005)
Manufacturing Employment Share	0.529***	(0.064)	0.590***	(0.067)	0.034*	(0.019)	0.527***	(0.077)	0.585***	(0.081)	0.034*	(0.019)
Agriculture Employment Share	0.413***	(0.047)	0.451***	(0.047)	0.007	(0.019)	0.411***	(0.058)	0.448***	(0.059)	0.007	(0.019)
Share of Cities Above 20,000 (%)	-0.282***	(0.115)	-0.256***	(0.102)	0.060	(0.043)	-0.283***	(0.107)	-0.246***	(0.095)	0.062	(0.045)
Share of Population in Cities Above 20,000 (%)	0.076	(0.101)	0.137**	(0.067)	0.034	(0.032)	0.076	(0.104)	0.132*	(0.068)	0.035	(0.033)
Panel B. Province-Level Manufacturing Employment by Sector												
Agriculture	0.247	(0.185)	0.251	(0.178)	0.004	(0.041)	0.304	(0.232)	0.309	(0.223)	0.005	(0.051)
Chemicals	0.075	(0.394)	0.328	(0.296)	0.253	(0.216)	0.093	(0.487)	0.405	(0.366)	0.312	(0.265)
Construction	0.109	(0.154)	0.177	(0.161)	0.068	(0.065)	0.134	(0.190)	0.218	(0.206)	0.084	(0.086)
Metalworking	0.233	(0.205)	0.348	(0.234)	0.115**	(0.058)	0.287	(0.251)	0.429	(0.296)	0.141*	(0.083)
Mining	0.471	(0.378)	0.451	(0.295)	0.108	(0.176)	0.585	(0.516)	0.536	(0.370)	0.129	(0.203)
Textiles	0.197	(0.180)	0.188	(0.206)	-0.009	(0.060)	0.243	(0.227)	0.232	(0.255)	-0.011	(0.074)
Panel C. Province-Level Capital Indicators												
N. of Firms			0.190	(0.120)					0.234	(0.157)		
N. of Firms with Engine			0.198	(0.125)					0.244	(0.164)		
N. of Electrical Engines			0.127	(0.304)					0.156	(0.370)		
Electrical Horsepower			0.188	(0.310)					0.231	(0.380)		
N. of Mechanical Engines			0.204**	(0.103)					0.252*	(0.138)		
Mechanical Horsepower			0.106	(0.189)					0.131	(0.232)		

Notes. This Table displays the correlation between the observable district- and province-level variables, observed US emigration (columns 1–6), and the instrument (columns 7–12). In columns (1–2) and (7–8), the correlation is between the treatments and the variables in 1901; in columns (3–4) and (9–10), the variables are recorded in 1911. In columns (5–6) and (11-12), we report the correlation between the two treatments and the growth rate of each variable between 1901 and 1911. In Panel A, the units of observation are districts; in Panels B and C, the units of observation are provinces. Capital variables, displayed in Panel C, are observed in 1911, 1927, and 1937, so we cannot compute the growth rate between two consecutive pre-treatment periods. The outcome variable in each regression is standardized to facilitate the comparisons. Each regression controls for the number of emigrants and region-fixed effects. Units are weighed by their population in 1881. Standard errors are clustered at the district level and are displayed in parentheses. Referenced on pages: 15, 17, 20, 24, 27.

***: $p < 0.01$, **: $p < 0.05$, *: $p < 0.10$

Table III. The Response of Population

	Dependent Variable: log(Population)					
	OLS			2SLS		
	(1)	(2)	(3)	(4)	(5)	(6)
US Emigrants × Post	0.069***	0.067***		0.053**	0.062***	
	(0.021)	(0.021)		(0.022)	(0.021)	
I(US Emigrants) × Post			0.084**			0.217**
			(0.037)			(0.102)
Emigrants × Post	-0.080***	-0.065***	-0.054***	-0.069***	-0.060***	-0.088***
	(0.019)	(0.020)	(0.018)	(0.021)	(0.021)	(0.030)
District FE	Yes	Yes	Yes	Yes	Yes	Yes
Decade FE	Yes	–	Yes	Yes	–	Yes
Region-Decade FE	No	Yes	No	No	Yes	No
N. of Districts	203	202	203	203	202	203
N. of Observations	1,202	1,196	1,202	1,202	1,196	1,202
K-P *F*-stat				85.619	98.996	13.747
Mean Dep. Var.	12.250	12.244	12.250	12.250	12.244	12.250
Std. Beta Coef.	0.430	0.417	0.054	0.352	0.411	0.141

Notes. This Table reports the estimated effect of district-level exposure to the US Quota Acts on population. The unit of observation is a district observed at a census-decade frequency between 1881 and 1936. The dependent variable is log-population. In columns (1–2) and (4–5), the treatment is an interaction term between US emigration and a post-1921 term. In columns (3) and (6), the treatment is an interaction between a dummy equal to one for above-median US emigration districts and a post-1921 term. Columns (1–3) display the OLS estimates; columns (4–6) display the two-stage least squares estimates obtained using the instrument defined in (2). Each regression includes district and time fixed effects and controls for the emigration rate and an indicator variable for Southern regions; both interacted with a post-1921 indicator. In columns (2) and (5), we replace decade with region-decade fixed effects. The last row reports the coefficient of the interaction between the number of emigrants and a post-1921 indicator. The K-P *F*-statistic refers to the Kleibergen-Paap statistic for weak instruments. Units are weighed by their population in 1881. Standard errors clustered at the district level are reported in parentheses. Referenced on pages: 16, 18.

***: $p < 0.01$, **: $p < 0.05$, *: $p < 0.10$

Table IV. The Response of Capital and Technology Adoption

	Dependent Variable: Province-Level (log) Number of...					
	(1) Firms	(2) Firms with Engines	(3) Mechanical Engines	(4) Electrical Engines	(5) Mechanical Horsepower	(6) Electrical Horsepower
Panel A. OLS Estimates						
US Emigrants × Post	-0.209***	-0.133	-0.170***	-0.191	-0.232***	-0.135
	(0.059)	(0.115)	(0.057)	(0.214)	(0.072)	(0.162)
Emigrants × Post	0.111*	0.040	0.050	-0.046	0.067	-0.191
	(0.065)	(0.111)	(0.063)	(0.204)	(0.086)	(0.160)
Std. Beta Coef.	-1.339	-0.696	-1.193	-0.517	-1.183	-0.380
Panel B. 2SLS Estimates						
US Emigrants × Post	-0.206	-0.352**	-0.136	-0.745*	-0.457***	-0.611**
	(0.162)	(0.168)	(0.107)	(0.407)	(0.171)	(0.292)
Emigrants × Post	0.108	0.237	0.020	0.453	0.270*	0.238
	(0.150)	(0.166)	(0.105)	(0.389)	(0.158)	(0.281)
K-P *F*-stat	18.424	18.424	18.424	18.424	18.424	18.424
Std. Beta Coef.	-1.316	-1.838	-0.954	-2.012	-2.328	-1.713
Province FE	Yes	Yes	Yes	Yes	Yes	Yes
Decade FE	Yes	Yes	Yes	Yes	Yes	Yes
N. of Provinces	69	69	69	69	69	69
N. of Observations	206	206	206	206	206	206
Mean Dep. Var.	8.921	7.060	6.267	7.444	9.435	9.542

Notes. This Table reports the estimated effect of exposure to the US Quota Acts on capital investment. The unit of observation is a province observed at a census-decade frequency between 1911 and 1936. The dependent variables are the number of firms (column 1), the number of firms with at least one engine (column 2), the number of mechanical engines (column 3), the number of electrical engines (column 4), the horsepower generated by mechanical (column 5) and electrical (column 6) engines. All dependent variables are expressed in log. The treatment is an interaction between a post-Quota (1921) indicator variable and the number of US emigrants. Panel A displays the baseline OLS estimates; Panel B displays the two-stage least squares estimates obtained using the instrument defined in (2). Each regression controls for the number of emigrants and an indicator variable for Southern regions, both interacted with a post-1921 indicator. Regressions include province and decade-fixed effects. The K-P *F*-statistic refers to the Kleibergen-Paap statistic for weak instruments. Units are weighed by their population in 1881. Standard errors clustered at the province level are reported in parentheses. Referenced on page: 20.

***: $p < 0.01$, **: $p < 0.05$, *: $p < 0.10$

Table V. The Response of Manufacturing Employment

	Dependent Variable: log(Manufacturing Employment)					
	OLS			2SLS		
	(1)	(2)	(3)	(4)	(5)	(6)
US Emigrants × Post	0.069**	0.108***		0.100**	0.067	
	(0.026)	(0.022)		(0.039)	(0.045)	
I(US Emigrants) × Post			0.061			0.348**
			(0.054)			(0.168)
Emigrants × Post	-0.009	-0.066	0.022	-0.030	-0.031	-0.050
	(0.040)	(0.047)	(0.041)	(0.046)	(0.045)	(0.064)
District FE	Yes	Yes	Yes	Yes	Yes	Yes
Decade FE	Yes	–	–	Yes	–	Yes
Region-Decade FE	No	Yes	Yes	No	Yes	No
N. of Districts	203	202	203	203	202	203
N. of Observations	1,202	1,196	1,202	1,202	1,196	1,202
K-P *F*-stat				85.619	98.996	13.747
Mean Dep. Var.	9.978	9.969	9.978	9.978	9.969	9.978
Std. Beta Coef.	0.295	0.465	0.027	0.465	0.309	0.157

Notes. This Table reports the estimated effect of district-level exposure to the US Quota Acts on manufacturing employment. The dependent variable is log-manufacturing employment. The unit of observation is a district observed at a census-decade frequency between 1881 and 1936. In columns (1–2) and (4–5), the treatment is an interaction term between US emigration and a post-1921 term. In columns (3) and (6), the treatment is an interaction between a dummy equal to one for above-median US emigration districts and a post-1921 term. In column. Columns (1–3) display the OLS estimates; columns (4–6) display the two-stage least squares estimates obtained using the instrument defined in (2). Each regression includes district and time-fixed effects and controls for the emigration rate and an indicator variable for Southern regions; both interacted with a post-1921 indicator. In columns (2) and (5), we replace decade with region-decade fixed effects. The last row reports the coefficient of the interaction between the number of emigrants and a post-1921 indicator. The K-P *F*-statistic refers to the Kleibergen-Paap statistic for weak instruments. Units are weighed by their population in 1881. Standard errors clustered at the district level are reported in parentheses. Referenced on pages: 16, 21.

***: $p < 0.01$, **: $p < 0.05$, *: $p < 0.10$

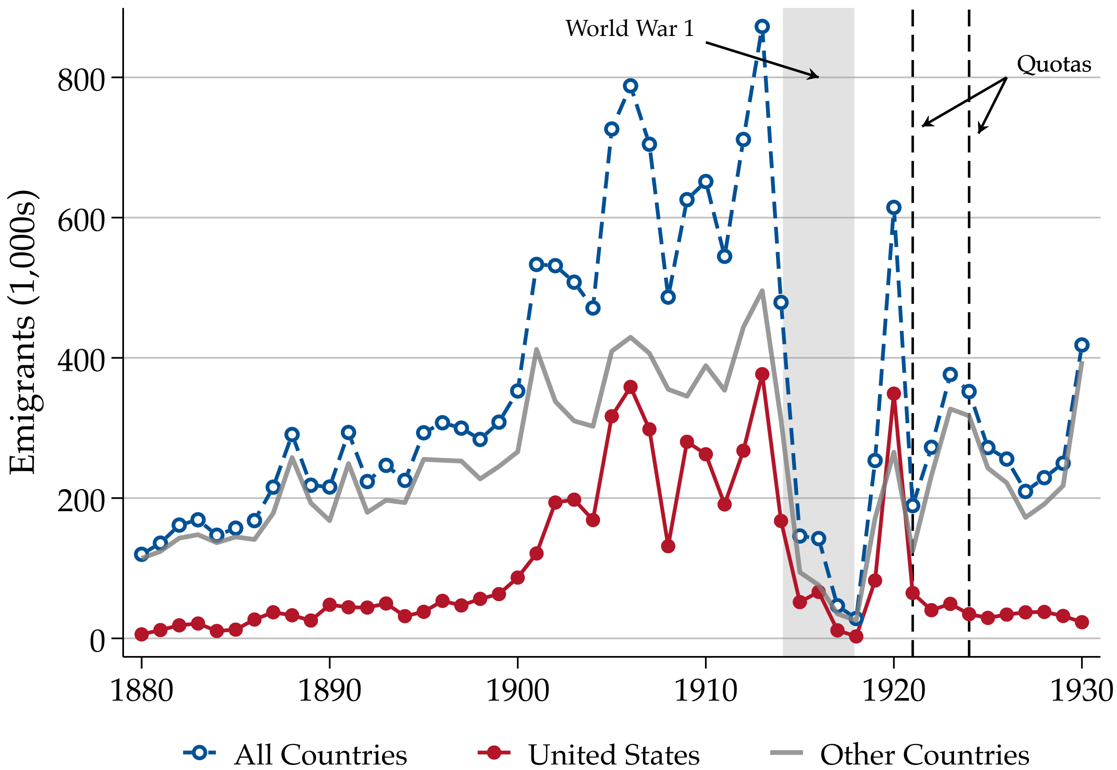
Table VI. The Response of Agriculture Employment

	Dependent Variable: log(Agriculture Employment)					
	OLS			2SLS		
	(1)	(2)	(3)	(4)	(5)	(6)
US Emigrants × Post	-0.010	-0.025		-0.031	-0.047	
	(0.027)	(0.019)		(0.041)	(0.055)	
I(US Emigrants) × Post			-0.021			0.026
			(0.053)			(0.138)
Emigrants × Post	-0.102***	-0.122**	-0.104***	-0.088**	-0.104***	-0.115**
	(0.034)	(0.057)	(0.038)	(0.034)	(0.039)	(0.047)
District FE	Yes	Yes	Yes	Yes	Yes	Yes
Decade FE	Yes	–	–	Yes	–	Yes
Region-Decade FE	No	Yes	Yes	No	Yes	No
N. of Districts	203	202	203	203	202	203
N. of Observations	1,000	995	1,000	1,000	995	1,000
K-P *F*-stat				87.036	99.680	14.490
Mean Dep. Var.	10.733	10.722	10.733	10.733	10.722	10.733
Std. Beta Coef.	-0.074	-0.187	-0.015	-0.238	-0.372	0.019

Notes. This Table reports the estimated effect of district-level exposure to the US Quota Acts on agricultural employment. The dependent variable is log-agriculture employment. The unit of observation is a district observed at a census-decade frequency between 1881 and 1936. In columns (1–2) and (4–5), the treatment is an interaction term between US emigration and a post-1921 term. In columns (3) and (6), the treatment is an interaction between a dummy equal to one for above-median US emigration districts and a post-1921 term. In column. Columns (1–3) display the OLS estimates; columns (4–6) display the two-stage least squares estimates obtained using the instrument defined in (2). Each regression includes district and time-fixed effects and controls for the number of emigrants and an indicator variable for Southern regions; both interacted with a post-1921 indicator. In columns (2) and (5), we replace decade with region-decade fixed effects. The last row reports the coefficient of the interaction between the number of emigrants and a post-1921 indicator. The K-P *F*-statistic refers to the Kleibergen-Paap statistic for weak instruments. Units are weighed by their population in 1881. Standard errors clustered at the district level are reported in parentheses. Referenced on page: 21.

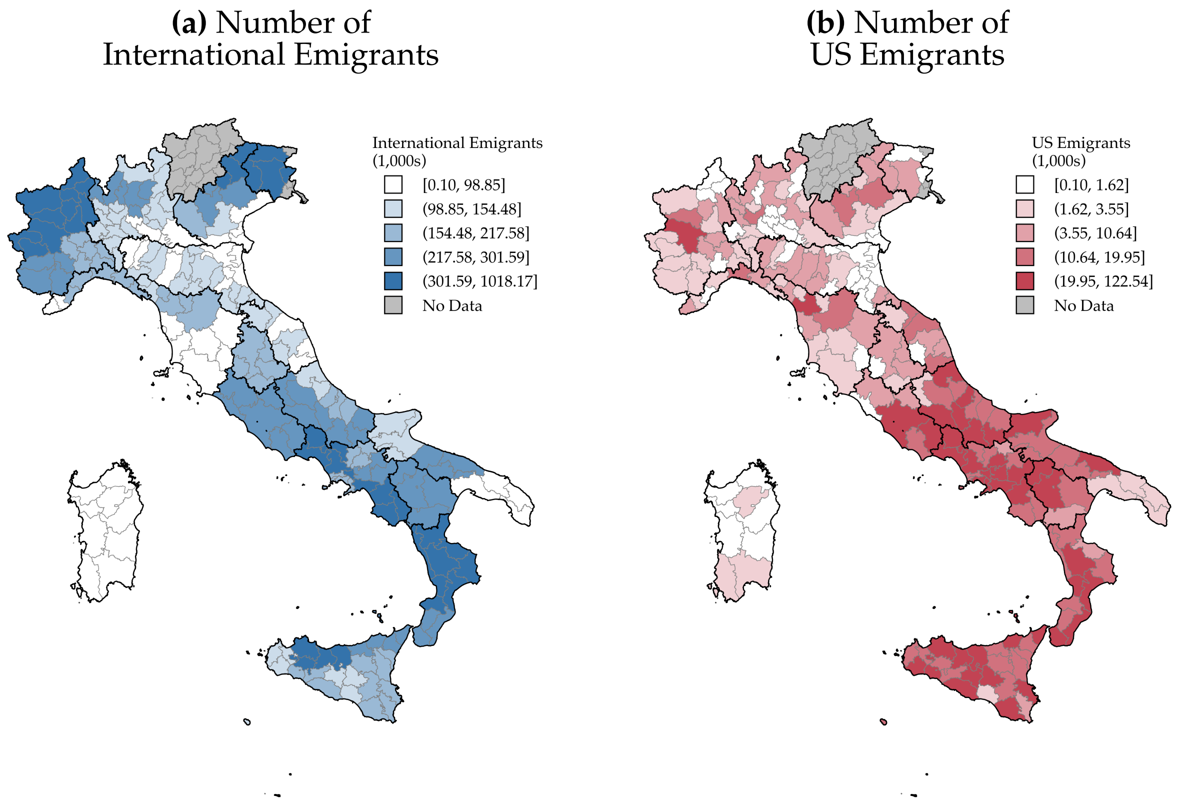
***: $p < 0.01$, **: $p < 0.05$, *: $p < 0.10$

Figure I. International Migrants, 1880–1930



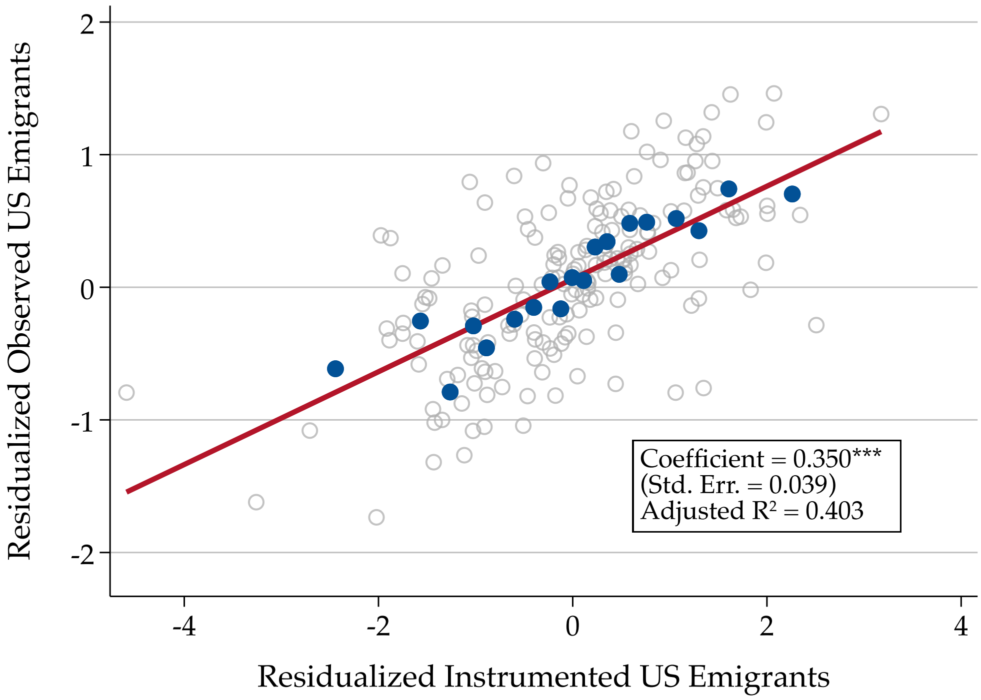
Notes. This figure reports the yearly outflow of international migrants from Italy between 1880 and 1930. Data are digitized from the *Annuario statistico dell'emigrazione italiana dal 1876 al 1925: con notizie sull'emigrazione negli anni 1869-1875* and from the *Statistica delle migrazioni da e per l'estero: anni 1928, 1929 e 1930 con confronti dal 1921 al 1927*, both edited by the Italian Statistical Office. The blue dashed line reports the overall number of international migrants; the red solid line reports the number of migrants to the United States, the single most important destination country over this period; the gray line reports emigrants to every other country. The shaded gray area marks the 1914–1918 war years; the dashed vertical black lines mark the 1921 and 1924 Emergency and Johnson-Reed Immigration Quota Acts, respectively. Referenced on pages: 1, 11, 19, 3.

Figure II. Spatial Distribution of International Emigration, 1892–1924



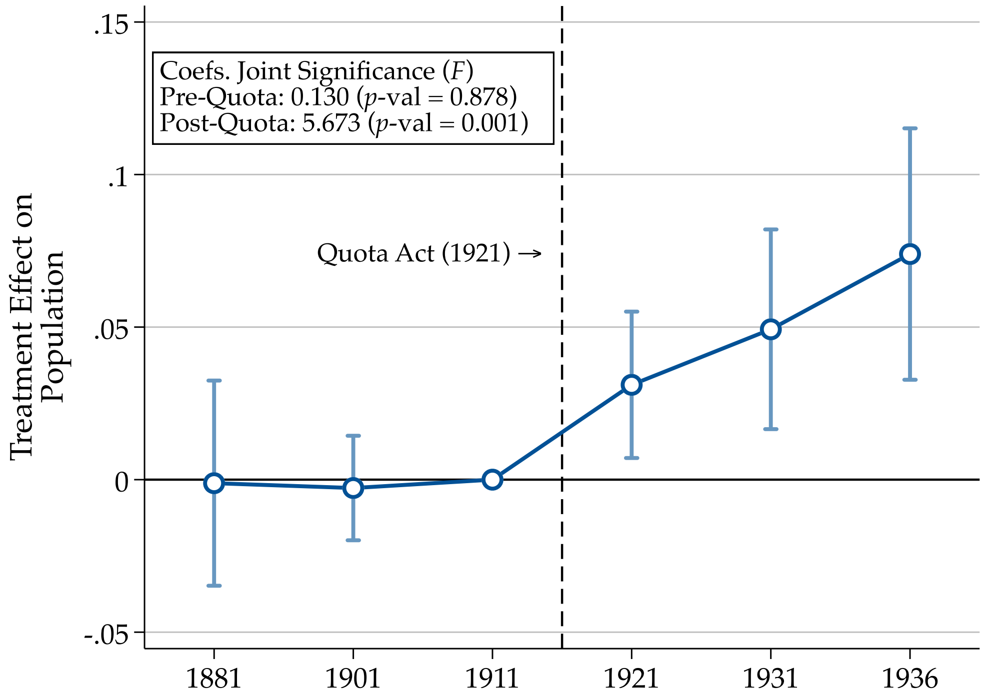
Notes. These figures report the spatial distribution of overall international emigrants (Figure IIa) and emigrants who settle in the United States (Figure IIb) across districts. In each map, the unit of observation is a district. Black lines superimpose region boundaries. The geography is at 1921 borders. The grey areas (today Trentino, Südtirol, Venezia-Giulia and Gorizia) were acquired after the Treaty of Versailles (1918) and are thus excluded from the sample. Lighter to darker shades indicate higher values. Panel IIa reports the number of emigrants who leave the Italian territory. The data is at the province level, hence the clustered rendering. Panel IIb displays the number of emigrants who settle in the US and are registered at the Ellis Island immigration station. The data are at the district level and constructed as detailed in the main text. The figures refer to 1892–1924; emigrants are expressed in thousand units. Referenced on page: 11.

Figure III. Correlation Between Measured and Predicted US Emigration



Notes. This Figure reports the "first stage" correlation between the observed Quota Exposure and the shift-share instrument. On the y- and x-axes, we report the residuals of observed US emigration and of the instrument. The residuals are obtained by regressing each variable against province fixed effects, which also control for an indicator for Southern regions and the number of emigrants, that respectively vary at the region and province level, and computing the regression residuals. The unit of observation is a district. Each gray dot refers to one district; the blue dots report the binned scatterplot of the residuals. The red line corresponds to the ordinary least-squares fit. The Figure reports the regression coefficient, its standard error clustered at the district level, and the adjusted coefficient of determination. Referenced on page: 17.

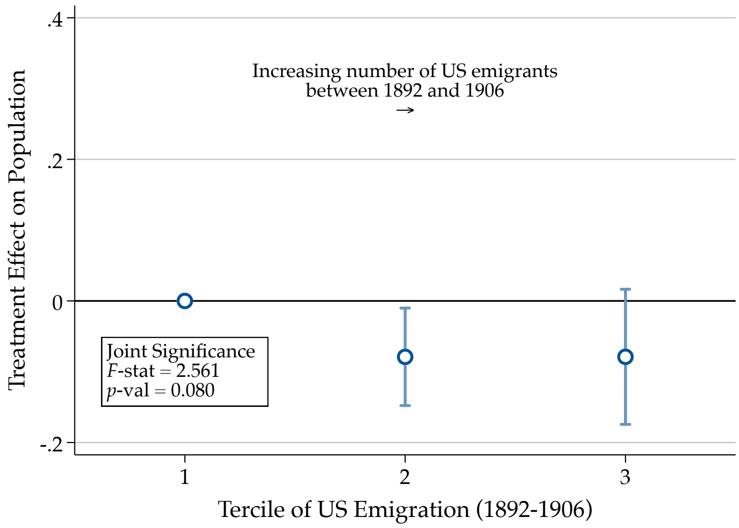
Figure IV. Effect of Exposure to the Quota Acts on Population



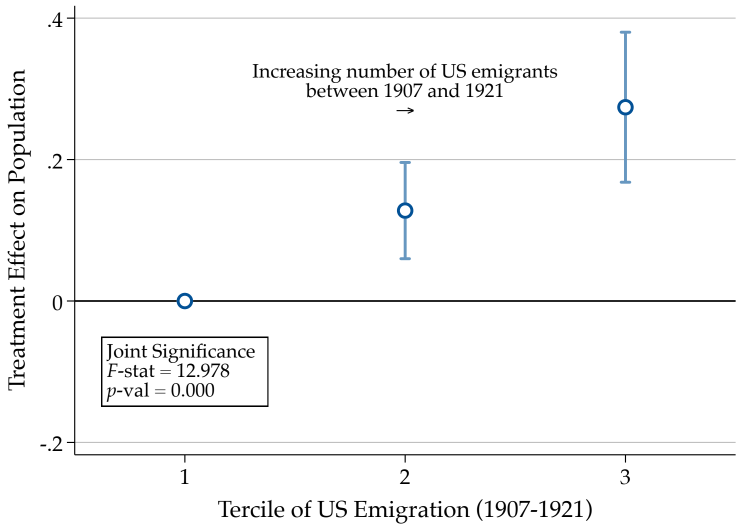
Notes. This figure reports the estimated effect of district-level exposure to the US Quota Acts on log-population. The unit of observation is a district observed at a census-decade frequency between 1881 and 1936. The outcome variable is population. Each dot reports the coefficient of an interaction term between period dummies and the cross-sectional instrument of US emigration. The estimates are obtained via two-stage least squares using the instrument defined in (2). The regression includes district and decade-fixed effects and controls for the volume of international migrants interacted with decade dummies and an indicator variable for Southern regions interacted with a post-1921 indicator. Units are weighed by their population in 1881. The bands report 95% confidence intervals from standard errors clustered at the district level. The graph reports the values of the Wald statistics of joint significance of pre-treatment and post-treatment coefficients and their respective p-values. Referenced on pages: 15, 19, 22, 24.

Figure V. Timing of US Emigration and Population Response to the Quota Acts

(a) First Half of the Sample Period: 1892–1906



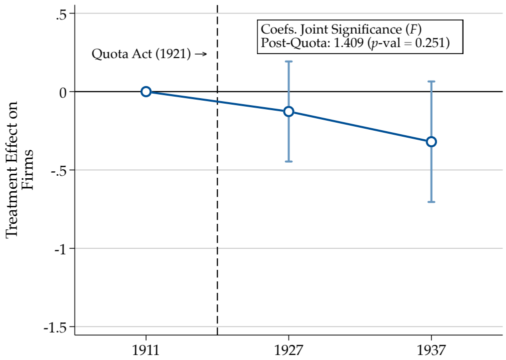
(b) Second Half of the Sample Period: 1907–1921



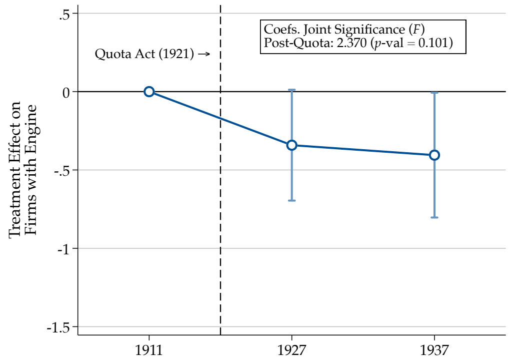
Notes. This figure reports the estimated effect of district-level exposure to the US Quota Acts on population, depending on the timing of US emigration. The unit of observation is a district observed at a census-decade frequency between 1881 and 1936. The outcome variable is log-population. Each dot reports the coefficient of an interaction term between the baseline Quota Exposure treatment, a post-quota indicator, and a set of indicator variables that code the quantile of US emigration over different time spans. In panel Va, the quantiles refer to the number of US emigrants in the first half of the analysis sample, i.e., between 1892 and 1906; in panel Vb, the quantiles are computed on the distribution of US emigrants in the second half of the sample. The estimates are obtained via two-stage least squares using the instrument defined in (2). The regression controls for the baseline treatment effect of Quota exposure. Standard errors are clustered at the district level; bands report 95% confidence intervals. Referenced on page: 19.

Figure VI. Effect of Exposure to the Quota Acts on Capital Investment

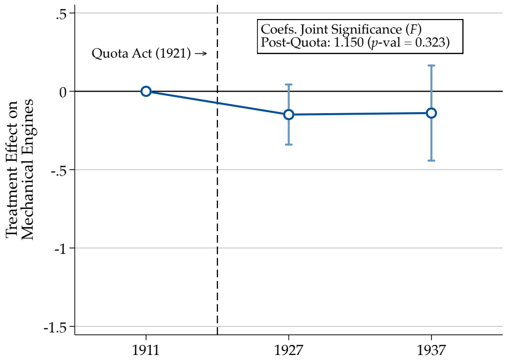
(a) Firms



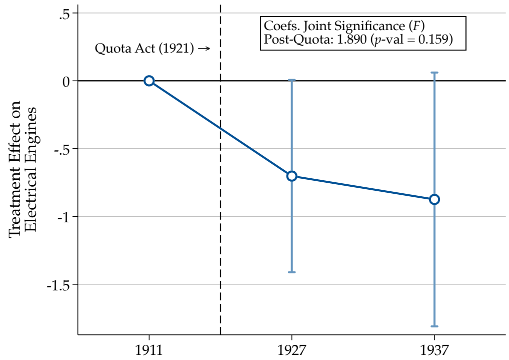
(b) Firms with an Engine



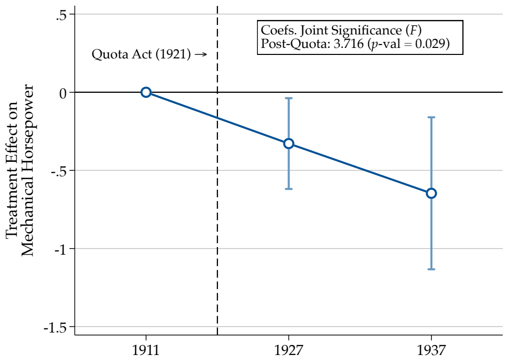
(c) Mechanical Engines



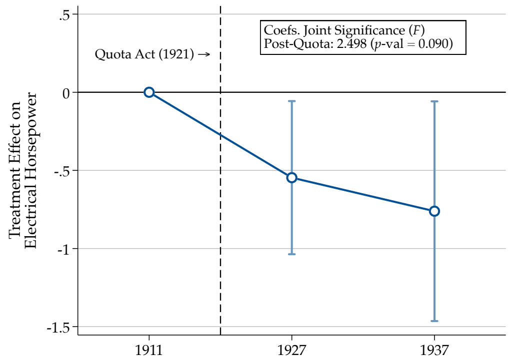
(d) Electrical Engines



(e) Mechanical Horsepower

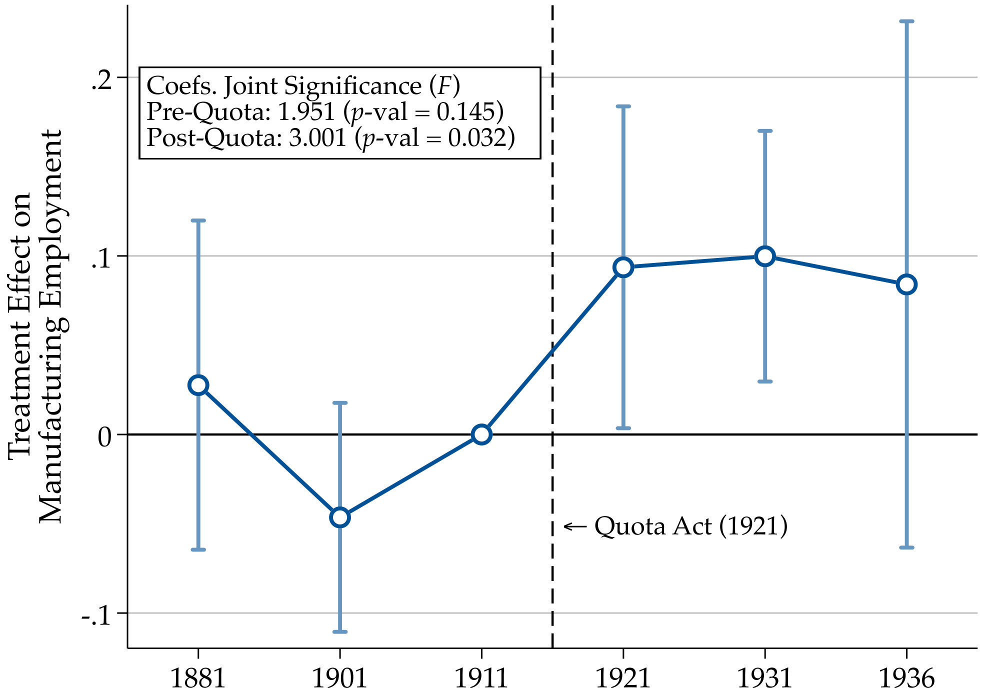


(f) Electrical Horsepower



Notes. These figures report the estimated effect of exposure to the US Quota Acts on capital investment over time. The unit of observation is a province observed at a census-decade frequency between 1911 and 1937. The outcome variables are firms (panel VIa), firms with an engine (panel VIb), mechanical (panel VIc) and electrical (panel VId) engines, mechanical (panel VIe) and electrical (panel VIf) horsepower. All dependent variables are expressed in log. Each dot reports the coefficient of an interaction term between period dummies and the cross-sectional instrument of US emigration. The estimates are obtained via two-stage least squares using the instrument defined in (2). Each regression includes province and decade-fixed effects and controls for the volume of international migrants interacted with decade dummies and an indicator variable for Southern regions interacted with a post-1921 indicator. Units are weighed by their population in 1881. Bands report 95% confidence intervals from standard errors clustered at the province level. The graphs report the values of the Wald statistics of joint significance of pre-treatment and post-treatment coefficients and their respective *p*-values. Referenced on page: 20.

Figure VII. Effect of Exposure to the Quota Acts on Manufacturing Employment



Notes. This Figure reports the effect of exposure to the Quota Acts on manufacturing employment. The unit of observation is a district observed at a census-decade frequency between 1881 and 1936. The outcome variable is log-employment in manufacturing. Each dot reports the coefficient of an interaction term between period dummies and the cross-sectional instrument of US emigration. The estimates are obtained via two-stage least squares using the instrument defined in (2). Each regression includes district and decade-fixed effects and controls for the volume of international migrants interacted with decade dummies and an indicator variable for Southern regions interacted with a post-1921 indicator. Units are weighed by their population in 1881. The bands report 95% confidence intervals from standard errors clustered at the district level. The graph reports the values of the Wald statistics of joint significance of pre-treatment and post-treatment coefficients and their respective p-values. Referenced on pages: 15, 22.

ONLINE APPENDIX

Emigration Restrictions and Economic Development: Evidence from the Italian Mass Migration to the U.S.

Davide M. Coluccia & Lorenzo Spadavecchia

November, 2025

TABLE OF CONTENTS

A DATA APPENDIX

A.I Data Sources

This section lists the sources, methodology, and coverage of the data that we assemble. We defer a more detailed description of the emigration data to Appendix A.III. Table B.1 summarizes the content of this section.

Population Censuses. We extract information from population censuses in 1881, 1901, 1911, 1921, 1931, and 1936. No census was taken in 1891. We digitize district-level information on the number of people employed in all major sectors in the 1881–1921 censuses. The same information is available at the municipality level for 1931 and 1936. Hence, we aggregate it at the district level. To assign municipalities to districts, we geo-code each municipality and overlay its coordinates with the 1921 district shapefile. From population censuses, we also extract information on employment by industry. This information is available by districts until 1921, but only by provinces in 1931 and 1936. We thus aggregate industry employment to provinces. Population data have been tabulated by ISTAT for each municipality. We aggregate them at the district and province levels. We also code two types of urbanization variables: the district-level share of the population living in cities with at least k-thousand inhabitants and the district-level share of municipalities with at least k-thousand inhabitants. The ISTAT population tables report information on the area, altitude, and access to the sea for each municipality.

Manufacture Censuses. Manufacture censuses were taken in 1911, 1927, and 1937. They report province-level information on the universe of manufacturing firms. We digitize data on a set of proxies for capital investment: the number of firms, the number of firms with at least one installed engine, the number of installed mechanical and electrical engines, and the number of installed mechanical and electrical horsepower.

World War One Deaths. WWI deaths are from `cadutigrandeguerra.it`, a dataset maintained by ISTORECO, a branch of the *Associatione Nazionale Partigiani d'Italia* (ANPI). The underlying data were collected by the Fascist regime for propaganda purposes. The honor roll call contains information on 570,355 Italian soldiers who lost their lives during the war. The data appear to be comprehensive since most estimates place the total death toll at 650,000. For each individual, we know their name and surname, birth year, municipality of origin (which we map to the municipalities listed in the ISTAT data), military rank at death, regiment, the date, location, and cause of death, and any decorations received. Deaths span the war years (1915–1918). We aggregate them at the district and province levels to have a cross-sectional indicator of mortality.

Railways. We reconstruct the network of Italian railways between 1839 and 1926 from a volume curated by the Italian Statistical Office and edited in 1927. This is the first paper using these data. The unit of observation in the data is truck lines connecting two stations. We have information on the precise date each line opened, the distance it covered, and the name of the station it connected to. We geo-code the location of each station and generate a dummy variable indicating whether a given municipality has at least one station, as well as a count variable that returns the number of stations. We also define a variable that returns the number of kilometers that separate every given municipality from the closest transatlantic migration port. Calabrese (2017) suggests that railway access to Genoa, Palermo, or Naples–the only ports with ships sailing toward the United States–was a crucial condition to ignite migration movements. We thus compute the shortest path connecting each municipality with each transatlantic port, given the state of the railway network in every given year, and take the minimum among the three. We aggregate this variable by district and province, taking a population-weighted average of the shortest railway distances to emigration ports.

Internal Migrations. We digitize internal migration data from the reports of the Official Statistics of the Commissioner General for Emigration. The data are available at the region level at a decade frequency and cover the period 1901–1931. We know, for each region, the number of individuals by their region of origin. This allows us to construct a bilateral matrix that tracks the stock of individuals by their region of origin and residence. Additionally, the data contains information on the number of individuals living in their municipality, province, and region of origin, as well as the number of individuals living outside of their municipality, province, and region of origin.

Remittances. Remittances data are from the *Annuario Statistico dell'Emigrazione Italiana*, a publication curated by the Commissioner General for Emigration. They are available at the national level at a yearly frequency between 1902 and 1925. The underlying data are from international payments made by emigrants in the United States at the *Banco di Napoli*, which in 1901 was granted a monopoly by the Italian government to handle immigrant remittances collection and transmission (Barbiellini Amidei and Goldstein, 2007). We complement the remittances data with annual Italian GDP estimates produced by Baffigi (2013).

A.II Construction of the Sample

All variables that are not computed from geo-coded data are cross-walked to consistent 1921 district and border geographies using the method described by Eckert et al. (2020). To do so, we use GIS boundary files publicly provided by ISTAT for each census year. Even though this yields quantitatively very small corrections, it is important to ensure that geographies remain consistent because the Fascist regime undertook a revision of local government divisions, which ultimately led to the abolition of districts in 1927.

We are forced to exclude areas that were annexed as a consequence of World War I in 1918–Trento and Trieste, Südtirol, Istria, and Zara–because we do not observe pre-Quota outcomes and treatment in those regions.

Unlike for US emigrants, we do not have district-level data on overall emigration. In the baseline exercise, we control for the province-level number of emigrants. In robustness exercises presented in Table B.9, we impute district-level emigration using two apportioning rules. First, we apportion emigrants to each district in proportion to its US emigration rate relative to the districts in that province. This approach assumes that the overall emigration rate was proportional to the US emigration rate. Second, we assign emigrants to each district in proportion to the agricultural employment share in that district relative to the other districts in the province. Section III.B explains this approach in more detail. Both apportionment methods yield results that are quantitatively similar to those of the baseline province-level approach.

We assembled three distinct samples (Samples 1, 2, and 3). Sample 1, which comprises population and employment-by-sector data, is at the district level and covers the years 1881, 1901, 1911, 1921, 1931, and 1936. Agricultural employment was not available in 1931. All data in Sample 1 are either digitized from population censuses or are aggregated from data tabulated by ISTAT. Sample 2 runs at the province level, covers capital variables, and is available in 1911, 1927, and 1937. Capital variables are retrieved from manufacturing censuses. Sample 3 runs at the province level, covers employment-by-industry variables, and spans the years 1901, 1911, 1921, 1927, and 1936. For the first three decades, the data are from population censuses; for the latter two, we digitize them from manufacturing censuses. Since capital variables exhibit substantial skewness, we winsorize them at the 5% level.

A.III Details on the Emigration Data

A.III.1 Data Source and Motivation

In this section, we provide a detailed description of the emigration data we collect. The raw data can be found at `https://heritage.statueofliberty.org/`. First, we describe the methodology that we adopt to assemble the data. Second, we show how to validate this dataset with external sources. Last, we present some stylized facts that the new data allow us to document.

This dataset addresses a key limitation of commonly used US census data (Ruggles et al., 2021). Census records contain the country of origin of immigrants residing in the US, but they do not report where immigrants originated from within their home country. This issue applies to all countries. Hence, separate papers developed strategies to reconstruct such information for Norway (Abramitzky, Boustan and Eriksson, 2012) and England (Coluccia and Dossi, 2025). This paper looks at emigrants from Italy, a major emigration country among the so-called "second-wave" nations.

To query the online dataset, we extract the surnames of all Italian immigrants appearing in the US census. We run a separate search looking for all individuals with each surname. Overall, we run approximately 700,000 single searches. We collect individual-level information on the name and surname of the immigrants, their municipality of origin, their age, immigration year, and literacy.

A.III.2 Nationality Imputation

The surname search returns approximately 13,000,000 uniquely identified individuals. The majority of them, however, are not Italian, even though they have a surname that is carried by at least one Italian immigrant in the census.[29] In this section, we explain how we narrow the dataset to ensure that it only comprises Italian immigrants.

The Ellis Island dataset's nationality information relies on the place of origin of the immigrants. Our final analysis only leverages migrants with a non-missing place of origin that we can geo-code

[29] For example, several Italians from Northern regions have German-sounding surnames, while Spanish-sounding surnames are present in Southern regions.

to an Italian municipality. Hence, the sample implicitly only includes migrants from Italy without requiring any nationality imputation. However, this approach is not suitable for comparing the Ellis Island data coverage with external sources because several records have a missing place of origin, even though they plausibly pertain to Italian immigrants. We discuss the extent and implications of this limitation below in this section.

We thus employ state-of-the-art large language models (`GPT-5-mini`) to predict the nationality of origin of the immigrants based on their first and last names. The key advantage of a semi-automated imputation approach is that it reduces human cognitive biases of manual validation approaches. We use the following prompt to predict the nationality of origin.[30]

`GPT-5-mini` Prompt for Nationality Imputation

You are a native Italian speaker. Given a "name surname" pair, determine whether the person is likely to be ethnically Italian. To be considered Italian, both the first name and the surname must be Italian. Names typical of minority groups in Italy (e.g., Slavic, Albanian, Greek, Germanic, or recent immigrant communities) should NOT be considered Italian.
Return a strict `JSON` object with no markdown, no explanation, and the following field: "Italian", equal to

- 1 if the name is likely Italian (both first name and surname are Italian);
- 0 otherwise

Input:
`< Name, Surname >`

Out of the roughly 13,000,000 results that we obtain from the surname queries, `GPT` identifies 4,194,728 Italians (see Table B.4). This figure aligns with existing studies that estimate that between 4 and 4.5 million Italians entered the United States between 1892 and 1924 (Choate, 2008). Out of this grand total, the place of origin is not missing for 88% of the records.

We employ two methods to validate the `GPT`-based classifier. First, for each name and surname, we compute an "Italian name/surname score" equal to the share of individuals with that name who are first- or second-generation Italian immigrants in the US census. The score equals 100 if all individuals with a given name or surname are Italian, and zero otherwise. In Figure C.1a and Figure C.1b, we plot the distribution of the Italian name- and surname-sounding indices for the sample of individuals that `GPT` identifies as non-Italians (in blue, shown on the left y-axis) and Italians (in red, on the right y-axis). For both names and surnames, the distribution of the Italian-sounding index of non-Italians is concentrated at zero and presents virtually zero mass above 20. By contrast, the distribution of Italians has a mass above 80. This sharp difference and the small overlap between the distributions of Italians and non-Italians provide compelling evidence that the `GPT` classifier successfully identifies

[30] It is worth noting that we only employ the imputed nationality to evaluate the coverage of the dataset. In the analysis, we only retain records with a non-missing municipality of origin.

Italians based on the plausibility that their name and surname are Italian-sounding.

As a complementary validation method, we focus on the subsample of records with a non-missing municipality of origin. As we discuss in the next section, we geo-reference them to their municipality of origin. We are thus sure that the individuals within this geo-references subsample originate from Italy. In Figure C.1c, we report the share of records from an Italian municipality that the `GPT`-based classifier correctly identifies as Italian. The average accuracy rate of the LLM classifier is 92%, implying that, out of 100 randomly selected individuals originating from an Italian municipality, `GPT` correctly identifies 92 as Italian. The accuracy rate remains high and stable throughout the sample period.

We use the imputed nationality to validate the dataset. Most importantly, it allows us to measure the probability that an Italian immigrant in the Ellis Island dataset does not have a reported municipality of origin. Because the nationality in the original Ellis Island dataset is based on the reported municipality, any record with a missing municipality would be automatically discarded, thus invalidating the assessment of the extent of missing observations. We do not, however, use the imputed nationality in the econometric analysis, because to measure Quota exposure, we require data on the municipality of origin of the immigrants.

A.III.3 Geo-Coding the Municipality of Origin of the Immigrants

The municipality of origin is recorded consistently only between 1892 and 1924. We document the coverage of the recorded origin municipality in Appendix A.III.5. Municipalities are frequently coded with spelling errors, possibly because they were recorded by American enumerators. In 29.8% of the cases, the listed origin is recorded without mistakes, so we pair the exact origin string with an existing municipality.

For the remaining municipality entries, we adopt a semi-automated approach. We employ `GPT-5-mini` to pair the recorded municipality with the actual municipality using the following prompt.

`GPT-5-mini` Prompt for Municipality Name Cleaning

You are given a string that might represent the name of an Italian municipality (comune).
It may contain spelling errors, abbreviations (e.g. "s." for "san" or "santo" or "santa"), or transcription errors (e.g., due to similar calligraphy).
However, it may also refer to a foreign place or not be a municipality.
Return a strict JSON object with the following fields (no markdown, no extra commentary):

- `municipality_name`: the most likely actual Italian municipality (e.g., "Forenze" → "Firenze") if one can be reliably inferred. If the input contains information about a region, province, or nearby town (e.g., "Italy," "Toscana," or "Napoli"), use that context to allow more permissive matching. If no plausible match can be identified, return null.

- `is_in_italy`: 1 if the string refers to an Italian location, 0 otherwise (e.g., "Roma," "Sicily," and "Italy" are in Italy, but "New York" or "Argentina" are not).

- `is_high_confidence`: 1 if the match is highly probable based on spelling and/or contextual cues (such as regions or nearby towns), 0 otherwise. Minimize false positives in the absence of contextual clues.

- `is_municipality`: 1 if the matched name corresponds to a real Italian municipality (comune), 0 otherwise (e.g., "Roma" is a municipality, but "Lazio" or "Italy" are not).

Input:
`< Municipality >`

To check the plausibility of the matches delivered by `GPT`, we compute (i) the Jaro-Winkler similarity between the recorded municipality and the municipality imputed by `GPT`, and (ii) the maximal similarity between the match delivered by `GPT` and all municipalities (i.e., the municipality with the highest JW similarity to the raw string).

In 98% of the $2,339,409 + 10,003 = 2,349,412$ cases where the origin location is not located outside of Italy, the municipality predicted by `GPT` coincides with the most similar municipality. Moreover, in 70% (resp. 32%) of the cases, the Jaro-Winkler similarity between the municipality predicted by `GPT` and the most similar municipality is 0.8 (resp. 0.9). Relatively larger spelling mistakes are typically due to errors in prefixes (e.g., "s." in place of "santo" or "santa") that the string similarity measure does not adequately correct for.

Then, we geo-code the remaining entries using the Google Maps API commercial software. In Appendix A.III.5, we comment on the performance of the geo-coding routine, which successfully assigns latitude and longitude coordinates 99.8% of the municipalities. We assess the plausibility of this matching exercise in Appendix A.III.6. In the analysis, the dataset is aggregated by district or province, depending on the part of the analysis, using boundaries in 1921 from historical shape files provided by ISTAT.

A.III.4 Gender Imputation

To explore the mechanisms that could explain the documented drop in capital investment in response to the Quotas, we need to measure the share of males among US emigrants in the Ellis Island dataset. Gender is not part of the variables available in the original data. We thus follow the standard practice in the literature and impute it based on migrants' first names (e.g., see Gallen and Wasserman, 2023; Adams, Hara, Milland and Callison-Burch, 2025). Specifically, we identify a name as indicating that an individual is a man (resp. woman) if at least 80% of those with the same name as the individual are listed as men (resp. women) in the US censuses 1900–1930. Names are coded with misspellings and abbreviations. We touch on this issue in Appendix A.III.8. To impute the gender, we match each name to the closest name that appears in the census in terms of their Jaro-Winkler string similarity, provided that their similarity is above 0.95.

Figure C.4 reports the resulting district-level share of males among US emigrants between 1892 and 1924. As evidenced in historical scholarship, the Italian transatlantic migration to the United

States was dominated by men. On average, our imputation exercise indicates that 73% of US emigrants were men. In no districts did women exceed men as a proportion of the overall migrant stock. There is, however, considerable heterogeneity, as the share of male US emigrants ranges from 60% to 90%.

A.III.5 Internal Validation

To evaluate the coverage of the Ellis Island dataset, we rely on the nationality imputed as described above. In column (2) of Table B.4, we tabulate the total number of Italians in the dataset. We extracted 4,194,728 records. Historians estimate that between 4 and 4.5 million Italians immigrated to the United States between 1892 and 1924 (Choate, 2008). Approximately 95% passed through the Ellis Island immigration station (Unrau, 1984), implying that the hypothetical complete coverage of the data ranges between 3.8 and 4.275 million individuals. The overall number of records in the dataset is thus in line with the near-universe of Italians that immigrated to the US over this period.

The key advantage of the Ellis Island dataset, compared to existing data, is that it allows us to trace individual migrants to their municipality of origin. Municipalities, however, are not available for the universe of records.[31] In Table B.4, we show the coverage of municipality data in relative (column 1) and absolute (column 2) terms. Approximately 510,000 records, or 12% of the dataset, have a missing municipality or list a place of origin that is not a municipality (e.g., "Italia" or "Calabria"). Approximately 31%, equal to 1,335,246 records, are matched to an origin place that is not in Italy. An immigrant at the Ellis Island station would report a place in the United States if they were returning to the United States. It is thus difficult to distinguish between returnees, that is, Italians who had already migrated to the US, temporarily returned to Italy, and then moved back to the US, from enumeration error. Wyman (1993) documents that temporary trips back to their origin country were relatively common among immigrants. However, a proportion of these records may be transcription errors of immigration officers. Out of the 56% of remaining matches, we geo-reference 55.77% to an Italian municipality. We are thus able to correctly geo-code 2,339,406 immigrants to accurate latitude and longitude coordinates.[32] These are the records that we ultimately use to compute district- and province-level exposure to the Quotas in the analysis.

In Figure C.2, we report the immigrants with a missing origin (blue), that are geocoded to an Italian municipality (in red) and that are geocoded to a place outside of Italy (in gray) both in absolute (Figure C.2a) and relative (Figure C.2b) terms over time. The share of immigrants that we successfully geocode to an Italian municipality remains stable around 60% between 1896 and 1921, except for the war years, when, however, the number of emigrants was very low. Before 1896, during World War I, and after 1921, the share of immigrants with a missing municipality of origin increases. These are, however, periods when the overall number of emigrants contracts, as shown by the gray area in the background, which displays the total number of immigrants entering the United States according to

[31] A quick check indicates that municipalities are missing for a vast part of the Ellis Island records before 1892 and after 1924. Additionally, migrants from Eastern European countries frequently indicate coarser origin places (e.g., "Poland") or list illegible origin places, possibly because immigration officers had issues understanding their language.

[32] A residual 0.24%, equal to 10,003 records, cannot be matched to coordinates by the geocoding routine.

US official statistics. Overall, the coverage of the Ellis Island dataset appears to remain consistently high throughout the sample period.

A possible concern with the municipality of origin listed in the Ellis Island data is that, since immigrants were asked about their last place of residence, they may have indicated the port city from which they sailed. Throughout the mass migration period, transatlantic ships sailed from three ports: Genoa, Naples, and Palermo. If immigrants listed the port of departure as their origin municipality, one would expect that those areas would have relatively higher emigration rates. To explore this possibility, in Figure C.5, we plot the distribution of the emigration rate, computed as the number of US emigrants between 1892 and 1924 divided by population, for all the 7,492 *municipalities*. We highlight the districts of the three transatlantic emigration ports, and within each district, we display the border of the port's municipality in red. Port municipalities do not display higher US emigration rates compared to their surrounding areas. If anything, the share of US emigrants is slightly lower. In Table B.6, we provide more systematic evidence by regressing the US emigration rate against a dummy equal to one for municipalities with a port (columns 1–2) or for municipalities in the same province of a port (columns 3–4). In columns (1) and (3), we report the unconditional correlation, while in columns (2) and (4), we control for region and emigration year fixed effects. We do not find a statistically significant association between the presence of a transatlantic port, or proximity to a port, and the share of US emigrants. This pattern indicates that immigrants plausibly listed their origin municipality, rather than the port of departure, at the Ellis Island immigration station.

A related concern is that migrants may have reported a large or significant city close to their municipality of origin instead of their true place of origin. We assess the plausibility of this concern in Table B.7. The dependent variable is the emigration rate normalized by population. In column (1), we ask whether regional capital cities (e.g., Palermo and Naples) display disproportionately higher emigration rates compared to cities in the same region. In column (2), we repeat the exercise with provincial capital cities. In columns (3) and (4), we compute the correlation between the US emigration rate and population in 1881 when comparing cities in the same region (column 3) and in the same province (column 4). None of these factors displays a statistically significant association with the US emigration rate, indicating that migrants were not differentially more likely to report regional capital cities, provincial capital cities, or larger cities as their municipality of origin.

A.III.6 External Validation

The granular nature of the dataset implies that we cannot validate it with existing data at similar levels of aggregation. Our strategy, instead, is to aggregate it at the regional level and compare it with data from official statistics on Italian emigration to the United States collected by the Commissioner General for Emigration. These span the period 1877–1925 and are available by region.

In Table B.5, we report the correlation between the Ellis Island dataset and US emigration as recorded in official statistics. In columns (1–5), we report the correlation between the raw series, while in columns (6–10), we take logs. We find a positive and large unconditional correlation between the two (columns 1 and 6). In particular, Ellis Island migration explains more than 80% of the region-level variation in US emigration as measured in official statistics. This correlation remains conditioning on year (2 and 7), region (3 and 8), and year and region (4 and 9) fixed effects. Importantly, it holds within the sub-sample period we use to compute the treatment (5 and 10). Figure C.3 displays the unconditional correlation between the two series (Figure C.3a) and the one absorbing region and year fixed effects (Figure C.3b). These exercises document a positive, large, and statistically significant correlation between the Ellis Island US emigration and data from official statistics.

In Figure C.6, we check that the correlation remains high in each year of the observation sample. Each dot in the figure reports the correlation between the two series in one year between 1892 and 1924. The correlation remains consistently above 0.9 and statistically significant throughout the sample period.

A.III.7 Stylized Facts

Dissecting the specific features of mass emigration to the United States is beyond the scope of this paper. Instead, we present two suggestive facts.

First, in Table B.8, we list the districts that were relatively more exposed to the US migration. In columns (1–3), districts are ranked by the absolute number of emigrants. In columns (4–6), we rank them by the emigration rate, expressed as the ratio between overall US emigrants and the 1921 population. The vast majority of top-ranked districts are located in Southern regions. Emigration rates are higher in Sicily and Campania. The district of Palermo alone accounts for almost 90,000 emigrants out of a population of 850,000.

We then provide evidence supporting the S-hypothesis advanced by Gould (1980) and recently analyzed by Spitzer and Zimran (2023). This maintains that local out-migration patterns followed a logistic-type dynamic, with initially low uptake, large increases in a relatively short time period, and subsequent plateau. Gould (1980) connects these dynamics to information diffusion within the population. To test this hypothesis, we mark the beginning of the mass migration in each district when the US emigration rate exceeded 0.1‰. This generates a setting akin to a staggered treatment roll-out. We then use the method of (Borusyak, Jaravel and Spiess, 2024) to estimate the dynamics of US out-migration. Importantly, this approach ensures that we compare emigration districts with areas where the migration had not already begun. We find that emigration followed Gould's S-shaped pattern as documented by Spitzer and Zimran (2023). The event-study figures associated with the resulting model are listed in Figure C.7 and show that out-migration follows an S-shaped pattern as argued by Gould (1980). We interpret this finding as additional evidence supporting the quality of the underlying data.

A.III.8 Linked Sample

To produce the instrumental variable for Quota Exposure, we require information on the origin district and province of Italian immigrants by US county. This information is not available in the US census or reported in the Ellis Island data. To address this limitation, we link the full-count non-anonymized US population census (Ruggles et al., 2021) and the Ellis Island records. To the best of our knowledge, ours is the first attempt to produce a linked sample between these two uniquely rich sources. The algorithm builds on similar automated linking procedures (e.g., Abramitzky, Boustan, Eriksson, Feigenbaum and Pérez, 2021). First, we translate the names of Italian-born individuals recorded in the US census to their Italian versions.[33] To perform the translation, we apply `GPT-5-mini` to the universe of recorded names. The linking algorithm is as follows.

Consider individual j in the Ellis Island dataset born in year b_j and immigrated in year t_j. Let $\mathcal{M}^j = \{m \text{ s.t. } |t_m - t_j| \leq 1 \wedge |b_m - b_j| \leq 1\}$ denote the set of Census entries of Italians with reported year of birth within two years of b_j and immigration year within two years of t_j. We refine $\mathcal{M}^j$ to only include records with the same NYSIIS-adjusted initial name and surname letters of j.[34] For all $m \in \mathcal{M}^j$, we compute the Jaro-Winkler similarity between j and m^j's name and surname. Let $d_N(j,m)$ and $d_S(j,m)$ denote them. We define the total similarity between j and m as $\delta^j_m = d_N(j,m) + d_S(j,m)$. Let $\bar{\delta}^j = \max_{m \in \mathcal{M}^j} \delta^j_m$ be the maximal similarity for j across all possible matches. The set of *candidate* matches is $\overline{\mathcal{M}}^j = \{m \in \mathcal{M}^j \text{ s.t. } \delta^m_j = \bar{\delta}^j\}$, that is, it contains the records that attain the maximal similarity with record j. Finally, the set of *accepted* matches is $\widetilde{\mathcal{M}}^j_\tau = \{m \in \overline{\mathcal{M}}^j \text{ s.t. } d_N(j,m) > \tau \wedge d_S(j,m) > \tau\}$, where $\tau \in (0,1)$ defines the minimal similarity threshold above which we accept a match in the linked sample. In our application, we set $\tau = 0.95$ to ensure that we only include high-quality matches, following the standard practice in the literature (Abramitzky et al., 2021).

We evaluate the distance between two strings i and j in terms of their Jaro-Winkler similarity $d(i,j)$:

$$d(i,j) \equiv \widehat{d}(i,j) + \ell p \left[1 - \hat{d}(i,j)\right] \tag{A.1}$$

where

$$\widehat{d}(i,j) \equiv \begin{cases} 0 & \text{if } m = 0 \\ \frac{1}{3}\left(\frac{m}{|i|} + \frac{m}{|j|} + \frac{m-t}{m}\right) & \text{else} \end{cases} \tag{A.2}$$

where m is the number of matching characters, $|A|$ is the length of string A, t is half the number of transpositions, ℓ is the length of the common or eventual common prefix no longer than four characters between i and j, and $p = 0.1$ is a constant scaling factor. Two characters are matching only if they are the same and are not farther than $\left\lfloor \frac{\max(|i|,|j|)}{2} \right\rfloor - 1$. Half the number of matching characters in a different sequence order is the number of transpositions. The Jaro-Winkler string similarity metric is commonly employed in the census-linking literature because it allows for fuzzy matches by penalizing string dissimilarities in the first part of the string more than in the latter portion of the string (Abramitzky et al., 2021).

In Figure C.8, we report the distribution of name and surname similarities for the sample of individuals with at least one potential match. There is substantial mass at 1, where matches are literal. In Figure C.9, we report the share of Ellis Island records with at least one accepted match, in blue. There are several reasons why someone recorded at Ellis Island may not appear in the 1900 census.

[33] For instance, we convert "Peter" to "Pietro." This procedure ensures that Anglicizations of Italian names that occur in the US census but not in the Ellis Island records do not artificially deflate our matching rate.

[34] This refinement, which is also applied by Abramitzky et al. (2021), is computationally useful, but it is unlikely to be empirically important, because a manual inspection indicates that reporting errors are frequent in the second half of names and surnames rather than at the beginning. The NYSIIS adjustment implies that "Carlo" and "Karlo" would be allowed to match because their first letters are phonetically analogous.

First, that person may have left before 1900. Second, women could change their surname. Moreover, Italians could choose to change their surname as an assimilation effort. If the name change did not simply consist of a translation, we would fail to detect this practice. Finally, the immigration year in the census may be coded with errors. The resulting $\approx$ 6.2% average matching rate is not very distant from the benchmark rate of Abramitzky et al. (2012), which is the only paper linking the census records of immigrants from the US census—in their case, from Norway—to external microdata—in their case, the Norwegian census records of the immigrants before they moved.

A key concern for the empirical strategy is that the probability of matching individuals from the Ellis Island data is correlated with their area of origin. This possibility would induce selection in the resulting linked sample, implying that the linked sample is not representative of the overall immigrant population. While we provide quantitative evidence to support the relevance of the first stage, we can check whether any systematic selection pattern emerges directly from the linked data. In Figure C.10, we report the correlation between the probability of matching and the origin region of immigrants. There appears to be no systematic selection pattern. Individuals from Calabria and Sicily are marginally more likely to be matched, even though this difference likely reflects a larger sample size of emigrants from those regions. In Figure C.11, we show that there is a positive and highly statistically significant correlation between the district-level US emigrant outflows in the Ellis Island dataset and in the linked Ellis Island-census dataset. These patterns indicate that the geographic coverage of the linked dataset closely matches that of the full migrant dataset, thereby demonstrating geographical representativeness.

B ADDITIONAL TABLES

Table B.1. Summary of the Data Sources and Coverage

Variable (1)	Observation Unit (2)	Source (3)	Observed Years (4)
Panel A. Demographics			
Population	District (1881-1921), Municipality (1931-1936)	Population Censuses	1881-1936, excl.1891
Area	District (1881-1921), Municipality (1931-1936)	Population Censuses	1881-1936, excl.1891
Urbanization	District (1881-1921), Municipality (1931-1936)	Population Censuses	1881-1936, excl.1891
Literacy	Municipality	Population Censuses	1881-1936, excl.1891
Panel B. Employment, by Sector			
Manufacture	District (1881-1921), Municipality (1931-1936)	Population Censuses	1881-1936, excl.1891
Agriculture	District (1881-1921), Municipality (1931-1936)	Population Censuses	1881-1936, excl.1891
Trade	District (1881-1921), Municipality (1931-1936)	Population Censuses	1881-1936, excl.1891
Liberal Professions	District (1881-1921), Municipality (1931-1936)	Population Censuses	1881-1936, excl.1891
Public Administration	District (1881-1921), Municipality (1931-1936)	Population Censuses	1881-1936, excl.1891
Panel C. Capital			
Firms	Province	Manufacture Censuses	1911, 1927, 1937
Firms with Engine	Province	Manufacture Censuses	1911, 1927, 1937
Mechanical Engines	Province	Manufacture Censuses	1911, 1927, 1937
Electrical Engines	Province	Manufacture Censuses	1911, 1927, 1937
Mechanical Horsepower	Province	Manufacture Censuses	1911, 1927, 1937
Electrical Horsepower	Province	Manufacture Censuses	1911, 1927, 1937
Panel D. Manufacturing Employment, by Industry			
Agriculture	District (1901–1921), Province (1927–1936)	Population Censuses, Manufacture Census	1901–1936
Chemicals	District (1901–1921), Province (1927–1936)	Population Censuses, Manufacture Census	1901–1936
Construction	District (1901–1921), Province (1927–1936)	Population Censuses, Manufacture Census	1901–1936
Metalworking	District (1901–1921), Province (1927–1936)	Population Censuses, Manufacture Census	1901–1936
Mining	District (1901–1921), Province (1927–1936)	Population Censuses, Manufacture Census	1901–1936
Textiles	District (1901–1921), Province (1927–1936)	Population Censuses, Manufacture Census	1901–1936
Panel E. Emigration			
US Emigration	Municipality	Ellis Island Data	1892-1924
Overall Emigration	Province	Official Statistics of the Commissioner General	1877-1925
Internal Migrations	Region	Official Statistics of the Commissioner General	1877-1925
Panel F. Other			
WW1 deaths	Municipality	ISTORECO, ANPI	1915-1918
Railways	Municipality	ISTAT	1839-1926
US GDP	National	Maddison (2007)	1890–1924
Italian GDP	National	Baffigi (2013)	1890–1940
Remittances	National	Official Statistics of the Commissioner General	1877–1925
Panel G. GIS Files			
Shapefiles	District, Provinces	ISTAT	1881–1936, excl. 1891

Notes. This table reports all variables used in the paper. Column (2) returns the level of aggregation at which the variable is measured. Column (3) displays the type of source the raw data are extracted from. Further references to original sources can be found in the text's main body. Column (4) reports the years when the raw data is available; "excl." indicates the years when the data do not exist. ISTAT: Italian Statistical Office or previous denominations. ISTORECO: Istituto per la storia della Resistenza e della società contemporanea, part of the Associazione Nazionale Partigiani Italiani (ANPI). Referenced on pages: 9, A2.

Table B.2. Regional US and Overall Emigration, 1876–1925

	Emigrants to US					Emigrants to all destinations					Share
	(1) 76-87	(2) 88-99	(3) 00-12	(4) 13-25	(5) Total	(6) 76-87	(7) 88-99	(8) 00-12	(9) 13-25	(10) Total	(11)
Abruzzi e Molise	26.9	68.0	371.0	161.6	627.4	58.3	164.1	585.7	241.6	1049.7	59.8
Basilicata	28.4	53.3	108.1	38.5	228.3	74.1	106.5	179.8	70.5	431.0	53.0
Calabria	15.0	58.5	457.7	125.1	656.3	74.1	178.5	539.8	253.6	1046.1	62.7
Campania	44.3	157.5	637.8	241.5	1081.2	131.3	339.6	871.0	360.7	1702.6	63.5
Emilia Romagna	1.3	8.4	62.0	24.0	95.8	60.5	137.7	422.4	178.7	799.2	12.0
Lazio	0.02	2.3	109.4	50.1	161.9	0.4	14.0	151.4	72.9	238.6	67.8
Liguria	8.2	10.8	27.2	10.6	56.8	63.0	51.1	89.0	92.9	296.1	19.2
Lombardia	4.4	11.0	56.7	28.6	100.8	237.9	259.7	675.8	441.6	1615.2	6.2
Marche	0.2	2.0	62.0	30.6	94.8	12.7	48.0	280.6	131.1	472.3	20.1
Piemonte	5.2	12.3	109.8	43.4	170.8	353.3	332.5	697.2	527.9	1910.8	8.9
Puglie	1.3	12.9	164.7	107.9	286.9	8.1	37.2	283.4	172.4	501.2	57.2
Sardegna	0.01	0.03	8.5	5.7	14.2	1.3	6.2	72.8	43.9	124.1	11.5
Sicilia	12.6	117.2	687.7	356.1	1173.6	26.8	170.9	946.5	516.4	1660.6	70.7
Toscana	3.3	12.9	89.6	42.0	147.8	110.7	157.5	412.4	230.6	911.2	16.2
Umbria	0.1	0.5	24.1	11.8	36.6	0.5	6.0	129.9	59.4	195.7	18.7
Veneto	1.0	6.0	52.7	48.4	108.1	486.3	1197.6	1298.2	651.0	3633.1	3.0
Total	152.1	533.9	3029.1	1326.0	5041.3	1699.3	3206.9	7635.8	4045.4	16587.4	30.4

Notes. This Table reports regional emigration towards the US and total emigration from 1876 to 1925. Figures are in thousands. Column (11) indicates the percentage of total emigrants towards the US relative to all emigrants from the given region in the whole period 1876-1925. Referenced on page: 6.

Source: our elaboration on data from the *Annuario statistico dell'emigrazione italiana dal 1876 al 1925: con notizie sull'emigrazione negli anni 1869-1875*, Italian Statistical Office (ISTAT), Roma, 1926.

Table B.3. Internal and International Migrations, 1921–1931

	Population	Internal Migrants			International Migrants	
		Within-Region		Across Regions	International	United States
		Other Municipality	Other Province			
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. Absolute Number						
Abruzzi e Molise	1,545,403	15,754	23,059	41,192	942,979	259,623
Basilicata	513,712	10,633	7,480	19,928	320,850	84,490
Calabria	1,723,426	54,701	22,990	32,751	921,610	254,185
Campania	3,508,774	7,752	11,119	54,060	1,480,747	515,384
Emilia Romagna	3,267,285	-14,049	63,015	124,611	718,309	60,147
Lazio	2,348,699	14,691	140,257	49,907	238,036	104,955
Liguria	1,422,596	-87,944	38,139	25,520	213,875	44,023
Lombardia	5,595,915	-103,501	213,408	85,909	1,275,719	84,331
Marche	1,239,863	10,404	16,640	45,518	450,618	61,198
Piemonte	3,523,691	-222,507	113,271	63,526	1,434,611	106,448
Puglia	2,508,305	5,548	51,164	85,060	473,104	151,062
Sardegna	983,760	-6,288	27,445	16,681	121,518	12,541
Sicilia	3,896,509	-110,236	96,293	57,499	1,593,647	634,392
Toscana	2,913,935	-131,462	73,242	67,109	751,345	89,022
Umbria	695,663	-11,849	17,104	11,828	194,572	21,838
Veneto	3,487,109	-24,658	152,542	213,602	1,582,862	66,545
Panel B. Shares of Population (%)						
Abruzzi e Molise		1.019	1.492	2.665	61.018	16.800
Basilicata		2.070	1.456	3.879	62.457	16.447
Calabria		3.174	1.334	1.900	53.475	14.749
Campania		0.221	0.317	1.541	42.201	14.688
Emilia Romagna		-0.430	1.929	3.814	21.985	1.841
Lazio		0.625	5.972	2.125	10.135	4.469
Liguria		-6.182	2.681	1.794	15.034	3.095
Lombardia		-1.850	3.814	1.535	22.797	1.507
Marche		0.839	1.342	3.671	36.344	4.936
Piemonte		-6.315	3.215	1.803	40.713	3.021
Puglia		0.221	2.040	3.391	18.862	6.022
Sardegna		-0.639	2.790	1.696	12.352	1.275
Sicilia		-2.829	2.471	1.476	40.899	16.281
Toscana		-4.511	2.514	2.303	25.785	3.055
Umbria		-1.703	2.459	1.700	27.969	3.139
Veneto		-0.707	4.374	6.125	45.392	1.908

Notes. This Table reports internal migration and out-migration flows over the period 1921-1931. Column (1) reports the population in 1881. Column (2) reports the population from each region living in a municipality other than their origin municipality; column (3) replicates column (2) but reports the population living in a different province. Column (4) reports the population originating from each region that is living in a region other than their region of origin. Since the source reports the *stock* of the overall population in each category, we compute the *flow*, i.e., migration flows, as the difference between 1931, the first post-treatment period, and 1921. Positive (negative) figures thus imply a net population loss (gain) in each category between 1921 and 1931 due to internal migrations. Column (5) reports the number of international emigrants. Column (6) reports the number of US migrants. Panel A reports absolute values; Panel B reports the shares relative to the region population in 1881, in percentage terms. Referenced on page: 26.

Source. Our elaboration on data from the *Annuario statistico dell'emigrazione italiana dal 1876 al 1925: con notizie sull'emigrazione negli anni 1869-1875*, Italian Statistical Office (ISTAT), Roma, 1926, and from *Censimento della Popolazione Italiana*, Italian Statistical Office (ISTAT), Roma, 1921 and 1931.

Table B.4. Coverage of Ellis Island Data: Pooled Statistics

	Coverage of Origin String	
	Share (%) (1)	Total (2)
Missing Origin	12.160	510,070
Origin Place Not in Italy	31.832	1,335,246
Origin Placed Matched to Municipality	55.770	2,339,409
Failed Geocoding	0.238	10,003
Universe of Name-Based Italians	100.000	4,194,728

Notes. This Table reports the coverage of the Ellis Island string of origin. The sample is defined as all immigrants defined as "Italian" according to the algorithm described in the main text. "Missing Origin" refers to records where the origin municipality is missing; "Origin Place No in Italy" refers to records whose string of origin is geocoded to a location outside of Italy; "Origin Placed Matched to Municipality" refers to records whose string of origin is geocoded to a municipality of origin in Italy; "Failed Geocoding" refers to records whose string of origin cannot be georeferenced. Column (1) reports the percentage share of entries relative to the universe of records; column (2) reports the total number. Referenced on pages: 10, A5, A8.

Table B.5. Correlation Between Ellis Island and Official Statistics US Emigration

	Official Statistics Emigrants					ln(1+Official Statistics Emigrants)				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Ellis Island Migrants	0.917***	0.917***	0.547***	0.547***	0.536***					
	(0.108)	(0.112)	(0.063)	(0.066)	(0.087)					
ln(1+Ellis Island Migrants)						0.874***	0.922***	0.773***	0.574***	0.601***
						(0.041)	(0.062)	(0.039)	(0.045)	(0.128)
Region FE	No	No	Yes	Yes	Yes	No	No	Yes	Yes	Yes
Year FE	No	Yes	No	Yes	Yes	No	Yes	No	Yes	Yes
Sample Years	All	All	All	All	1900–1914	All	All	All	All	1900–1914
N. of Regions	16	16	16	16	16	16	16	16	16	16
N. of Observations	527	527	527	527	240	527	527	527	527	240
R^2	0.841	0.841	0.971	0.971	0.983	0.854	0.914	0.897	0.959	0.971

Notes. This Table compares the number of US emigrants recorded in Italian official statistics with data from the Ellis Island Foundation dataset. The unit of observation is a region at a yearly frequency. The sample period spans 1892 to 1925. In columns (1–5), the dependent and independent variables are the number of emigrants recorded in official statistics and at Ellis Island, respectively. In columns (6–10), both variables are taken as log(1+). Columns (1) and (6) display the unconditional correlation; in columns (2) and (7), (3) and (8), and (4) and (9), we include year, region, and year and region fixed effects. In columns (5) and (10), we restrict the observation sample to the years 1900–1914, which is the period with a lower share of missing municipalities. Standard errors are always clustered at the region level and are displayed in parentheses. Referenced on pages: 11, A9.

***: $p < 0.01$, **: $p < 0.05$, *: $p < 0.10$

Table B.6. Over-Representation of Transatlantic Emigration Ports in the Ellis Island Dataset

	Dep. Var.: US Emigration Rate (‰)			
	(1)	(2)	(3)	(4)
Municipality with Transatlantic Port	0.323	-1.909*		
	(0.939)	(1.017)		
Province with Transatlantic Port			2.219	-0.012
			(1.525)	(1.470)
Region FE	No	Yes	No	Yes
Year FE	No	Yes	No	Yes
N. of Municipalities	7,492	7,492	7,492	7,492
N. of Observations	224,850	224,850	224,850	224,850
R^2	0.000	0.238	0.010	0.236
Mean Dep. Var.	2.511	2.511	2.511	2.511

Notes. This Table reports the correlation between the US emigration rate and the presence of Transatlantic Ports. The unit of observation is a municipality observed at a yearly frequency between 1892 and 1921. The dependent variable is the number of emigrants to the United States as a share of the population, in per-thousand units. The explanatory variable is an indicator equal to one if the municipality (columns 1–2) or the province (columns 3–4) has an emigration port (Genoa, Naples, or Palermo), and zero otherwise. In columns (1) and (3), we report the unconditional correlation; columns (2) and (4) include region and year fixed effects. Units are clustered by their population in 1881. Standard errors are clustered at the municipality (columns 1–2) or province (columns 3–4) level and are reported in parentheses. Referenced on pages: 10, A9.

Table B.7. Over-Representation of Capital and Large Cities in the Ellis Island Dataset

	Dep. Var.: US Emigration Rate (‰)			
	(1)	(2)	(3)	(4)
Region Capital City	0.217			
	(0.788)			
Province Capital City		0.605*		
		(0.312)		
Population in 1881			-0.154	-0.149
			(0.213)	(0.170)
Region FE	Yes	–	Yes	–
Province FE	No	Yes	No	Yes
Year FE	Yes	Yes	Yes	Yes
N. of Municipalities	7,820	7,820	7,820	7,820
N. of Observations	225,180	225,180	225,180	225,180
R^2	0.177	0.198	0.177	0.198
Mean Dep. Var.	2.260	2.260	2.260	2.260

Notes. This Table reports the correlation between the US emigration rate, region (column 1), and province (column 2), city status, and population (columns 3–4). The unit of observation is a municipality observed at a yearly frequency between 1892 and 1921. The dependent variable is the number of emigrants to the United States as a share of the population, in per-thousand units. The explanatory variable is an indicator equal to one if the municipality is a region capital city (column 1), a province capital city (column 2), and population (in 10,000 units) in 1881 (columns 3–4). To compare each municipality with their surrounding areas, in column (1) (resp. 2), we include region (resp. province) fixed effects. Column (3) and (4) respectively include region and province fixed effects. All regressions include year fixed effects. Standard errors are clustered at the municipality level and are reported in parentheses. Referenced on pages: 11, A9.

Table B.8. Most Common Emigration Districts

	Most Common Origin Districts					
	Number of Emigrants			Emigration Rate		
	(1) District	(2) Emigrants	(3) Emigration Rate (‰)	(4) District	(5) Emigrants	(6) Emigration Rate (‰)
1	Palermo	128,066	20.161	Termini Imerese	37,483	37.899
2	Salerno	56,694	17.840	Sciacca	24,837	37.029
3	Caserta	56,119	16.551	Primiero	4,089	34.390
4	Bari delle Puglie	56,075	12.548	Sulmona	30,771	29.740
5	Cosenza	53,088	22.801	Cefalù	30,342	27.841
6	Napoli	47,848	4.816	Mistretta	17,466	27.132
7	Avellino	47,622	24.092	Campobasso	33,233	26.361
8	Frosinone	46,642	20.784	Sant'Angelo de' Lombardi	34,024	25.465
9	Girgenti	46,455	16.356	Avellino	47,622	24.092
10	Messina	45,525	15.433	Melfi	24,800	23.610
11	Termini Imerese	37,483	37.899	Piedimonte d'Alife	11,639	23.570
12	Lucca	35,765	9.635	Mazara del Vallo	27,086	23.526
13	Sant'Angelo de' Lombardi	34,024	25.465	Cosenza	53,088	22.801
14	Campobasso	33,233	26.362	Nicastro	29,011	22.699
15	Roma	33,183	3.388	Corleone	13,383	22.541
16	Catanzaro	32,842	18.562	Sala Consilina	17,933	22.537
17	Sulmona	30,771	29.740	Nola	24,847	21.997
18	Potenza	30,730	19.642	Bovino	12,069	21.747
19	Cefalù	30,342	27.841	Isernia	30,062	21.616
20	Siracusa	30,174	16.688	Cerreto Sannita	18,880	21.511
21	Gaeta	30,070	16.608	Bivona	17,413	21.376
22	Isernia	30,062	21.616	Campagna	21,994	20.914
23	Gerace Marina	29,535	19.198	Frosinone	46,642	20.784
24	Reggio di Calabria	29,290	15.723	Palermo	128,066	20.161
25	Caltanissetta	29,011	17.600	Benevento	25,529	20.010
26	Nicastro	29,011	22.699	Potenza	30,730	19.642
27	Chieti	28,761	17.171	Gerace Marina	29,535	19.198
28	Mazara del Vallo	27,086	23.526	Patti	24,419	18.904
29	Sora	26,636	14.047	Catanzaro	32,842	18.562
30	Teramo	25699	12.7864	Matera	20510	18.0056

Notes. This Table reports the districts with the largest number of US emigrants (columns 1–3) and US emigration rates relative to the 1921 population (columns 4–6). We list the top 30 origin districts in each category. Emigration rates are expressed in per-thousand units. Referenced on page: A10.

Table B.9. Imputation of Emigrants to Districts

	Apportioning Variable:					
	US Emigration Rate			Agricultural Employment Rate		
	(1) Population	(2) Manufacturing Employment	(3) Agricultural Employment	(4) Population	(5) Manufacturing Employment	(6) Agricultural Employment
Panel A. OLS Estimates						
US Emigrants × Post	0.073***	0.084**	-0.053	0.057***	0.082***	-0.039
	(0.024)	(0.034)	(0.043)	(0.019)	(0.024)	(0.027)
Std. Beta Coef.	0.453	0.361	-0.382	0.353	0.352	-0.283
Panel B. 2SLS Estimates						
US Emigrants × Post	0.052*	0.122**	-0.095	0.038**	0.109***	-0.067*
	(0.027)	(0.055)	(0.069)	(0.019)	(0.033)	(0.039)
K-P F-stat	43.156	43.156	43.796	77.162	77.162	78.116
Std. Beta Coef.	0.322	0.526	-0.692	0.234	0.468	-0.484
District FE	Yes	Yes	Yes	Yes	Yes	Yes
Decade FE	Yes	Yes	Yes	Yes	Yes	Yes
N. of Districts	203	203	203	203	203	203
N. of Observations	1,202	1,202	1,000	1,202	1,202	1,000
Mean Dep. Var.	12.250	9.978	10.733	12.250	9.978	10.733

Notes. This Table reports the estimated effect of district-level exposure to the US Quota Acts on the district-level outcomes, when province-level emigration is imputed to districts. The unit of observation is a district observed at a census-decade frequency between 1881 and 1936. The dependent variable is log-population (columns 1 and 4), log-manufacturing employment (columns 2 and 5), and log-agricultural employment (columns 3 and 6). The treatment is an interaction term between US emigration and a post-1921 term. Panel A displays the OLS estimates; Panel B displays the two-stage least squares estimates obtained using the instrument defined in (2). Each regression includes district and time fixed effects and controls for the emigration rate and an indicator variable for Southern regions; both interacted with a post-1921 indicator. To calculate the district-level number of emigrants, let $E_{p(d)}$ denote the province-level number of emigrants for district d. Let x_d be the *apportioning variable*. We compute E_d in proportion to the apportioning variable: $E_d \equiv (x_d / \sum_{d' \in p(d)} x_{d'}) \times E_{p(d)}$. In columns (1–3), the apportioning variable is the US emigration rate, defined as the number of US emigrants divided by population in 1881; in columns (4–6), the apportioning variable is the agricultural employment rate in 1881. The K-P F-statistic refers to the Kleibergen-Paap statistic for weak instruments. Units are weighed by their population in 1881. Standard errors clustered at the district level are reported in parentheses. Referenced on pages: 16, A3.
***: $p < 0.01$, **: $p < 0.05$, *: $p < 0.10$

Table B.10. First-Stage Regressions

	Dependent Variable: Measured US Emigrants							
	Weighed				Unweighed			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Predicted US Emigrants	0.471***	0.299***	0.310***	0.375***	0.459***	0.262***	0.301***	0.350***
	(0.048)	(0.048)	(0.043)	(0.051)	(0.047)	(0.045)	(0.034)	(0.042)
Controls	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Region FE	No	No	Yes	–	No	No	Yes	–
Province FE	No	No	No	Yes	No	No	No	Yes
N. of Districts	187	187	186	169	191	191	191	176
R^2	0.459	0.646	0.777	0.861	0.458	0.678	0.795	0.861
Mean Dep. Var.	9.063	9.063	9.064	9.161	9.063	9.063	9.063	9.165
Std. Beta Coef.	0.678	0.431	0.447	0.552	0.661	0.378	0.434	0.516

Notes. This Table reports the correlation between the number of US emigrants and the shift-share instrument constructed from equation (2). The units of observation are districts. Each district is observed once, and the outcome and the dependent variables are aggregated over time. The outcome variable is measured exposure to the US Quota Acts. In columns (1–4), districts are weighed by 1881 population; in columns (5–8), districts are not weighed. Columns (1) and (5) report the unconditional correlation; in columns (2) and (6), we include the number of emigrants and an indicator for southern regions as controls; in columns (3) and (7), we include region fixed effects; columns (4) and (8) control for province fixed effects. Standard errors clustered at the district level are reported in parentheses. Referenced on page: 17.
***: $p < 0.01$, **: $p < 0.05$, *: $p < 0.10$

Table B.11. Recentered Instrumental Variable Estimates: District-Level Estimates

	Population	Employment in:	
	(1)	(2) Manufacture	(3) Agriculture
US Emigrants × Post (Recentered IV)	0.053**	0.089**	-0.029
	(0.023)	(0.041)	(0.043)
Emigrants × Post	-0.069***	-0.023	-0.090**
	(0.022)	(0.046)	(0.035)
District FE	Yes	Yes	Yes
Decade FE	Yes	Yes	Yes
N. of Districts	196	196	196
N. of Observations	1,160	1,160	965
R^2	0.117	0.192	0.030
K-P F-stat	89.620	89.620	91.905
Mean Dep. Var.	12.263	9.997	10.741
Std. Beta Coef.	0.331	0.382	-0.209

Notes. This Table reports the estimated effect of district-level exposure to the US Quota Acts on the district-level variables. The unit of observation is a district observed at a census-decade frequency between 1881 and 1936. The dependent variable is log-population (column 1), log-manufacturing employment (column 2), and log-agricultural employment (column 3). The Table displays the two-stage least squares estimates obtained using the instrumental variable (2), net of the expected value of the instrument, following the "recentering" approach of Borusyak and Hull (2023). Each regression includes district and time fixed effects and controls for the emigration rate and an indicator variable for Southern regions; both interacted with a post-1921 indicator. The last row reports the coefficient of the interaction between the number of emigrants and a post-1921 indicator. The K-P F-statistic refers to the Kleibergen-Paap statistic for weak instruments. Units are weighed by their population in 1881. Standard errors clustered at the district level are reported in parentheses. Referenced on page: 18.
***: $p < 0.01$, **: $p < 0.05$, *: $p < 0.10$

Table B.12. Recentered Instrumental Variable Estimates: Province-Level Estimates

	Dependent Variable:					
	(1) Firms	(2) Firms with Engine	(3) Mechanical Engines	(4) Electrical Engines	(5) Mechanical Power	(6) Electric Power
US Emigrants × Post (Recentered IV)	-0.283*	-0.377**	-0.135	-0.732*	-0.478***	-0.628**
	(0.147)	(0.173)	(0.101)	(0.429)	(0.171)	(0.306)
Province FE	Yes	Yes	Yes	Yes	Yes	Yes
Decade FE	Yes	Yes	Yes	Yes	Yes	Yes
N. of Provinces	67	67	67	67	67	67
N. of Observations	200	200	200	200	200	200
R^2	0.121	-0.016	0.070	-0.001	0.011	-0.031
K-P *F*-stat	32.229	32.229	32.229	32.229	32.229	32.229
Mean Dep. Var.	8.935	7.079	6.279	7.476	9.458	9.575
Std. Beta Coef.	-1.825	-1.983	-0.950	-1.989	-2.455	-1.772

Notes. This Table reports the estimated effect of province-level exposure to the US Quota Acts on the province-level variables. The unit of observation is a province observed at a census-decade frequency between 1911 and 1936. The dependent variables are the number of firms (column 1), the number of firms with at least one engine (column 2), the number of mechanical engines (column 3), the number of electrical engines (column 4), the horsepower generated by mechanical (column 5) and electrical (column 6) engines. All dependent variables are expressed in log. The Table displays the two-stage least squares estimates obtained using the instrumental variable (2), net of the expected value of the instrument, following the "recentering" approach of Borusyak and Hull (2023). Each regression includes province and time fixed effects and controls for the emigration rate and an indicator variable for Southern regions; both interacted with a post-1921 indicator. The K-P *F*-statistic refers to the Kleibergen-Paap statistic for weak instruments. Units are weighed by their population in 1881. Standard errors clustered at the province level are reported in parentheses. Referenced on page: 18.
***: $p < 0.01$, **: $p < 0.05$, *: $p < 0.10$

Table B.13. Exposure to the US Quota Acts and Population

	OLS Estimates					2SLS Estimates				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
US Emigrants × Post (Increasing weight)	0.062***					0.046**				
	(0.018)					(0.018)				
US Emigrants × Post (Decreasing weight)		0.062***					0.055**			
		(0.018)					(0.022)			
US Emigrants × Post (Between 1892 and 1924)			0.064***					0.049**		
			(0.019)					(0.020)		
US Emigrants × Post (Between 1892 and 1920)				0.064***					0.049**	
				(0.019)					(0.020)	
US Emigrants × Post (Between 1892 and 1914)					0.064***					0.048**
					(0.019)					(0.020)
District FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Decade FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N. of Districts	203	203	203	203	203	203	203	203	203	203
N. of Observations	1,202	1,202	1,202	1,202	1,202	1,202	1,202	1,202	1,202	1,202
R^2	0.113	0.121	0.118	0.119	0.116	0.106	0.120	0.113	0.113	0.110
K-P *F*-stat						128.930	61.415	93.260	95.878	104.304
Mean Dep. Var.	12.250	12.250	12.250	12.250	12.250	12.250	12.250	12.250	12.250	12.250
Std. Beta Coef.	0.326	0.343	0.400	0.399	0.391	0.240	0.303	0.305	0.303	0.293

Notes. This Table reports the estimated effect of district-level exposure to the US Quota Acts on population. The unit of observation is a district observed at a census-decade frequency between 1881 and 1936. The dependent variable is log-population. Each column represents the estimates of a regression where the treatment is an interaction between a post-Quota (1921) indicator variable and the district-level number of US emigrants. We use five different approaches to compute US emigration, constructed using different weighting of the number of emigrants to the US or computing the number of US emigrants over different periods. In columns (1) and (2), we assign a higher weight to emigrants who departed closer to the Quota Acts date and much earlier: for both measures, we adopt an exponential weighting of factor 0.9. In columns (3–5), we construct the Quota Exposure using the number of emigrants to the US who departed between 1892 and 1924, between 1892 and 1920, and between 1892 and 1914. Each regression controls for the number of emigrants and an interaction term between an indicator variable for Southern regions and a post-1921 indicator. Columns (1–5) display the OLS estimates; columns (6–10) display the two-stage least squares estimates obtained using the instrument defined in (2) and adapted to the different time weighting schemes. The K-P *F*-statistic refers to the Kleibergen-Paap statistic for weak instruments. Units are weighed by their population in 1881. Standard errors clustered at the district level are reported in parentheses. Referenced on page: 23.
***: $p < 0.01$, **: $p < 0.05$, *: $p < 0.10$

Table B.14. Exposure to the US Quota Acts and Capital Investment

	OLS Estimates						2SLS Estimates					
	(1) Firms	(2) Firms with Engines	(3) Mechanical Engines	(4) Electrical Engines	(5) Mechanical Horsepower	(6) Electrical Horsepower	(7) Firms	(8) Firms with Engines	(9) Mechanical Engines	(10) Electrical Engines	(11) Mechanical Horsepower	(12) Electrical Horsepower
Panel A. Increasing weight												
US Emigrants × Post	-0.190***	-0.124	-0.166***	-0.196	-0.234***	-0.167	-0.162	-0.263*	-0.134	-0.609*	-0.392**	-0.501**
	(0.061)	(0.107)	(0.061)	(0.210)	(0.071)	(0.161)	(0.133)	(0.133)	(0.102)	(0.323)	(0.150)	(0.232)
R^2	0.101	0.049	0.069	0.132	0.051	0.086	0.099	0.026	0.068	0.057	0.035	0.035
K-P *F*-stat							38.524	38.524	38.524	38.524	38.524	38.524
Panel B. Decreasing weight												
US Emigrants × Post	-0.188***	-0.070	-0.127**	-0.123	-0.162**	-0.073	-0.241	-0.388**	-0.181	-0.932*	-0.529**	-0.725**
	(0.055)	(0.101)	(0.059)	(0.194)	(0.063)	(0.149)	(0.183)	(0.193)	(0.159)	(0.488)	(0.260)	(0.363)
R^2	0.106	0.037	0.058	0.123	0.034	0.076	0.100	-0.092	0.055	-0.193	-0.062	-0.141
K-P *F*-stat							11.837	11.837	11.837	11.837	11.837	11.837
Panel C. US emigrants between 1892 and 1924												
US Emigrants × Post	-0.190***	-0.091	-0.149**	-0.137	-0.196***	-0.098	-0.206	-0.334**	-0.159	-0.784*	-0.468**	-0.623**
	(0.058)	(0.107)	(0.059)	(0.202)	(0.065)	(0.155)	(0.161)	(0.165)	(0.132)	(0.412)	(0.204)	(0.302)
R^2	0.102	0.041	0.064	0.124	0.041	0.078	0.102	-0.029	0.063	-0.064	-0.008	-0.052
K-P *F*-stat							21.199	21.199	21.199	21.199	21.199	21.199
Panel D. US emigrants between 1892 and 1920												
US Emigrants × Post	-0.190***	-0.092	-0.151**	-0.137	-0.199***	-0.100	-0.200	-0.328**	-0.159	-0.757*	-0.470**	-0.612**
	(0.058)	(0.107)	(0.058)	(0.202)	(0.066)	(0.155)	(0.157)	(0.161)	(0.129)	(0.401)	(0.200)	(0.294)
R^2	0.102	0.041	0.064	0.124	0.041	0.078	0.102	-0.025	0.064	-0.048	-0.007	-0.045
K-P *F*-stat							22.275	22.275	22.275	22.275	22.275	22.275
Panel E. US emigrants between 1892 and 1914												
US Emigrants × Post	-0.183***	-0.094	-0.156***	-0.125	-0.206***	-0.093	-0.190	-0.313**	-0.157	-0.711*	-0.461**	-0.585**
	(0.057)	(0.107)	(0.058)	(0.199)	(0.067)	(0.153)	(0.152)	(0.154)	(0.122)	(0.385)	(0.186)	(0.280)
R^2	0.097	0.041	0.066	0.122	0.043	0.078	0.097	-0.015	0.066	-0.030	0.000	-0.036
K-P *F*-stat							26.297	26.297	26.297	26.297	26.297	26.297
Province FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Decade FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N. of Provinces	69	69	69	69	69	69	69	69	69	69	69	69
N. of Observations	206	206	206	206	206	206	206	206	206	206	206	206
Mean Dep. Var.	8.921	7.060	6.267	7.444	9.435	9.542	8.921	7.060	6.267	7.444	9.435	9.542

Notes. This Table reports the estimated effect of exposure to the US Quota Acts on capital investment. The unit of observation is a province observed at a census-decade frequency between 1911 and 1936. The dependent variables are the number of firms (column 1), the number of firms with at least one engine (column 2), the number of mechanical engines (column 3), the number of electrical engines (column 4), the horsepower generated by mechanical (column 5) and electrical (column 6) engines. All variables are expressed in logs. The treatment is an interaction between a post-Quota (1921) indicator variable and US emigration. In the five panels, we construct US emigration using different approaches. In Panel A and B we respectively assign a higher weight to emigrants departed closer to the Quota Acts date and much earlier in time: for both measures we adopt an exponential weighting of factor 0.9. In Panel C, D, and E, we respectively construct the Quota Exposure using the number of emigrants to the US who departed, respectively, between 1892 and 1924, between 1892 and 1920, and between 1892 and 1914. Each regression controls for the number of emigrants and an interaction term between an indicator variable for Southern regions and a post-1921 indicator. Columns (1–6) display the OLS estimates; columns (7–12) display the two-stage least squares estimates obtained using the instrument defined in (2) and adapted to the different time weighting schemes. The K-P *F*-statistic refers to the Kleibergen-Paap statistic for weak instruments. Units are weighed by their population in 1881. Standard errors clustered at the province level are reported in parentheses. Referenced on page: 23.
***: $p < 0.01$, **: $p < 0.05$, *: $p < 0.10$

Table B.15. Exposure to the US Quota Acts and Manufacture Employment

	OLS Estimates					2SLS Estimates				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
US Emigrants × Post (Increasing weight)	0.052					0.077**				
	(0.035)					(0.035)				
US Emigrants × Post (Decreasing weight)		0.043					0.098**			
		(0.033)					(0.043)			
US Emigrants × Post (Between 1892 and 1924)			0.046					0.091**		
			(0.034)					(0.039)		
US Emigrants × Post (Between 1892 and 1920)				0.047					0.090**	
				(0.034)					(0.039)	
US Emigrants × Post (Between 1892 and 1914)					0.046					0.088**
					(0.034)					(0.038)
District FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Decade FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N. of Districts	203	203	203	203	203	203	203	203	203	203
N. of Observations	1,202	1,202	1,202	1,202	1,202	1,202	1,202	1,202	1,202	1,202
R^2	0.193	0.192	0.192	0.192	0.192	0.192	0.185	0.188	0.188	0.188
K-P *F*-stat						128.930	61.415	93.260	95.878	104.304
Mean Dep. Var.	9.978	9.978	9.978	9.978	9.978	9.978	9.978	9.978	9.978	9.978
Std. Beta Coef.	0.188	0.164	0.201	0.201	0.194	0.280	0.372	0.393	0.389	0.377

Notes. This Table reports the estimated effect of exposure to the US Quota Acts on manufacturing employment. The unit of observation is a district observed at a census-decade frequency between 1881 and 1936. The dependent variable is log-manufacturing employment. Each column represents the estimates of a regression where the treatment is an interaction between a post-Quota (1921) indicator variable and the number of US emigrants. We use five different approaches to compute the number of US emigrants. In columns (1–2), we assign a higher weight to emigrants who departed closer to the Quota Acts date and much earlier: for both measures, we adopt an exponential weighting of factor 0.9. In columns (3–5), we construct the Quota Exposure using the number of emigrants to the US who departed between 1892 and 1924, between 1892 and 1920, and between 1892 and 1914. Each regression controls for the number of emigrants and an interaction term between an indicator variable for Southern regions and a post-1921 indicator. Columns (1–5) display the OLS estimates; columns (6–10) display the two-stage least squares estimates obtained using the instrument defined in (2) and adapted to the different time weighting schemes. The K-P *F*-statistic refers to the Kleibergen-Paap statistic for weak instruments. Units are weighed by their population in 1881. Standard errors clustered at the district level are reported in parentheses. Referenced on page: 23.
***: $p < 0.01$, **: $p < 0.05$, *: $p < 0.10$

Table B.16. Exposure to the Quota Acts and Agriculture Employment

	OLS Estimates					2SLS Estimates				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
US Emigrants × Post (Increasing weight)	-0.040					-0.029				
	(0.043)					(0.036)				
US Emigrants × Post (Decreasing weight)		-0.044					-0.035			
		(0.042)					(0.044)			
US Emigrants × Post (Between 1892 and 1924)			-0.042					-0.032		
			(0.044)					(0.040)		
US Emigrants × Post (Between 1892 and 1920)				-0.041					-0.031	
				(0.043)					(0.040)	
US Emigrants × Post (Between 1892 and 1914)					-0.038					-0.030
					(0.044)					(0.039)
District FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Decade FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N. of Districts	203	203	203	203	203	203	203	203	203	203
N. of Observations	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000
R^2	0.037	0.039	0.038	0.037	0.037	0.037	0.039	0.037	0.037	0.036
K-P *F*-stat						131.644	61.984	94.752	97.506	106.081
Mean Dep. Var.	10.733	10.733	10.733	10.733	10.733	10.733	10.733	10.733	10.733	10.733
Std. Beta Coef.	-0.244	-0.281	-0.304	-0.298	-0.277	-0.176	-0.226	-0.230	-0.223	-0.220

Notes. This Table reports the estimated effect of exposure to the US Quota Acts on agricultural employment. The unit of observation is a district observed at a census-decade frequency between 1881 and 1936. The dependent variable is log-agriculture employment. Each column represents the estimates of a regression where the treatment is an interaction between a post-Quota (1921) indicator variable and the number of US emigrants. We use five approaches to compute the number of US emigrants. In columns (1–2), we assign a higher weight to emigrants who departed closer to the Quota Acts date and much earlier: for both measures, we adopt an exponential weighting of factor 0.9. In columns (3–5), we construct the Quota Exposure using the number of emigrants to the US who departed between 1892 and 1924, between 1892 and 1920, and between 1892 and 1914. Each regression controls for the number of emigrants and an interaction term between an indicator variable for Southern regions and a post-1921 indicator. Columns (1–5) display the OLS estimates; columns (6–10) display the two-stage least squares estimates obtained using the instrument defined in (2) and adapted to the different time weighting schemes. The K-P *F*-statistic refers to the Kleibergen-Paap statistic for weak instruments. Units are weighed by their population in 1881. Standard errors clustered at the district level are reported in parentheses. Referenced on page: 23.
***: $p < 0.01$, **: $p < 0.05$, *: $p < 0.10$

Table B.17. Exposure to the US Quota Acts and Population

	OLS Estimates						2SLS Estimates					
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
US Emigrants × Post	0.067***	0.053***	0.074***	0.068***	0.076***	0.069***	0.045**	0.047**	0.047**	0.051**	0.064***	0.053**
	(0.018)	(0.020)	(0.021)	(0.021)	(0.020)	(0.021)	(0.018)	(0.020)	(0.021)	(0.022)	(0.021)	(0.022)
Literacy × Post	0.130						0.138					
	(0.084)						(0.091)					
Urbanization Rate × Post		0.135***						0.144***				
		(0.040)						(0.045)				
Altitude × Post			-0.049***						-0.047***			
			(0.009)						(0.009)			
Railway × Post				0.040						0.048**		
				(0.026)						(0.023)		
WW1 Death Rate × Post					5.039**						4.562	
					(2.521)						(2.839)	
US GDP Growth × Emigrants × Post						-0.001						-0.001
						(0.000)						(0.000)
District FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Decade FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N. of Districts	203	203	203	203	203	203	203	203	203	203	203	203
N. of Observations	1,202	1,202	1,202	1,202	1,202	1,202	1,202	1,202	1,202	1,202	1,202	1,202
R^2	0.139	0.148	0.177	0.122	0.136	0.121	0.130	0.147	0.163	0.117	0.133	0.115
K-P *F*-stat							88.766	104.439	89.558	85.630	85.149	85.623
Mean Dep. Var.	11.818	11.818	11.818	11.818	11.818	11.818	11.818	11.818	11.818	11.818	11.818	11.818
Std. Beta Coef.	0.449	0.360	0.500	0.459	0.510	0.467	0.301	0.315	0.318	0.345	0.428	0.354

Notes. This Table reports the estimated effect of district-level exposure to the US Quota Acts on population. The dependent variable is log-population. Each column includes the interaction between the post-quota indicator variable and a district- or country-specific variable as one further control. In column (1), we use the district's literacy rate in 1901; in column (2), the urbanization rate in 1901, measured as the share of people living in towns with more than 5,000 inhabitants in the district; in column (3), the altitude at which the district is located; in column (4) an indicator variable specifying whether the district was connected to the railway network before 1901; in column (5) district's number of deaths caused by WW1. In column (6), we use the US GDP growth interacted with the volume of international emigrants. Each regression controls for the number of emigrants and an interaction term between an indicator variable for Southern regions and a post-1921 indicator. Columns (1–6) display the OLS estimates; columns (7–12) display the two-stage least squares estimates obtained using the instrument defined in (2). The K-P *F*-statistic refers to the Kleibergen-Paap statistic for weak instruments. Units are weighed by their population in 1881. Standard errors clustered at the district level are reported in parentheses. Referenced on page: 23.
***: $p < 0.01$, **: $p < 0.05$, *: $p < 0.10$

Table B.18. Exposure to the US Quota Acts and Capital Investment

	OLS Estimates						2SLS Estimates					
	(1) Firms	(2) Firms with Engines	(3) Mechanical Engines	(4) Electrical Engines	(5) Mechanical Horsepower	(6) Electrical Horsepower	(7) Firms	(8) Firms with Engines	(9) Mechanical Engines	(10) Electrical Engines	(11) Mechanical Horsepower	(12) Electrical Horsepower
Panel A. Control: Literacy Rate × Post												
US Emigrants × Post	-0.212***	-0.122	-0.167***	-0.181	-0.225***	-0.130	-0.117	-0.296**	-0.118	-0.538**	-0.421***	-0.457**
	(0.038)	(0.099)	(0.060)	(0.163)	(0.065)	(0.130)	(0.115)	(0.128)	(0.092)	(0.259)	(0.149)	(0.196)
R^2	0.231	0.085	0.071	0.295	0.061	0.177	0.214	0.046	0.069	0.232	0.035	0.126
K-P *F*-stat							18.178	18.178	18.178	18.178	18.178	18.178
Panel B. Control: Urbanization Rate × Post												
US Emigrants × Post	-0.187***	-0.163	-0.205***	-0.181	-0.251***	-0.174	-0.224	-0.317*	-0.111	-0.723*	-0.431**	-0.556*
	(0.060)	(0.122)	(0.050)	(0.216)	(0.068)	(0.169)	(0.144)	(0.165)	(0.108)	(0.400)	(0.163)	(0.289)
R^2	0.129	0.066	0.090	0.132	0.053	0.092	0.127	0.041	0.081	0.012	0.034	0.029
K-P *F*-stat							28.856	28.856	28.856	28.856	28.856	28.856
Panel C. Control: Altitude × Post												
US Emigrants × Post	-0.236***	-0.107	-0.171**	-0.256	-0.219**	-0.128	-0.233	-0.356*	-0.132	-0.890**	-0.476**	-0.672**
	(0.070)	(0.132)	(0.071)	(0.254)	(0.100)	(0.199)	(0.180)	(0.194)	(0.137)	(0.407)	(0.205)	(0.309)
R^2	0.125	0.057	0.071	0.144	0.051	0.082	0.125	-0.006	0.069	-0.010	0.014	-0.037
K-P *F*-stat							16.155	16.155	16.155	16.155	16.155	16.155
Panel D. Control: Railway Access × Post												
US Emigrants × Post	-0.214***	-0.141	-0.167***	-0.200	-0.224***	-0.162	-0.217	-0.377**	-0.128	-0.780*	-0.443**	-0.687**
	(0.059)	(0.121)	(0.059)	(0.218)	(0.071)	(0.171)	(0.164)	(0.172)	(0.105)	(0.414)	(0.170)	(0.298)
R^2	0.119	0.059	0.072	0.135	0.054	0.115	0.119	-0.005	0.070	-0.010	0.024	-0.011
K-P *F*-stat							18.433	18.433	18.433	18.433	18.433	18.433
Panel E. Control: WW1 Death Rate × Post												
US Emigrants × Post	-0.217***	-0.120	-0.176***	-0.171	-0.217***	-0.115	-0.223	-0.438*	-0.131	-0.994	-0.541**	-0.817*
	(0.071)	(0.134)	(0.062)	(0.256)	(0.081)	(0.191)	(0.240)	(0.243)	(0.144)	(0.597)	(0.250)	(0.428)
R^2	0.116	0.052	0.071	0.132	0.051	0.083	0.116	-0.044	0.069	-0.110	-0.005	-0.103
K-P *F*-stat							9.497	9.497	9.497	9.497	9.497	9.497
Panel F. Control: US GDP Growth × Emigrants × Post												
US Emigrants × Post	-0.209***	-0.134	-0.170***	-0.191	-0.233***	-0.135	-0.206	-0.352**	-0.136	-0.745*	-0.457***	-0.611**
	(0.059)	(0.116)	(0.057)	(0.214)	(0.072)	(0.163)	(0.162)	(0.169)	(0.107)	(0.408)	(0.172)	(0.292)
R^2	0.115	0.056	0.076	0.132	0.051	0.084	0.115	0.001	0.074	-0.002	0.018	-0.020
K-P *F*-stat							18.336	18.336	18.336	18.336	18.336	18.336
Province FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Decade FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N. of Provinces	68	68	68	68	68	68	68	68	68	68	68	68
N. of Observations	204	204	204	204	204	204	204	204	204	204	204	204
Mean Dep. Var.	8.667	6.800	6.124	7.055	9.166	9.143	8.667	6.800	6.124	7.055	9.166	9.143

Notes. This Table reports the effect of the Quotas on capital investment. All variables are expressed in logs. In the five panels, we add a different control, given by the interaction between the post-quota indicator variable and a province-specific variable. We use, respectively, the literacy rate in 1901, the urbanization rate in 1901 (share of people in towns with more than 5'000 inhabitants), the altitude, the share of municipalities connected to the railway network before 1901, the number of deaths caused by WW1; the US GDP growth interacted with the province's number of emigrants. Each regression controls for the number of emigrants and an interaction term between an indicator variable for Southern regions and a post-1921 indicator. Columns (1–6) display the OLS estimates; columns (7–12) display the two-stage least squares estimates obtained using the instrument defined in (2). The K-P *F*-statistic refers to the Kleibergen-Paap statistic for weak instruments. Units are weighted by their population in 1880. Standard errors clustered at the province level are reported in parentheses. Referenced on page: 23.
***: $p < 0.01$, **: $p < 0.05$, *: $p < 0.10$

Table B.19. Exposure to the US Quota Acts and Manufacture Employment

	OLS Estimates						2SLS Estimates					
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
US Emigrants × Post	0.057***	0.087***	0.069**	0.066**	0.067***	0.069**	0.068**	0.108***	0.100**	0.098**	0.101**	0.099**
	(0.022)	(0.025)	(0.027)	(0.027)	(0.025)	(0.026)	(0.035)	(0.040)	(0.039)	(0.040)	(0.042)	(0.040)
Literacy × Post	0.553***						0.549***					
	(0.114)						(0.112)					
Urbanization Rate × Post		-0.155						-0.183*				
		(0.113)						(0.103)				
Altitude × Post			-0.001						-0.003			
			(0.022)						(0.022)			
Railway × Post				0.093						0.078		
				(0.057)						(0.059)		
WW1 Death Rate × Post					-0.889						0.439	
					(4.717)						(5.035)	
US GDP Growth × Emigrants × Post						-0.006						-0.006
						(0.007)						(0.007)
District FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Decade FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N. of Districts	203	203	203	203	203	203	203	203	203	203	203	203
N. of Observations	1,202	1,202	1,202	1,202	1,202	1,202	1,202	1,202	1,202	1,202	1,202	1,202
R^2	0.226	0.199	0.196	0.197	0.196	0.200	0.226	0.198	0.194	0.195	0.194	0.198
K-P F-stat							88.766	104.439	89.558	85.630	85.149	85.623
Mean Dep. Var.	9.443	9.443	9.443	9.443	9.443	9.443	9.443	9.443	9.443	9.443	9.443	9.443
Std. Beta Coef.	0.267	0.405	0.320	0.307	0.315	0.320	0.319	0.503	0.466	0.457	0.473	0.463

Notes. This Table reports the estimated effect of district-level exposure to the US Quota Acts on manufacturing employment. The dependent variable is log-manufacturing employment. In each column, we control for the interaction between the post-quota indicator variable and a district- or country-specific variable. In column (1), we use the district's literacy rate in 1901; in column (2), the urbanization rate in 1901, measured as the share of people living in towns with more than 5'000 inhabitants in the district; in column (3), the altitude at which the district is located; in column (4) an indicator variable specifying whether the district was connected to the railway network before 1901; in column (5) district's number of deaths caused by WW1. In column (6), we use the US GDP growth interacted with the volume of international emigrants. Each regression controls for the number of emigrants and an interaction term between an indicator variable for Southern regions and a post-1921 indicator. Columns (1–6) display the OLS estimates; columns (7–12) display the two-stage least squares estimates obtained using the instrument defined in (2). Units are weighed by their population in 1881. Standard errors clustered at the district level are reported in parentheses. Referenced on page: 23.
***: $p < 0.01$, **: $p < 0.05$, *: $p < 0.10$

Table B.20. Exposure to the Quota Acts and Agriculture Employment

	OLS Estimates						2SLS Estimates					
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
US Emigrants × Post	-0.002	0.002	-0.008	-0.008	-0.009	-0.010	-0.003	-0.028	-0.033	-0.029	-0.030	-0.033
	(0.026)	(0.027)	(0.027)	(0.027)	(0.028)	(0.027)	(0.040)	(0.040)	(0.040)	(0.041)	(0.046)	(0.042)
Literacy × Post	-0.460***						-0.459***					
	(0.112)						(0.111)					
Urbanization Rate × Post		-0.104						-0.065				
		(0.131)						(0.105)				
Altitude × Post			-0.025						-0.023			
			(0.017)						(0.018)			
Railway × Post				-0.061						-0.051		
				(0.037)						(0.040)		
WW1 Death Rate × Post					0.980						0.141	
					(3.521)						(4.191)	
US GDP Growth × Emigrants × Post						-0.010*						-0.010*
						(0.006)						(0.006)
District FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Decade FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N. of Districts	203	203	203	203	203	203	203	203	203	203	203	203
N. of Observations	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000
R^2	0.073	0.034	0.033	0.032	0.031	0.048	0.072	0.031	0.031	0.030	0.030	0.046
K-P *F*-stat							90.943	107.885	90.062	86.973	86.194	87.263
Mean Dep. Var.	10.435	10.435	10.435	10.435	10.435	10.435	10.435	10.435	10.435	10.435	10.435	10.435
Std. Beta Coef.	-0.012	0.014	-0.054	-0.057	-0.061	-0.066	-0.024	-0.194	-0.228	-0.200	-0.209	-0.230

Notes. This Table reports the estimated effect of district-level exposure to the US Quota Acts on agricultural employment. The dependent variable is log-agricultural employment. In each column, we control for the interaction between the post-quota indicator variable and a district- or country-specific variable. In column (1), we use the district's literacy rate in 1901; in column (2), the urbanization rate in 1901, measured as the share of people living in towns with more than 5'000 inhabitants in the district; in column (3), the altitude at which the district is located; in column (4) an indicator variable specifying whether the district was connected to the railway network before 1901; in column (5) district's number of deaths caused by WW1. In column (6), we use the US GDP growth interacted with the volume of international emigrants. Each regression controls for the number of emigrants and an interaction term between an indicator variable for Southern regions and a post-1921 indicator. Columns (1–6) display the OLS estimates; columns (7–12) display the two-stage least squares estimates obtained using the instrument defined in (2). The K-P *F*-statistic refers to the Kleibergen-Paap statistic for weak instruments. Units are weighed by their population in 1881. Standard errors clustered at the district level are reported in parentheses. Referenced on page: 23.
***: $p < 0.01$, **: $p < 0.05$, *: $p < 0.10$

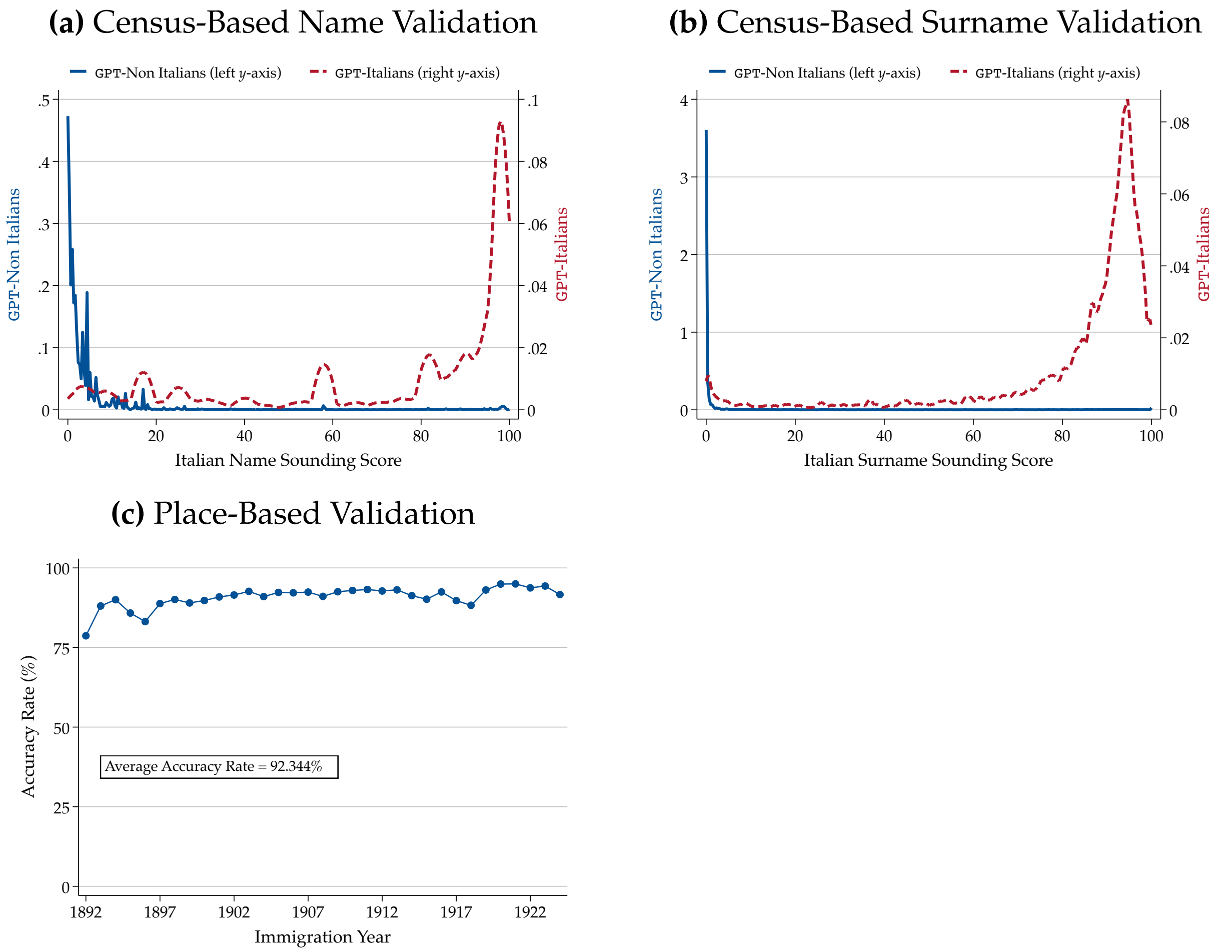
Table B.21. Internal Migration Dynamics in the Aftermath of the Quotas

	Share (%) of Inhabitants Living in Other:					
	(1) Municipality	(2) Province	(3) Region	(4) Municipality	(5) Province	(6) Region
Post Quotas	-2.722***	-1.591***	1.611***			
	(0.494)	(0.450)	(0.269)			
US Emigrants × Post				1.271	-1.283	-0.661
				(0.726)	(1.005)	(0.532)
Emigrants × Post				-1.254*	0.949	0.675
				(0.627)	(0.826)	(0.461)
Region FE	Yes	Yes	Yes	Yes	Yes	Yes
Decade FE	–	–	–	Yes	Yes	Yes
N. of Regions	16	16	16	16	16	16
N. of Observations	48	64	64	48	64	64
R^2	0.940	0.451	0.761	0.944	0.457	0.767
Mean Dep. Var.	17.642	5.451	5.048	17.642	5.451	5.048

Notes. This Table displays the dynamics of internal migrations after the 1921 Quota Acts. The unit of observation is a region, observed at a decade frequency between 1901 and 1931. The dependent variable is the share of the population of each region that lives: in a municipality other than their municipality of origin (columns 1 and 4), in a province other than their province of origin (columns 2 and 5), or in another region (columns 3 and 6). The share of the population living in municipalities other than their origin municipality is not available in 1901. In columns (1–3), we perform a simple pre-post Quota comparison; hence, the main explanatory variable is a dummy variable equal to one in 1921 and 1931 and zero otherwise. The regressions include decade fixed effects. In columns (4–6), we estimate difference-in-differences regressions where the main treatment is an interaction term between the number of US emigrants from each region and a post-Quota indicator. We further control for an interaction term between the number of overall emigrants and a post-Quota indicator, an interaction between an indicator variable for Southern regions and a post-Quota indicator, region fixed effects, and decade fixed effects. Standard errors, displayed in parentheses, are clustered at the region level. Referenced on page: 26.
***: $p < 0.01$, **: $p < 0.05$, *: $p < 0.10$

C ADDITIONAL FIGURES

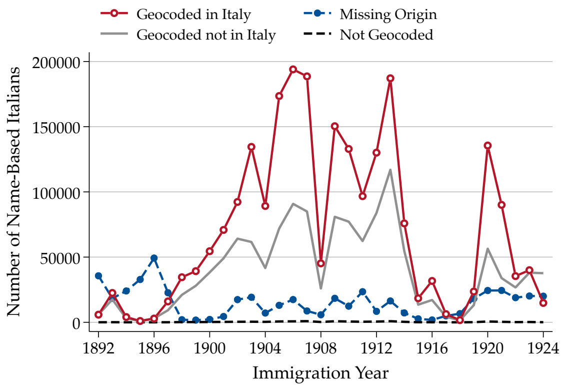
Figure C.1. Validation of the Nationality-Assignment Algorithm



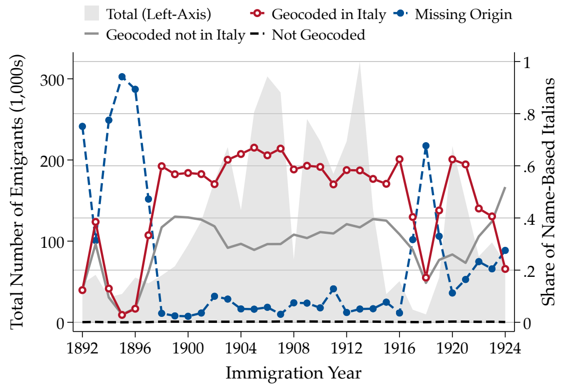
Notes. This Figure displays two alternative validation approaches to the algorithm to impute the nationality of the records extracted from the Ellis Island data. In Panels C.1a and C.1b, we display the distribution of the Italian-sounding name and surname index for the subsample of GPT-Italians (dashed red, right y-axis) and all other Ellis Island records that GPT does not identify as Italians (solid blue, left y-axis). In Panel C.1c, we focus on the subsample geocoded within Italy, and report the percentage share of Italians identified by the GPT algorithm by their immigration year, as well as the average accuracy rate. Referenced on pages: A5, A6.

Figure C.2. Coverage of the Ellis Island Dataset

(a) Number of US Immigrants



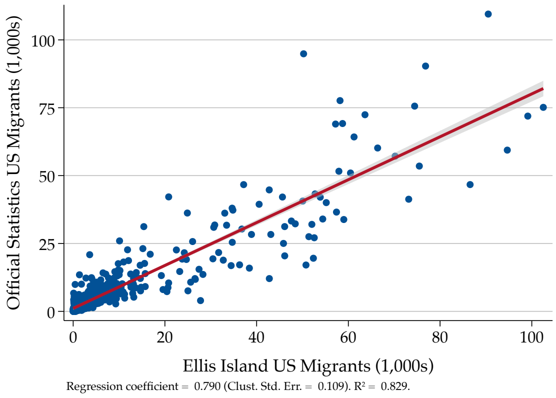
(b) Share of US Immigrants



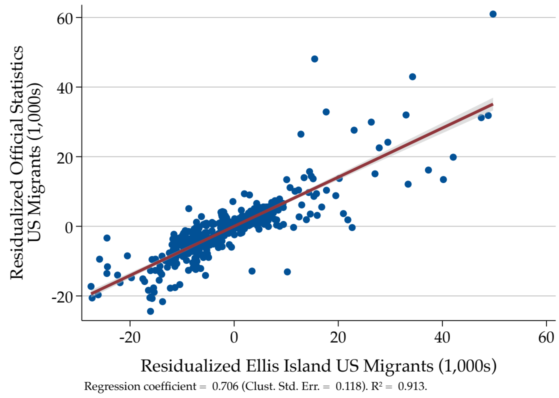
Notes. This Figure reports data on the coverage of the Ellis Island dataset. The sample comprises all records that we identify as Italian based on the name-based algorithm described in the main text. The Panels display migrants geocoded to an Italian municipality (red), the number of migrants geocoded to places outside of Italy (gray), migrants with no reported place of origin (blue), and records that we could not geocode (black dashed). Panel C.2a reports the total number of migrants; Panel C.2b reports the share relative to the total number of records in each year. In Panel C.2a, the gray area in the background reports the overall sample size (on the left *y*-axis). Referenced on pages: 10, A8.

Figure C.3. Correlation between Official Statistics and Ellis Island US Emigration

(a) Unconditional Correlation

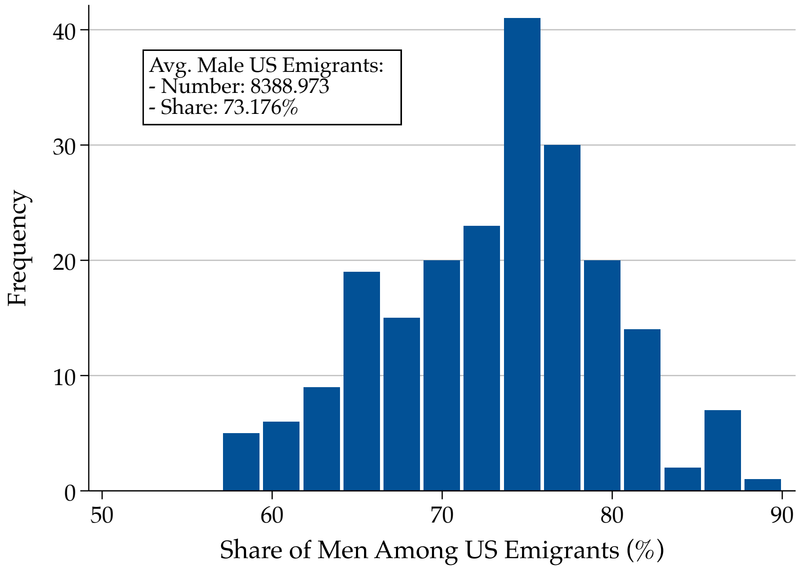


(b) Conditional Correlation



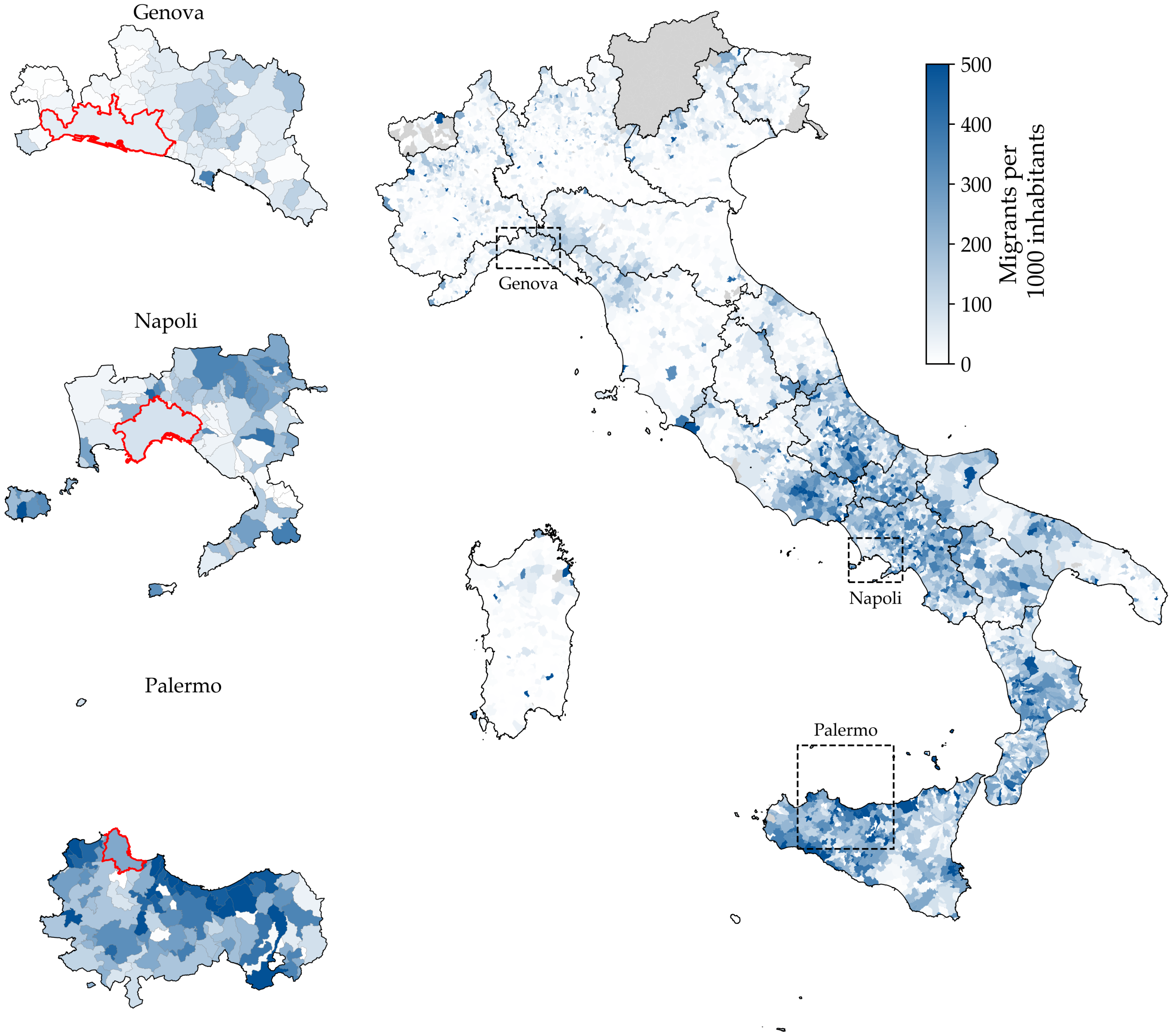
Notes. This Figure reports the correlation between the number of US emigrants recorded in the Italian official statistics and the number of Italian immigrants recorded in the Ellis Island dataset. The unit of observation is a region at a yearly frequency between 1892 and 1925. In Panel C.3a, we report the unconditional correlation. Panel C.3b displays the correlation between the variables after residualizing region and year-fixed effects. Both graphs report the fitted values of a linear regression between the variables. Referenced on pages: 11, A9.

Figure C.4. Distribution of the Share of Men Among US Emigrants



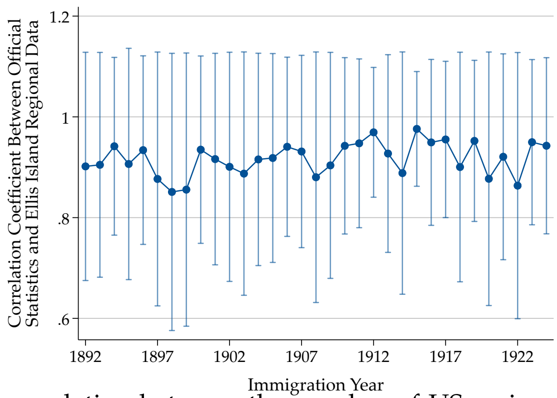
Notes. This Figure reports the district-level distribution of the number of men among US emigrants, relative to the total number of US emigrants, in the Ellis Island dataset. To impute the gender of US migrants, we adopt the algorithm explained in the main text: we identify an individual as a man (resp. woman) if, in the US census, at least 70% of those with the same name are men (resp. women), and discard individuals with ambiguous names. The Figure further reports the average number of male US emigrants (approximately 8,400), and the average share (approximately 73%). Referenced on pages: 24, 25, A7.

Figure C.5. Municipality-Level US Emigration



Notes. This Figure reports the US emigration rate at the municipality level, defined as the total number of US emigrants between 1892 and 1924 relative to the population in 1881. Lighter to darker blue indicates a higher share of US emigrants per 1,000 inhabitants. Municipality borders are contemporary borders because historical municipality GIS data is not available. The black lines superimpose region borders. The map highlights the provinces with a transatlantic emigration port: Genoa, Naples, and Palermo. Within each province, the municipality with the transatlantic port is highlighted in red. Referenced on pages: 10, A9.

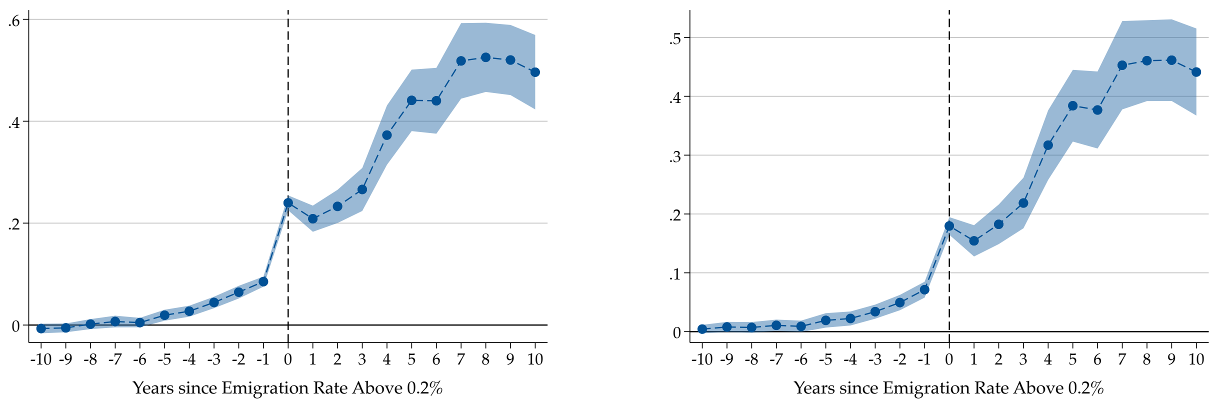
Figure C.6. Year-by-Year Correlation Between Ellis Island and Official Statistics US Emigrants



Notes. This Figure reports the correlation between the number of US emigrants recorded in the Italian official statistics and the number of Italian immigrants recorded in the Ellis Island dataset. Each dot reports the correlation between the two variables in one specific year between 1892 and 1925. In each dot, the unit of observation is a region. Both variables are standardized to have zero mean and unitary standard deviation in each year for readability. The bars report 95% confidence intervals from standard errors clustered at the regional level. Referenced on pages: 11, A10.

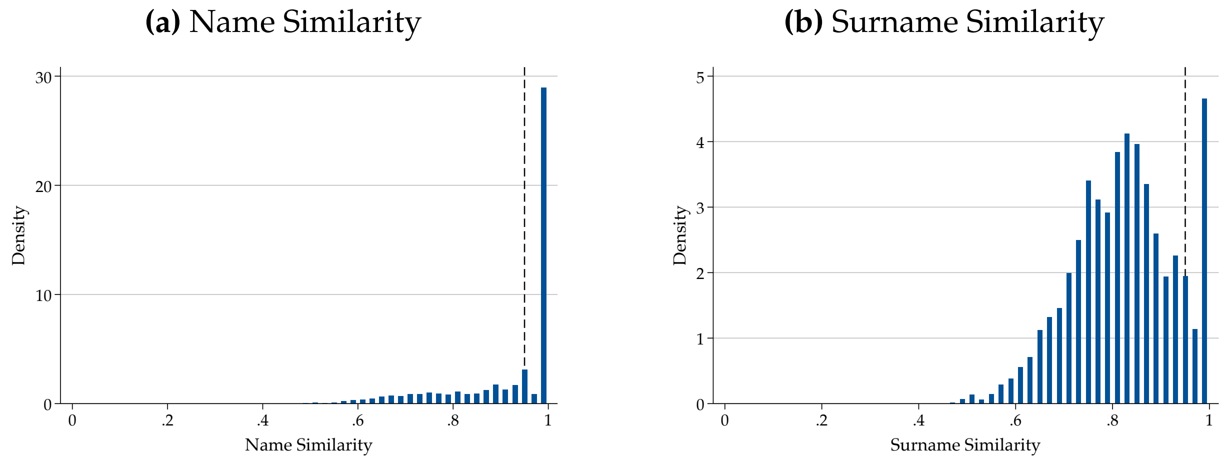
Figure C.7. Testing the S-Shaped Hypothesis

(a) Unconditional Emigration | **(b)** Emigration Net of Fixed Effects



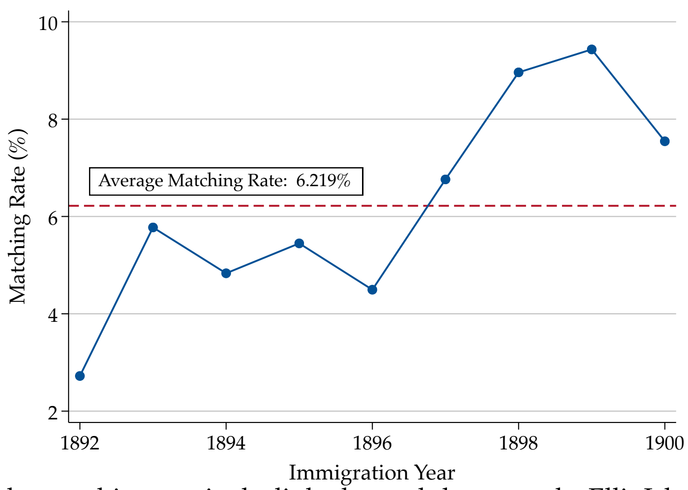
Notes. These figure test the S-shape emigration hypothesis of Gould (1980). The dependent variable is US emigration, observed at the district level every year between 1892 and 1924. For each district, we define a set of period dummies around the period when the ratio between US emigrants and population in 1921 exceeds 0.1%. We report the estimated difference-in-differences coefficients obtained using the method of Borusyak et al. (2024) around this threshold. Panel C.7a does not include any fixed effect in the regression; in Panel C.7b, we include district and year fixed effects. Standard errors are clustered at the district level; bands report 99% confidence intervals. Referenced on pages: 19, A10.

Figure C.8. Name and Surname Similarity in the Linked Ellis Island–US Census Sample



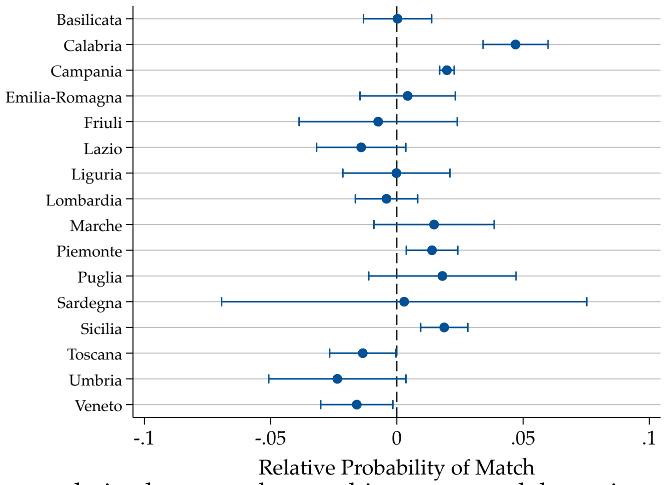
Notes. These figures report the name (panel C.8a) and surname (panel C.8b) similarities between the records of Ellis Island and those that appear in the US census. To compare name and surname matches, we adopt the Monge-Elkan method with the embedded Jaro-Winkler word measure. The resulting values are normalized between zero and one, with values closer to one indicating closer comparisons. For each name and surname recorded in the Ellis Island data, we select the individual in the US census with the highest name and surname similarity among those whose initial NYSIIS-adjusted letter is the same as the Ellis Island record to be matched. The dashed lines mark the quality thresholds below which we reject the matches as uncertain. Referenced on page: A11.

Figure C.9. Matching Rate in the Linked Ellis Island–US Census Sample



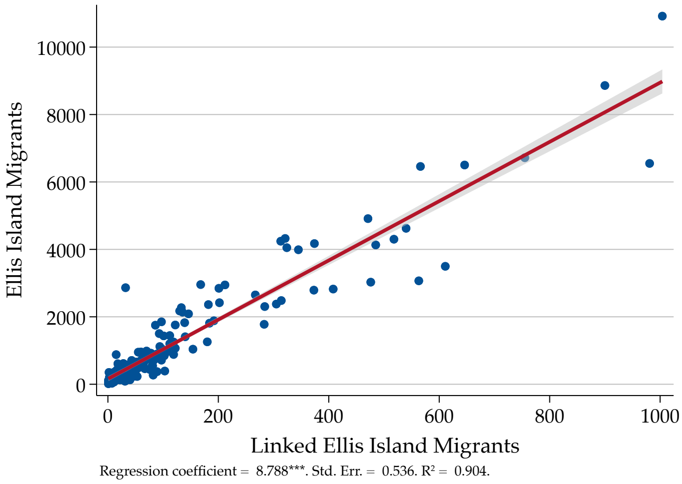
Notes. This Figure reports the matching rate in the linked sample between the Ellis Island data and the records of Italians living in the US as recorded in the population census. The blue dots report the share of individuals in the Ellis Island dataset with an accepted match in the US population census. The red line plots the average matching rate. Matching rates are expressed in percentage terms and by immigration year. Referenced on page: A11.

Figure C.10. Probability of Match: Breakdown by Origin of Ellis Island Immigrants



Notes. This Figure report the correlation between the matching status and the region of origin in the Ellis Island–US Census linked sample. The sample comprises all individuals who appear in the Ellis Island dataset who immigrated between 1892 and 1900. The dependent variable is an indicator equal to one if the individual has at least one accepted match in the linked sample and zero otherwise. The right-hand side consists of a series of indicator variables that tag regions of origin. Standard errors are clustered at the year level, and the bands report the associated 95% confidence intervals. Referenced on pages: 11, A12.

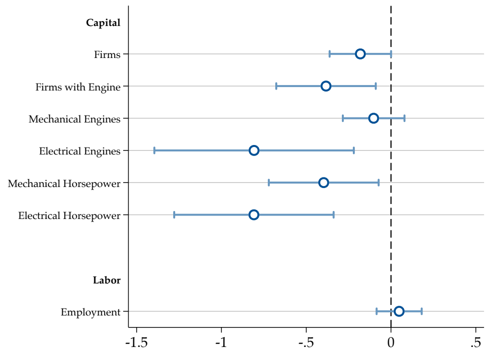
Figure C.11. Correlation Between Linked Sample and Full Sample of Ellis Island Dataset



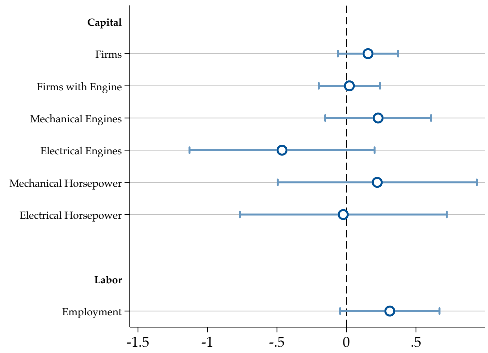
Notes. This Figure reports the district-level correlation between the number of migrants to the United States in the Ellis Island dataset, displayed on the y-axis, and in the linked Ellis Island-census dataset, displayed on the x-axis. The unit of observation is a district. The y-axis reports the number of migrants between 1892 and 1900, which is the same sample covered by the linked dataset. The graph displays the fitted values from a linear regression between the two, the regression coefficient, standard error clustered at the district level, and the R-squared of the regression. Referenced on pages: 11, A12.

Figure C.12. Effect of the Quota Acts on Capital and Employment by Sector

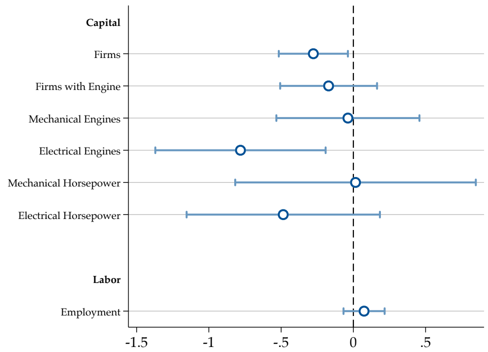
(a) Agriculture



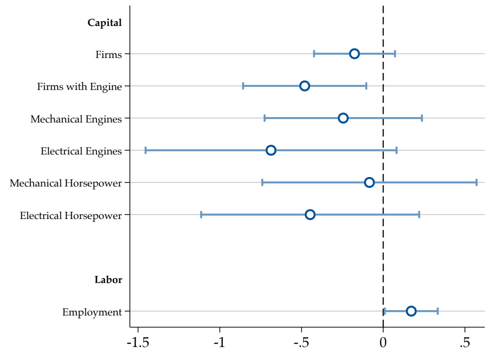
(b) Chemicals



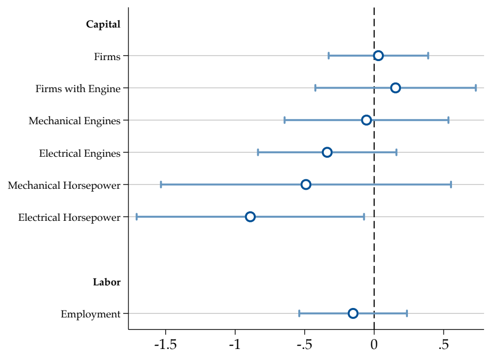
(c) Construction



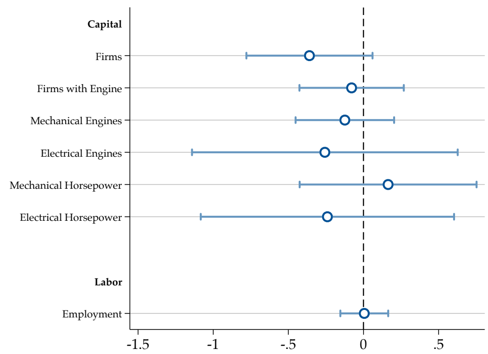
(d) Metalworking



(e) Mining

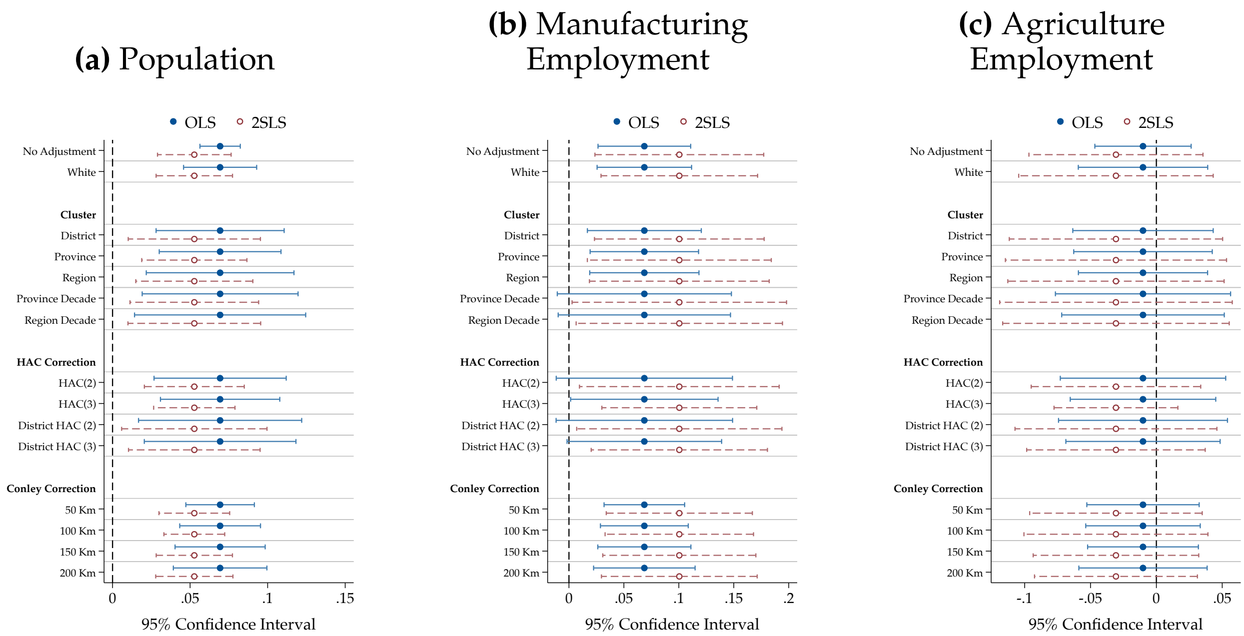


(f) Textiles



Notes. These figures report the estimated effect of exposure to the US Quota Acts on capital and employment by sector over time. The unit of observation is a province observed at a census-decade frequency between 1901 and 1936. Each panel reports the results for a specific manufacturing sector. The outcome variables are listed by row. Each dot reports the coefficient of an interaction between a post-1921 indicator and the instrument of US emigration. The estimates are obtained via two-stage least squares using the instrument defined in (2). Each regression includes province and decade-fixed effects and controls for the volume of emigration and an indicator variable for Southern regions, both interacted with a post-1921 indicator. Units are weighed by their population in 1881. Bands report 95% confidence intervals from standard errors clustered at the province level. Referenced on pages: 21, 22.

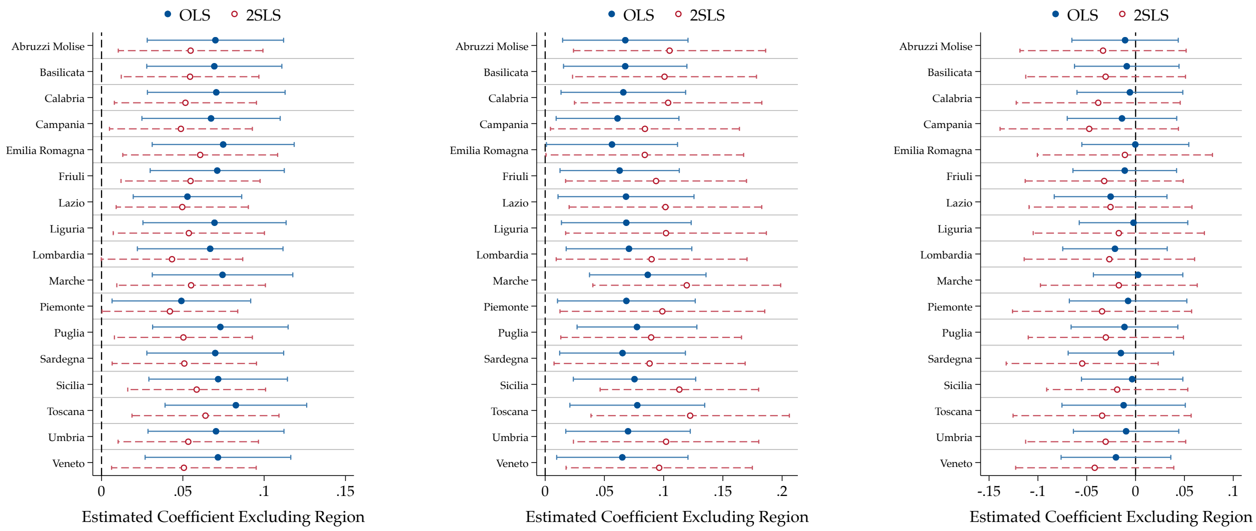
Figure C.13. Alternative Standard Errors: District-Level Regressions



Notes. This Figure reports alternative estimators of the standard errors in the baseline district-level regressions. The standard error estimators are: White (heteroskedastticity-consistent), cluster at different levels of aggregation, HAC, and Conley (1999) at different bandwidths between 50 and 200 km. The unit of observation is a district, at a decade frequency between 1881 and 1936. The dependent variables are population (Panel C.13a), manufacturing employment (Panel C.13b), and agriculture employment (Panel C.13c). All outcomes are expressed in logs. Regressions are estimated through OLS and include district and decade fixed effects. The treatment is an interaction term between US emigration and a post-1921 term. Each regression controls for the number of emigrants and an interaction term between an indicator variable for Southern regions and a post-1921 indicator. The blue dots report the OLS estimates; the red dots report the two-stage least-squares obtained using the instrument defined in (2). Units are weighed by their population in 1881. Bands report 95% confidence intervals. Referenced on page: 23.

Figure C.14. Excluding one region at the time: District-Level Regressions

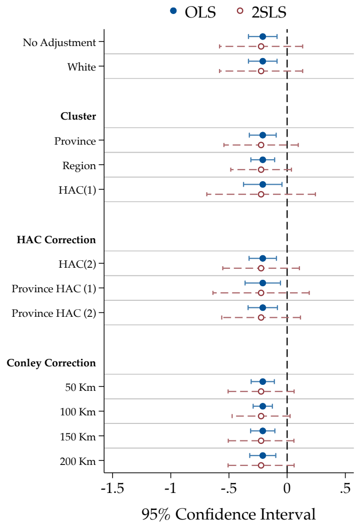
(a) Population **(b)** Manufacturing Employment **(c)** Agriculture Employment



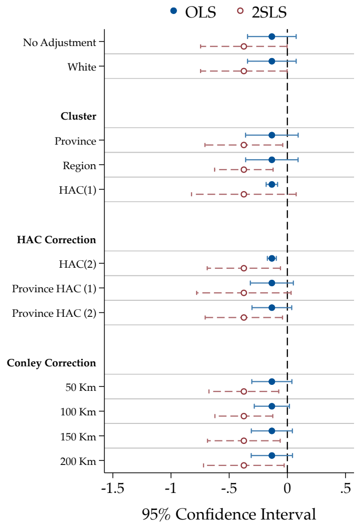
Notes. This Figure reports the estimates of the baseline district-level regressions when excluding one region. The unit of observation is a district, at a decade frequency between 1881 and 1936. The dependent variable is population (Panel C.14a), manufacturing employment (Panel C.14b), and agriculture employment (Panel C.14c). All outcomes are expressed in logs. Regressions include district and decade-fixed effects. The treatment is an interaction term between US emigration and a post-1921 term. Each regression controls for the number of emigrants and an interaction term between an indicator variable for Southern regions and a post-1921 indicator. The left axis indicates the excluded region. The blue dots report the OLS estimates; the red dots report the two-stage least-squares obtained using the instrument defined in (2). Units are weighed by their population in 1881. Bands report 95% confidence intervals. Standard errors clustered at the district level. Referenced on page: 23.

Figure C.15. Alternative Standard Errors: Province-Level Regressions

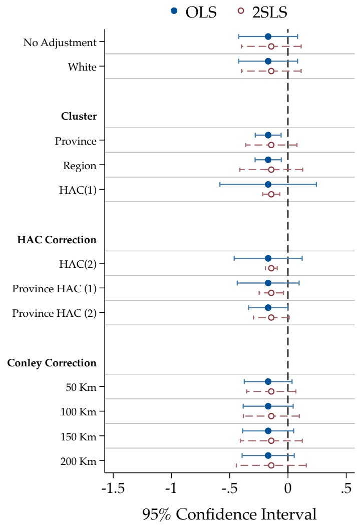
(a) Firms



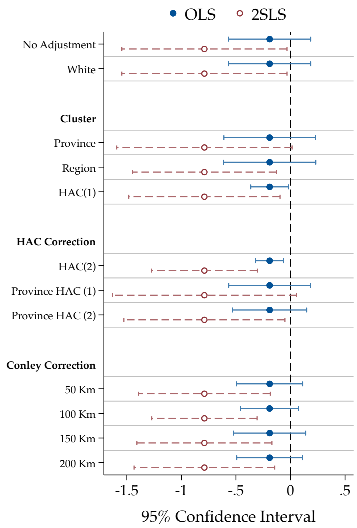
(b) Firms with Engine



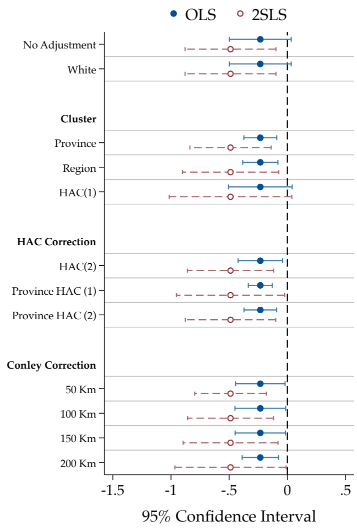
(c) Mechanical Engines



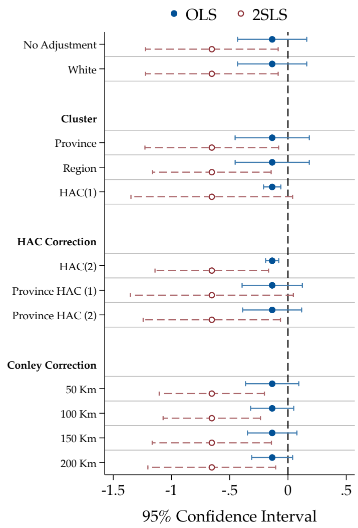
(d) Electrical Engines



(e) Mechanical Horsepower



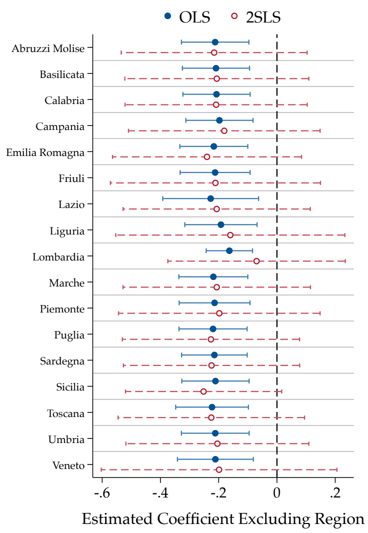
(f) Electrical Horsepower



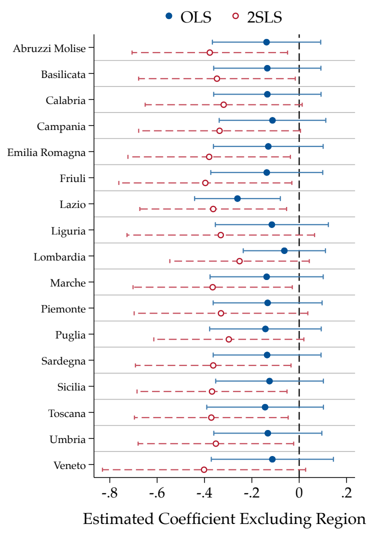
Notes. This Figure reports alternative estimators of the standard errors in the baseline province-level regressions. The standard error estimators are: White (heteroskedastticity-consistent), cluster at different levels of aggregation, HAC, and Conley (1999) at different bandwidths between 50 and 200 km. The unit of observation is a province, at a decade frequency between 1911 and 1937. All outcomes are expressed in logs. Regressions are estimated through OLS on the logged outcomes and include province and decade fixed effects. The treatment is an interaction term between US emigration and a post-1921 term. Each regression controls for the number of emigrants and an interaction term between an indicator variable for Southern regions and a post-1921 indicator. The blue dots report the OLS estimates; the red dots report the two-stage least-squares obtained using the instrument defined in (2). Units are weighed by their population in 1881. Bands report 95% confidence intervals. Referenced on page: 23.

Figure C.16. Excluding one region at the time: Province-Level Regressions

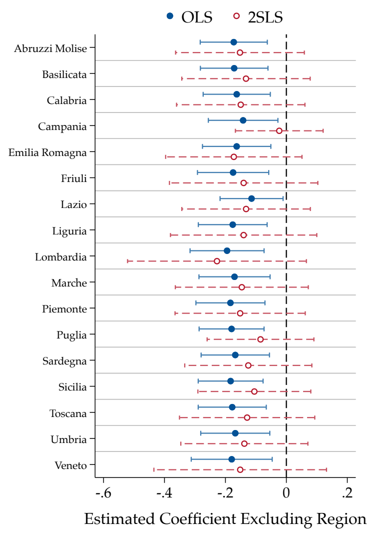
(a) Firms



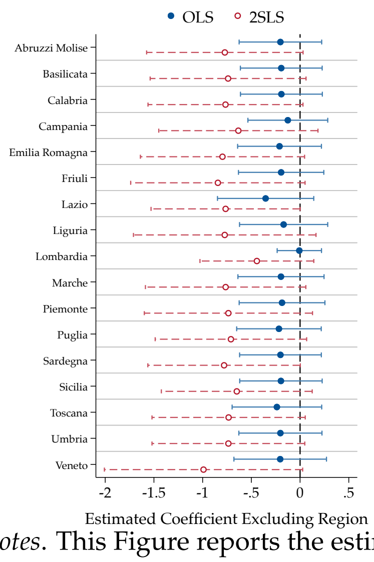
(b) Firms with Engine



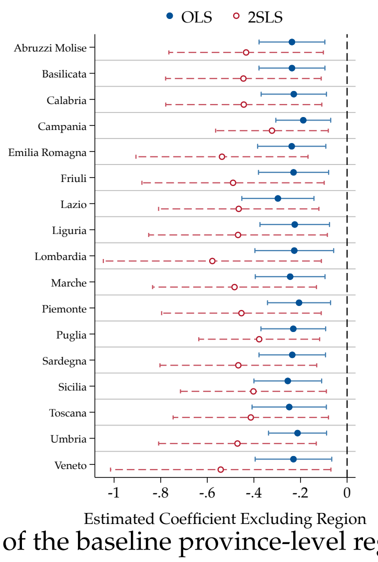
(c) Mechanical Engines



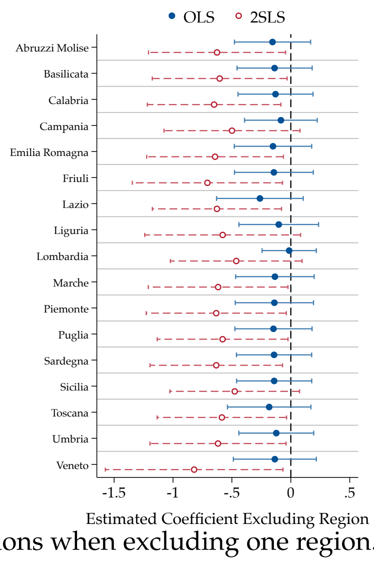
(d) Electrical Engines



(e) Mechanical Horsepower



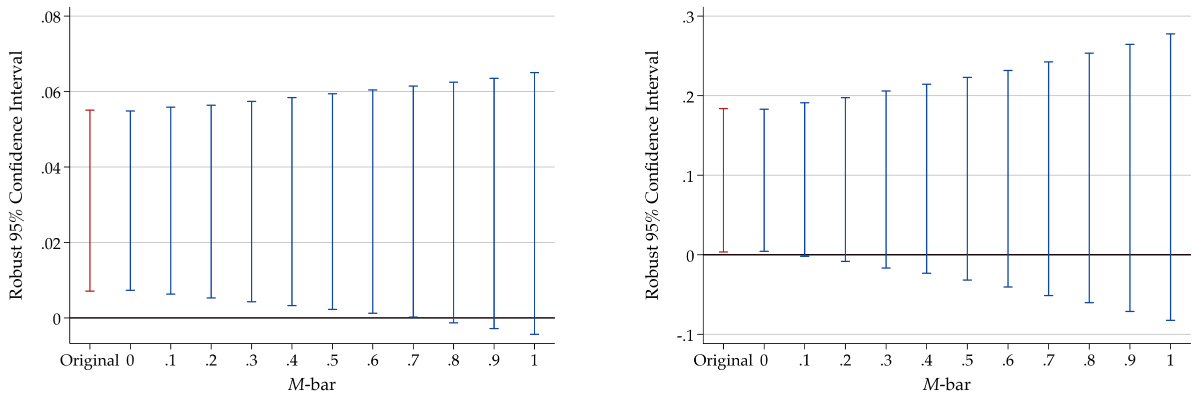
(f) Electrical Horsepower



Notes. This Figure reports the estimates of the baseline province-level regressions when excluding one region. The unit of observation is a province, at a decade frequency between 1911 and 1937. All outcomes are expressed in logs. Regressions include province and decade fixed effects. The left axis indicates the excluded region. The treatment is an interaction term between US emigration and a post-1921 term. Each regression controls for the number of emigrants and an interaction term between an indicator variable for Southern regions and a post-1921 indicator. The blue dots report the OLS estimates; the red dots report the two-stage least-squares obtained using the instrument defined in (2). Units are weighed by their population in 1881. Bands report 95% confidence intervals. Standard errors clustered at the province level. Referenced on page: 23.

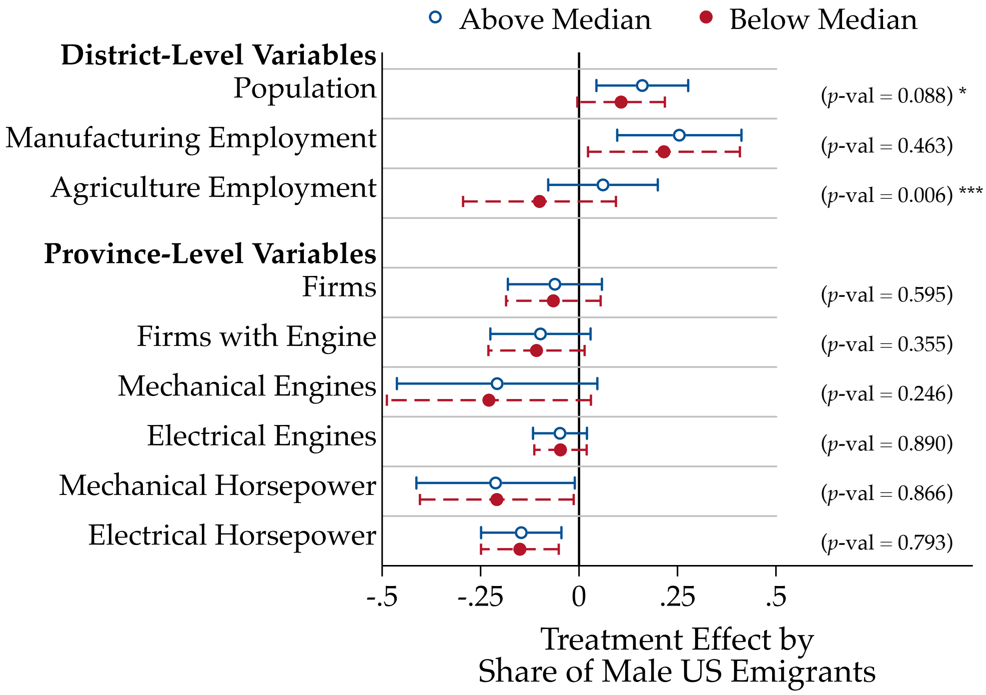
Figure C.17. Relative Magnitude of Parallel Trends Violation

(a) Population **(b)** Manufacturing Employment



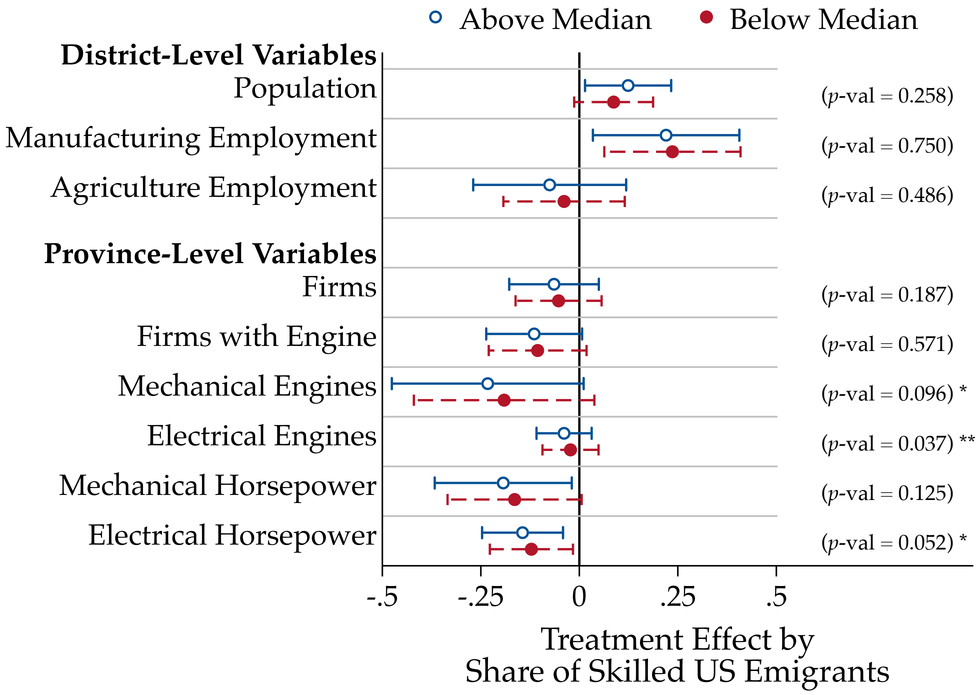
Notes. This Figure presents alternative confidence intervals for the post-Quota coefficient displayed in the event-study graphs for population (Panel C.17a) and manufacturing employment (Panel C.17b). All outcomes are expressed in logs. The bands report 95% robust confidence intervals constructed following the method developed by Rambachan and Roth (2023). In both cases, the unit of observation is a district observed at a census-decade frequency between 1881 and 1936. The regression includes district and decade-fixed effects and controls for the volume of international migrants and an indicator variable for Southern regions, both interacted with a post-1921 indicator. Units are weighed by their population in 1881. Referenced on pages: 15, 23.

Figure C.18. Heterogeneous Treatment Effects by Share of Men Among US Emigrants



Notes. This Figure reports the estimated effect of district- and province-level exposure to the US Quota Acts on the outcomes considered in the paper, by the share of men among US emigrants. The unit of observation is a district or a province observed at a census-decade frequency between 1881 and 1936 (for district-level variables) or between 1911 and 1937 (for province-level variables). The dependent variables are expressed in logs. The treatment is an interaction term between the baseline treatment, itself an interaction term between US emigration and a post-1921 term, and categorical variables coding units with below- and above-median share of men among US emigrants. The gender of the emigrants is imputed as described in the main text. The dots display the treatment effect for districts with above-median (resp. below-median) share of male US emigrants in blue (resp. red). All estimates are obtained via two-stage least squares estimates obtained using the instrument defined in (2). Each regression includes units (i.e., district or province) and time fixed effects and controls for the emigration rate and an indicator variable for Southern regions; both interacted with a post-1921 indicator. Units are weighed by their population in 1881. Standard errors clustered at the district level or at the province level depending on the level of aggregation; bands report 95% confidence intervals. The Figure displays the p-value of a test of equality between the treatment effects of districts with above- and below-median share of male US emigrants. Referenced on page: 25.

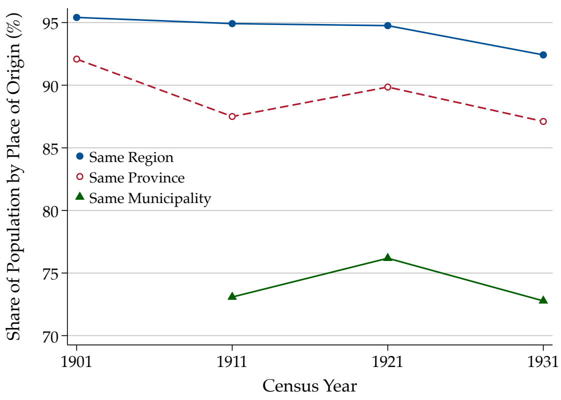
Figure C.19. Heterogeneous Treatment Effects by Share of Skilled Migrants Among US Emigrants



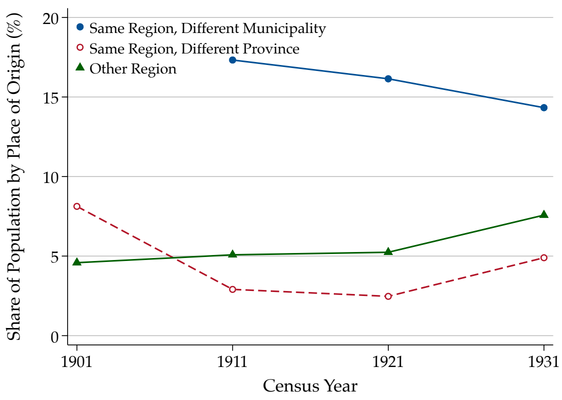
Notes. This Figure reports the estimated effect of district- and province-level exposure to the US Quota Acts on the outcomes considered in the paper, by the share of migrants employed in skilled occupations in the US. The unit of observation is a district or a province observed at a census-decade frequency between 1881 and 1936 (for district-level variables) or between 1911 and 1937 (for province-level variables). The dependent variables are expressed in logs. The treatment is an interaction term between the baseline treatment, itself an interaction term between US emigration and a post-1921 term, and categorical variables coding units with below- and above-median share of skilled migrants within the Ellis Island-Census linked dataset. A migrant is "skilled" if their occupation (OCC1950) in the census is one of "Professional, Technical," "Managers, Officials, and Proprietors," or "Craftsmen." The dots display the treatment effect for districts with above-median (resp. below-median) share of skilled US emigrants in blue (resp. red). All estimates are obtained via two-stage least squares estimates obtained using the instrument defined in (2). Each regression includes units (i.e., district or province) and time fixed effects and controls for the emigration rate and an indicator variable for Southern regions; both interacted with a post-1921 indicator. Units are weighed by their population in 1881. Standard errors clustered at the district level or at the province level depending on the level of aggregation; bands report 95% confidence intervals. The Figure displays the p-value of a test of equality between the treatment effects of districts with above- and below-median share of skilled US emigrants. Referenced on page: 28.

Figure C.20. Internal Migrations Over Time

(a) Stayers

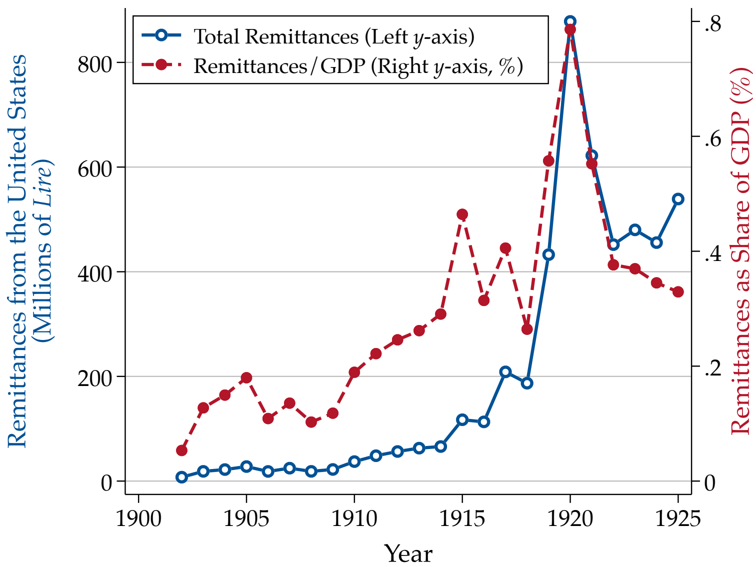


(b) Movers



Notes. This Figure reports the dynamics of internal migrations between 1901 and 1931. We aggregate region-level data from the population censuses. Panel C.20a reports the share of the population within each region that resides in their region of origin (blue), in their province of origin (red), and in their municipality of origin (green). In Panel C.20b, we focus on movers and display the share of the population that remains in their region of origin but moves municipality (blue), those that move to a different province within their region of origin (red), and those that move to a different region (green). Referenced on page: 26.

Figure C.21. Remittances from Italian Emigrants to the United States to Italy



Notes. This Figure displays the amount of financial remittances to Italy paid by Italian emigrants to the United States between 1902 and 1925. The blue dots, shown on the left *y*-axis, report the total nominal amount of remittances in millions of *lire*. The red dots, displayed on the right *y*-axis, report nominal remittances as a share of the Italian GDP, in percentage terms. Historical GDP estimates are from Baffigi (2013). Remittances data are from the *Annuario statistico dell'emigrazione italiana dal 1876 al 1925: con notizie sull'emigrazione negli anni 1869-1875*, Italian Statistical Office (ISTAT), Roma, 1926. Referenced on page: 27.

D A MODEL OF DIRECTED TECHNOLOGICAL ADOPTION

In this section, we develop a simple framework to rationalize our main findings in the context of labor-saving technical change theory. Proofs and further analytical insights on the baseline environment can be found in Appendix D.III.

D.I Theoretical Framework

In this section, we develop a simple analytical framework inspired by Zeira (1998) and San (2023) to clarify the empirical implications of directed technical change and adoption theory. The core assumption we make is that capital goods–hereafter, machines–substitute labor as a production input. We thus implicitly restrict technological progress to be labor-saving, as in Acemoglu (2002, 2007). The decision of the firm to adopt productivity-enhancing machines will depend on their price relative to the cost of labor. In the equilibrium, a labor supply shock–such as the one induced by IRPs–dampens the incentive to adopt machines because it pushes down the wage, hence prompting firms to substitute capital for labor.

Consider a closed economy with one consumption good and a representative household supplying labor. The consumption good is produced by a continuum of tasks $j \in [0, 1]$. Each task can be performed with either labor or machines. The number of machines in task j is denoted by $x(j)$, whereas the amount of labor employed is $e(j)$. Note that each task can be fulfilled with either machines or labor, but not both. This is intended to model, in a stylized manner, labor-saving machines. To simplify the analysis and following Zeira (1998), we assume that machines fully depreciate at the end of the period; hence, the model is essentially static.

The final consumption good is produced by identical, perfectly competitive firms with the following production function:

$$Y = A\left[\int_0^{\iota} m x(j)^{\alpha}\, dj + \int_{\iota}^{1} e(j)^{\alpha}\, dj\right], \tag{D.1}$$

where A is a technology parameter, m is the relative productivity of machines and $\alpha \in (0, 1)$ is a production parameter. We assume $m \in (0, 1)$ following San (2023), and restrict machines to be equally productive across tasks j. The choice variable $\iota \in [0, 1]$ denotes *industrialization* defined as the share of automated tasks, which are those fulfilled by machines. We assume that tasks are ordered by degree of complexity. Because the marginal cost of producing machines–which we define below–is increasing in complexity, the price of machines is non-decreasing in j. It is therefore without loss of generality to assume that the first ι tasks are automatized. This is because the final good producer will first automate tasks whose machine costs the least since the relative productivity of machines is constant across tasks. We assume that there is a fixed stock of labor $L > 0$ which is supplied inelastically by the household.

The problem of the representative final good producer is, therefore, to choose the industrialization level ι, and input quantities $x(j)$ and $e(j)$ for each task to maximize profits

$$\max_{\iota,\{x(j),e(j)\}_{j\in[0,1]}} Y - \int_0^{\iota} p(j)x(j)\,dj - w\int_{\iota}^{1} e(j)\,dj, \tag{D.2}$$

where $p(j)$ is the price of a machine for task j, w is the nominal wage, subject to the technology constraint (D.1). Note that the price of the consumption good is implicitly normalized to one. In Appendix D.III, we formally show that the demand schedules for machines and labor are given by the following demand schedules:

$$x(j) = p(j)^{-\frac{1}{1-\alpha}} (\alpha A m)^{\frac{1}{1-\alpha}} \quad \forall j \in [0,\iota], \tag{D.3a}$$

$$e(j) = w^{-\frac{1}{1-\alpha}} (\alpha A)^{\frac{1}{1-\alpha}} \qquad \forall j \in [\iota, 1]. \tag{D.3b}$$

Combining (D.3a)-(D.3b) with the first order condition for the industrialization rate, it follows that in the equilibrium ι^* is pinned down by the following:

$$m = \left[\frac{p(\iota^*)}{w}\right]^{\alpha}. \tag{D.4}$$

The economic intuition behind condition (D.4) is that at the marginal task, *i.e.* the last automatized task, the price of the machine fulfilling the task must be equal to the cost of labor, adjusted by the technology parameter and the relative productivity of machines.

Each machine is produced by a monopolist, following Zeira (1998). The machine producer will seek to set the monopoly price, which maximizes its profits subject to a demand for machines (D.3a). We assume that the marginal cost of machines $\psi(\cdot)$ is increasing in the complexity of tasks, *i.e.* $\psi'(\cdot) > 0$. Moreover, we assume that the marginal cost function satisfies basic Inada conditions.[35] This is intended to capture the idea that machines substituting low-skill tasks are not as expensive as those replacing tasks on the right side of the skill distribution of workers. The problem of the machine producer is therefore

$$\max_{p(j)} [p(j) - \psi(j)]\, x(j), \tag{D.5}$$

subject to (D.3a). In Appendix D.III, we show that the first-order conditions imply

$$p(j) = \min\left\{mw, \frac{\psi(j)}{\alpha}\right\}, \tag{D.6}$$

where the minimum descends from the observation that, because each task can be performed by labor

[35] In this setting, this simply boils down to $\lim_{j\uparrow 1}\psi(j) = +\infty$ and $\lim_{j\downarrow 0}\psi(j) = 0$. The economic intuition behind these is that it is never profitable for the representative firm to automate all tasks. Similarly, there is always at least one task that is automated. Note that while these assumptions are sufficient for the existence of an equilibrium, they are not necessary.

as well as by machines, setting a price greater than the productivity-adjusted wage simply pushes the final goods producer not to automate the task. We now obtain two technical results to ensure the existence and uniqueness of the equilibrium. The formal definition of the competitive equilibrium in this economy, as well as the proofs of all lemmas and propositions, can be found in Appendix D.III.

Lemma D.1. *In the equilibrium, the marginal task ι^* is such that $p(\iota^*) = \psi(\iota^*)/\alpha = wm^{1/\alpha}$.*

Combining this result with the equilibrium conditions of the final goods producer, we derive the following strong existence result.

Proposition D.1. *There exists one and only one $\iota^* \in [0,1]$ which solves the problem of the final good producer* (D.3a)-(D.3b)-(D.4) *as well as the problem of the machine producers* (D.6) *and verifies labor market clearing. In particular, the equilibrium industrialization ι^* is the solution to the following:*

$$\psi(\iota^*) = L^{\alpha-1}(1-\iota^*)^{1-\alpha}\alpha^2 A m^{1/\alpha}.$$

This concludes our analytical characterization of the environment. We now exploit the model to deliver a number of testable predictions that will guide our empirical analysis.

D.II Empirically Testable Implications

Having established the existence of the equilibrium, we can now derive two key empirical implications of this directed technical adoption setting. First, note that Lemma D.1 conveys the basic intuition of the model. In particular, we have $\psi(\iota^*) = \alpha m^{1/\alpha} w$, hence an increase in the nominal wage induces industrialization to rise because $\psi'(\cdot) > 0$ by assumption. The economic intuition behind this result is that if the cost of labor increases, then the final good producer will seek to automate more tasks in order to avoid paying the increase in wages. This is summarized in the following implication statement.

Implication D.1. Following an exogenous increase (resp. decrease) in the nominal wage w, the share of tasks performed by machines ι^* increases (resp. decreases).

A similar comparative static result follows considering an increase in the labor stock. To see it, notice that because the nominal wage is invariant across tasks, from (D.3b) and labor market clearing, the total labor stock L is evenly allocated across the $(1-\iota^*)$ non-automated tasks. Using this insight, we obtain the following empirical prediction.

Implication D.2. Following an exogenous increase (resp. decrease) in the labor supply stock L, the share of tasks performed by machines ι^* decreases (resp. increases).

This is the key implication of the model that we test in the paper. In our setting, we provide evidence that immigration restriction policies induce positive labor supply shocks, hence increasing the labor stock. We show that firms operating in districts that were more exposed to the Quota Acts decreased investment in machinery–Section IV.B–and increased employment—Section IV.C. These findings are fully in line with the empirical predictions D.2 of the model and hence provide evidence in favor of labor-saving directed technical adoption.

D.III Proofs of Analytical Results

Solution of the problem of the final good producer. Plugging the technology constraint into problem (D.2), the problem of the final good producer reads out as follows:

$$\max_{\iota,\{x(j),e(j)\}_{j\in[0,1]}} A\left[\int_0^{\iota} mx(j)^{\alpha}\,dj+\int_{\iota}^{1} e(j)^{\alpha}\,dj\right]-\int_0^{\iota} p(j)x(j)\,dj-w\int_{\iota}^{1} e(j)\,dj.$$

The–necessary and sufficient–first-order conditions with respect to labor and capital in the generic task j are

$$x(j)=p(j)^{-\frac{1}{1-\alpha}}(\alpha Am)^{\frac{1}{1-\alpha}} \quad \forall j\in[0,\iota],$$
$$e(j)=w^{-\frac{1}{1-\alpha}}(\alpha A)^{\frac{1}{1-\alpha}} \qquad \forall j\in[\iota,1].$$

To obtain the first-order condition for the optimal industrialization rate, apply the Leibniz integral rule with respect to ι to get:

$$x(\iota^*)\left[mx(\iota^*)^{\alpha-1}-p(\iota^*)\right]=e(\iota^*)\left[e(\iota^*)^{\alpha-1}-w\right].$$

Plugging (D.3a)-(D.3b) into the expression above we get $m=(p(\iota^*)/w)^{\alpha}$. □

Solution of the problem of the monopolist. The solution is trivial upon plugging (D.3a) into the objective function (D.5). □

Proof of Lemma D.1. From (D.6) and (D.4), it is

$$p(\iota^*)=\min\left\{\frac{\psi(\iota^*)}{\alpha},mw\right\},$$
$$p(\iota^*)=m^{1/\alpha}w$$

Hence, we have

$$m=\left[\frac{\min\left\{\frac{\psi(\iota^*)}{\alpha},mw\right\}}{w}\right]^{\alpha}.$$

We can distinguish two cases. Assume $mw\leq\psi(\iota^*)/\alpha$. This implies that $m=m^{\alpha}$, which is only verified if $m=1$ or $m=0$. Since by assumption $m\in(0,1)$, this can never hold. We are left with the case $mw>\psi(\iota^*)/\alpha$. We show that this is consistent with all the parameter restrictions. Note first that since $m \in (0,1)$, it must be $\psi(\iota^*)/\alpha < w$, since otherwise it would be $m \geq 1$. We therefore have $\psi(\iota^*)/\alpha < w$ and $\psi(\iota^*)/\alpha < mw$. Because $m < 1$, the only binding constraint is $\psi(\iota^*)/\alpha < mw$. It is

$$m = \left[\frac{\psi(\iota^*)}{\alpha} \cdot \frac{1}{w}\right]^{\alpha},$$

which implies $\psi(\iota^*)/\alpha = wm^{1/\alpha}$. Because $m \in (0,1)$, $m^{1/\alpha} < m$ since $\alpha \in (0,1)$, and therefore $\psi(\iota^*)/\alpha = wm^{1/\alpha} < wm$. This implies that the solution is acceptable. Hence, $p(\iota^*) = \psi(\iota^*)/\alpha$ and this concludes the proof. ☐

Proof of Proposition D.1. Because $w(j) = w$ for all $j \in [0,1]$, from (D.3b) we get that $e(j)$ does not depend on j and:

$$e(j) = e = w^{-\frac{1}{1-\alpha}}(\alpha A)^{\frac{1}{1-\alpha}} = \frac{L}{1-\iota^*},$$

where the last equality holds by labor market clearing, which requires $(1-\iota^*)e = L$. From Lemma D.1, it is $w = \psi(\iota^*)/(\alpha m^{1/\alpha})$. Plugging this into the previous equation, we get

$$\begin{aligned}
\left(\frac{\psi(\iota^*)}{\alpha m^{1/\alpha}}\right)^{-\frac{1}{1-\alpha}}(\alpha A)^{\frac{1}{1-\alpha}} &= \frac{L}{1-\iota^*} \\
\frac{\psi(\iota^*)}{\alpha m^{1/\alpha}}(\alpha A)^{-1} &= \left(\frac{L}{1-\iota^*}\right)^{-1+\alpha} \\
\psi(\iota^*)L^{\alpha-1} &= (1-\iota^*)^{1-\alpha}\alpha^2 A m^{1/\alpha}.
\end{aligned}$$

Because $\psi'(\cdot) > 0$, the left-hand side is strictly increasing in ι^*. Moreover, because $\alpha \in (0,1)$, the right-hand side is strictly decreasing in ι^*. By the Inada conditions, $\lim_{z\uparrow 1}\psi(z) = +\infty$ and $\lim_{z\downarrow 0}\psi(z) = 0$. If $\iota^* = 0$, the right-hand side is strictly positive, whereas it is zero if $\iota^* = 1$. Hence, because both are trivially continuous, by the intermediate value theorem, there exists at least one ι^* which verifies the equation. Since both are strictly monotone, ι^* is unique. ☐

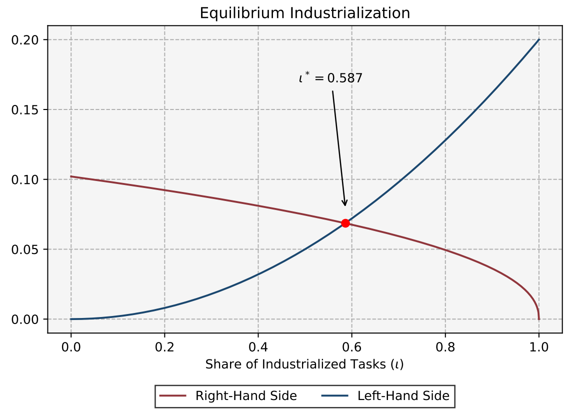
Proof of Implication D.1. From Lemma D.1, it is $m^{1/\alpha} = \psi(\iota^*)/(\alpha w)$, or

$$\alpha w m^{1/\alpha} = \psi(\iota^*).$$

Because $\psi'(\cdot) > 0$, an increase in w in the equilibrium implies an increase in $\psi(\iota^*)$, hence in ι^*. ☐

Proof of Implication D.2. First note that because w is invariant across tasks, then by (D.3b) $e(j) = e$ for all j. Moreover, since the productivity of labor is constant across tasks, it is optimal to divide evenly L across the $(1-\iota^*)$ non-automatized tasks. Therefore, by labor market clearing $e = L/(1-\iota^*)$. Plug

Figure D.1. Equilibrium of the Model



Notes. This figure plots the equilibrium of the model. The blue and red lines, respectively, display the left and right-hand sides of the final equation of the proof of Proposition D.1. We assume $\psi(j) = \gamma j^2$ even though quadratic costs do not verify the Inada conditions.
Parametrization: $\alpha = .55$, $\gamma = .2$, $A = .5$, $L = 1$, $m = .5$.

this in the left-hand side of (D.3b), yielding

$$w^{-\frac{1}{1-\alpha}}(\alpha A)^{\frac{1}{1-\alpha}} = \frac{L}{1-\iota^*}.$$

Using Lemma D.1 into the previous equation we get

$$\begin{aligned}\frac{\psi(\iota^*)}{\alpha m^{1/\alpha}} &= \left(\frac{L}{1-\iota^*}\right)^{\alpha-1}\alpha A \\ L^{1-\alpha} &= \frac{(1-\iota^*)^{1-\alpha}}{\psi(\iota^*)}\alpha^2 A m^{1/\alpha}.\end{aligned}$$

Because $\alpha \in (0,1)$ and $\psi'(\cdot) > 0$, the right-hand side is decreasing in ι^*. Therefore, an exogenous increase in L leads to an increase in the right-hand side, hence a decrease in ι^*. Following an increase in the labor supply, the share of automated tasks decreases. $\square$

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